

GPS Research Slides

2026-xx-xx

Link

Template

- template

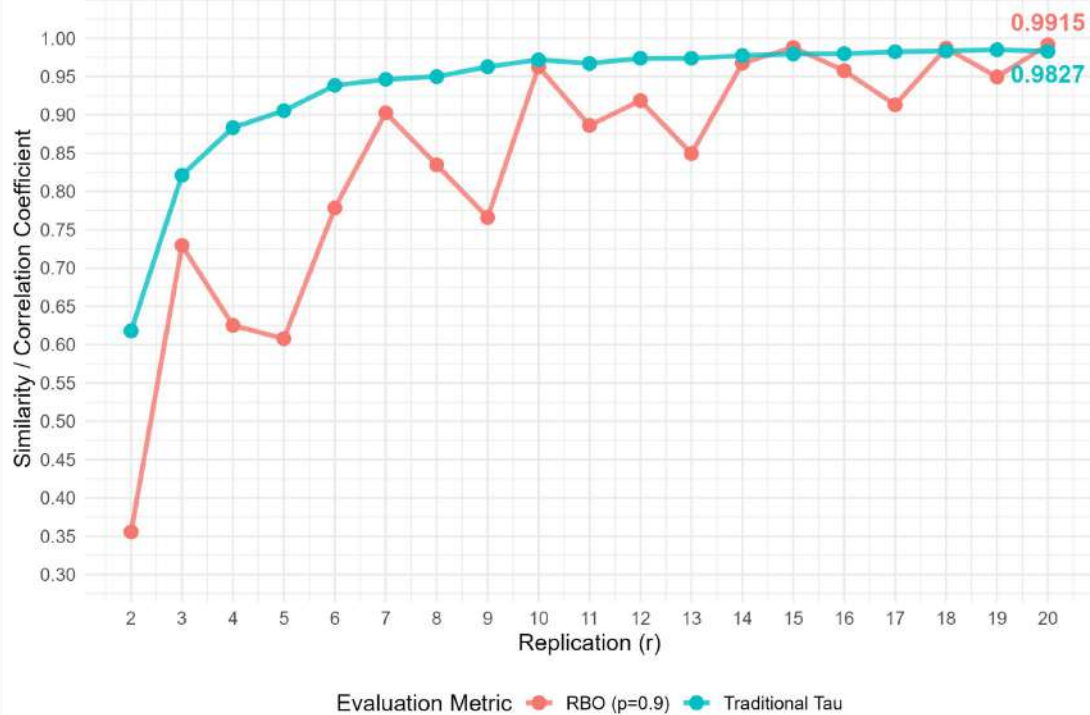
2026-7-7

Link

$t = 160$, $k = 4$, creative

Convergence of Global Ranking Over Replications

Similarity between GlobalRanking(r) and GlobalRanking($r-1$)



Link

$t = 160$, $k = 4$, creative

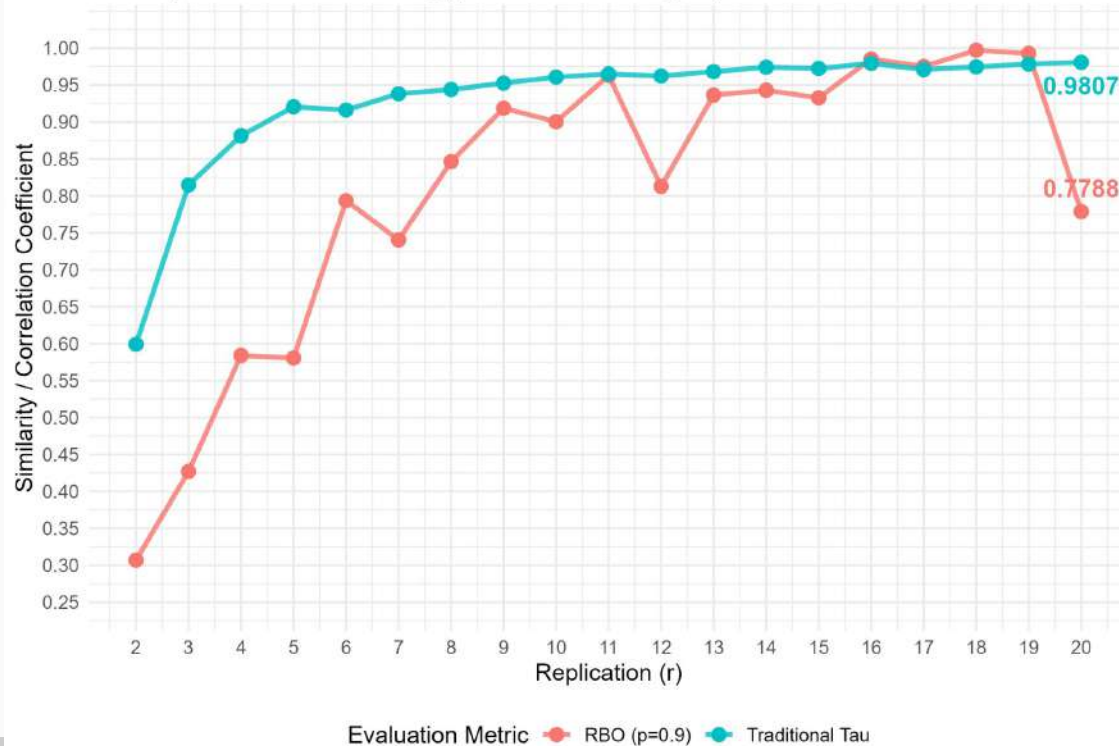
- [Creative ai check](#)

Link

$t = 152$, $k = 4$, practical

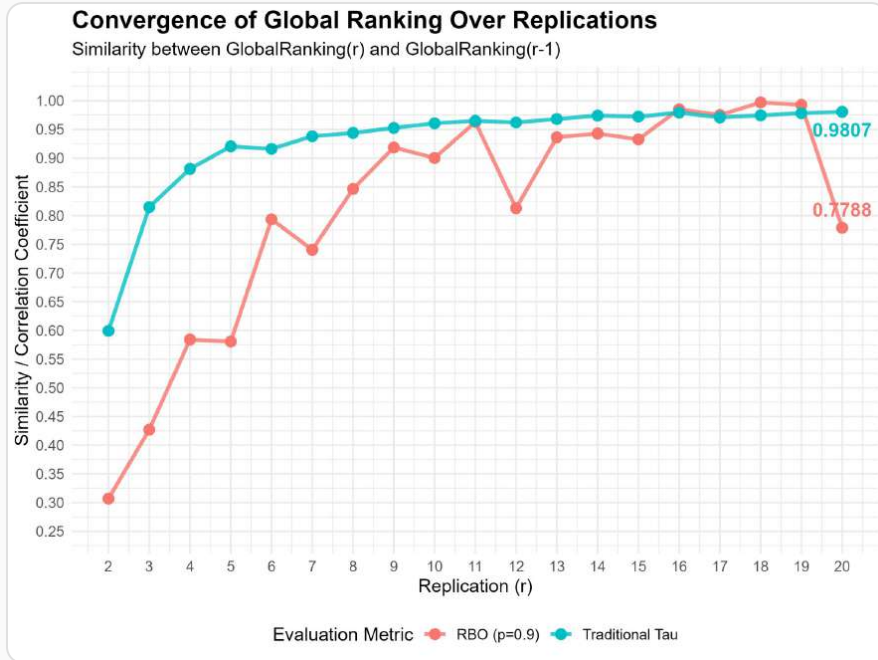
Convergence of Global Ranking Over Replications

Similarity between GlobalRanking(r) and GlobalRanking($r-1$)



Link

$t = 152$, $k = 4$, practical



1. The Anomaly (Metric Divergence)

- Kendall's Tau: Stable at **0.98** (Near-perfect correlation)
- RBO ($p=0.9$): Plummets to **0.77** (Severe penalty)

2. The Root Cause (Plackett-Luce Worth)

- At $r=19$: Item 49 dominated as **Rank 1** (Worth: 0.204)
- At $r=20$: Item 49 worth contracted by ~50% (0.111) to **Rank 4**; Item 18 usurped **Rank 1**

3. Key Conclusion

- **Cons of Tau** : #1 vs #4 flip creates only 3/12,720 discordant pairs (0.02% error), completely masking the disruption
- **Top-Heavy RBO is Essential**: Accurately detects and penalizes critical apex displacement for LLM top-k evaluation

Link

$t = 152$, $k = 4$, practical

ITEM_ID	Worth_r18	Rank_r18	Worth_r19	Rank_r19	Worth_r20	Rank_r20
18	0.15061	3	0.15122	3	0.17815	1
128	0.16295	2	0.16206	2	0.1689	2
143	0.12848	4	0.13056	4	0.14101	3
49	0.20405	1	0.20467	1	0.11119	4
98	0.05305	5	0.05325	5	0.06415	5
75	0.02982	8	0.02952	8	0.04517	6
16	0.02304	9	0.02362	9	0.0385	7
108	0.04933	6	0.04794	6	0.03464	8
139	0.01666	11	0.01725	11	0.02506	9
43	0.03489	7	0.03411	7	0.02384	10

Link

$t = 152$, $k = 4$, practical

- [Practical ai check](#)

Link

Outline

- <https://docs.google.com/document/d/1MqBOD99TDPUqTHu0LzITY7rvL7BXnEy7YB-9tJt7w50/edit?tab=t.0>

2026-06-23

Kendall's Rank Correlation

- Measure of Relative Agreement
- Evaluates how often two rankings agree on the relative order of any given pair of items.
- Treatment of Ties as Uncertainty
- A tie indicates an inability to distinguish between items, resulting in a loss of ordering information.
- Adjustment Mechanism
- To prevent artificially inflated scores, the denominator is dynamically reduced to exclude pairs that were tied (unresolved)

Kendall's Rank Correlation

- Eqn behind it

$$\tau_b(U, V) = \frac{\sum_{i < j} \text{sign}(u_j - u_i) \cdot \text{sign}(v_j - v_i)}{\sqrt{\sum_{i < j} \text{sign}(u_j - u_i)^2} \sqrt{\sum_{i < j} \text{sign}(v_j - v_i)^2}}, \quad (10)$$

Kendall's Rank Correlation

o Setup:

ing 3 items (A, B, C). Total possible pairs = 3.

ng U: 1st A, 2nd B, 3rd C (No ties)

ng V: 1st A, 2nd B & C (Tied!)

The Numerator (Pairwise Scoring)

(A, B): U says A wins (+1), V says A wins (+1). 👍 $1 * 1 = 1$

(A, C): U says A wins (+1), V says A wins (+1). 👍 $1 * 1 = 1$

(B, C): U says B wins (+1), V says Tie (0). 👍 $1 * 0 = 0$

Numerator = $1 + 1 + 0 = 2$

The Denominator (The Tau-b Adjustment)

ut adjustment (Naive scoring):

g by total pairs (3) gives a score of $2 / 3 = 0.67$. (Unfair, because Ranking V lost a pair to uncertainty, not to dis

Tau-b adjustment (sqrt formula):

left term (Ranking U): $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

right term (Ranking V): $\sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$

True Denominator = $\sqrt{3} * \sqrt{2} = \sqrt{6} \approx 2.45$

Link

Kendall's Rank Correlation via wdm Package

0.816

on, the denominator mathematically ignores the tied pairs (since $0^2 = 0$). It dynamically shrinks the denominator to "preserve information" and raising the score from an unfair 0.67 to an accurate 0.816.

Rank-Biased Overlap (RBO)

- Top-Heavy Weighting Mechanism
- Prioritizes agreement at the top of the ranking using a geometric decay parameter (p). Mistakes at rank 1 are penalized far more heavily than mistakes at rank 10.
- Handling Incomplete & Non-conjoint Lists
- Mathematically assumes rankings are prefixes of infinitely long lists, enabling robust comparison even when rankings contain different items.
- Resolves ties by computing the expected RBO score across all possible permutations of the tied items

Rank-Biased Overlap (RBO)

- Eqn

$$RBO_{S,L,p} = \frac{1-p}{p} \sum_{d=1}^{\infty} A_{S,L,d} \cdot p^d, \quad (3)$$

Rank-Biased Overlap (RBO)

$$(1 - p) * \text{Sum} [A_d * p^{(d-1)}]$$

at Depth d)

evaluating the whole list at once, RBO calculates the overlap proportion step-by-step from the top.

At Depth 2, if List U has [A, B] and List V has [A, C], the overlap is 1 out of 2. Thus, $A_2 = 50\%$.

Top-Heavy Weighting

the "Top-Heavy" rule. The parameter p (e.g., 0.9) ensures that mismatches at the top are penalized far more heavily.

Depth 1 is weighted at $0.9^0 = 1.0$ (100% weight). Depth 3 is weighted at $0.9^2 = 0.81$. A mismatch at rank 1 immediately

Normalization Constant

Information theoretically extends to infinity, the raw total could grow into a large arbitrary number.

Multiplying by the constant $(1 - p)$ mathematically compresses the infinite geometric series, guaranteeing the final score

is identical to 1.0 (identical).

Score Upgrade (Fractional Overlap)

If List V ties items B and C for 2nd place, it's impossible to determine which item truly occupies rank 2.

The algorithm converts ties into probabilities. At Depth 2, it assumes a 50% chance for B and a 50% chance for C. The

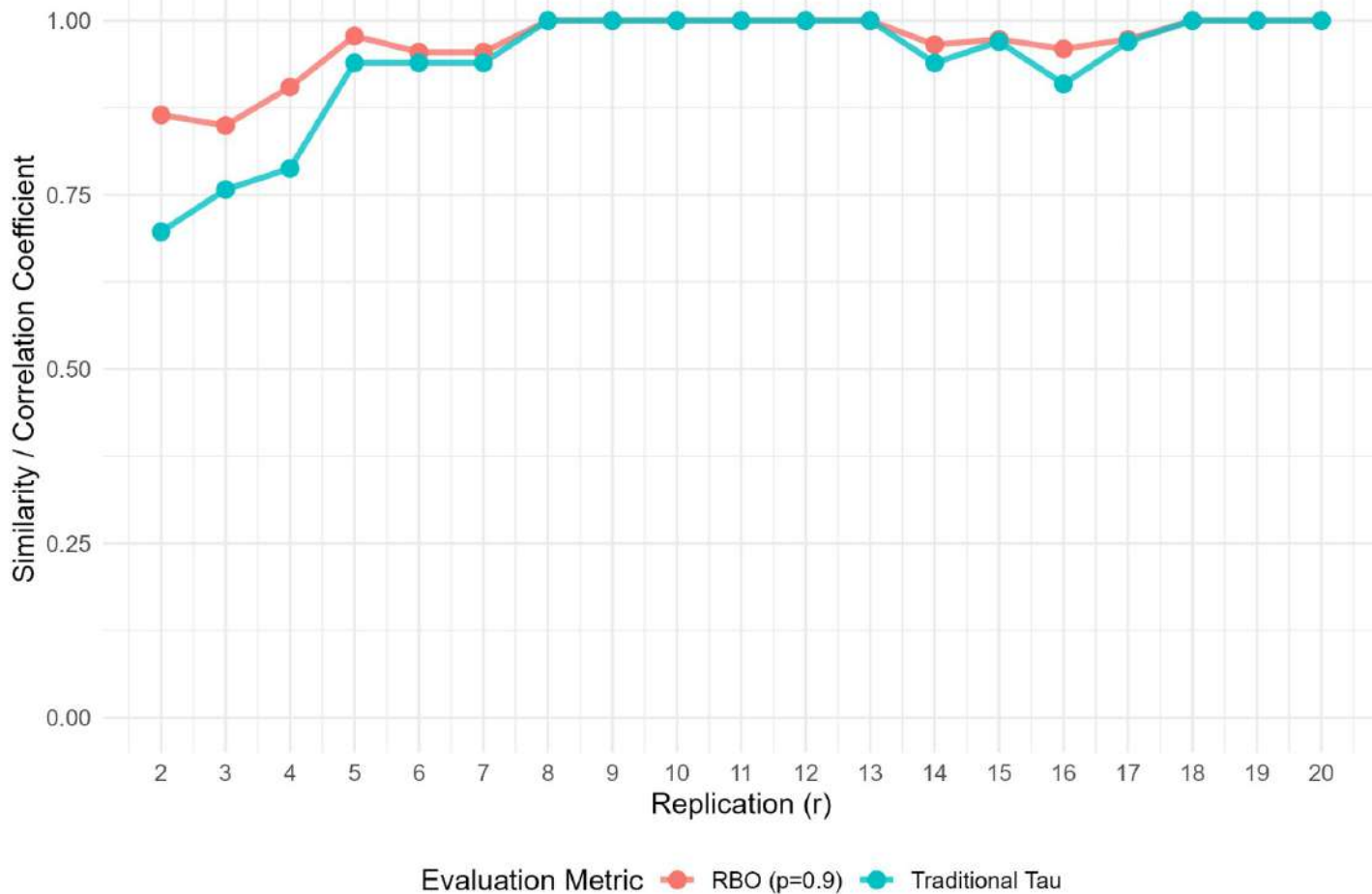
weight is 0.5 rather than 1. This elegantly computes the exact average of all possible permutations without brute-force calculation.

Link

- $t = 12$

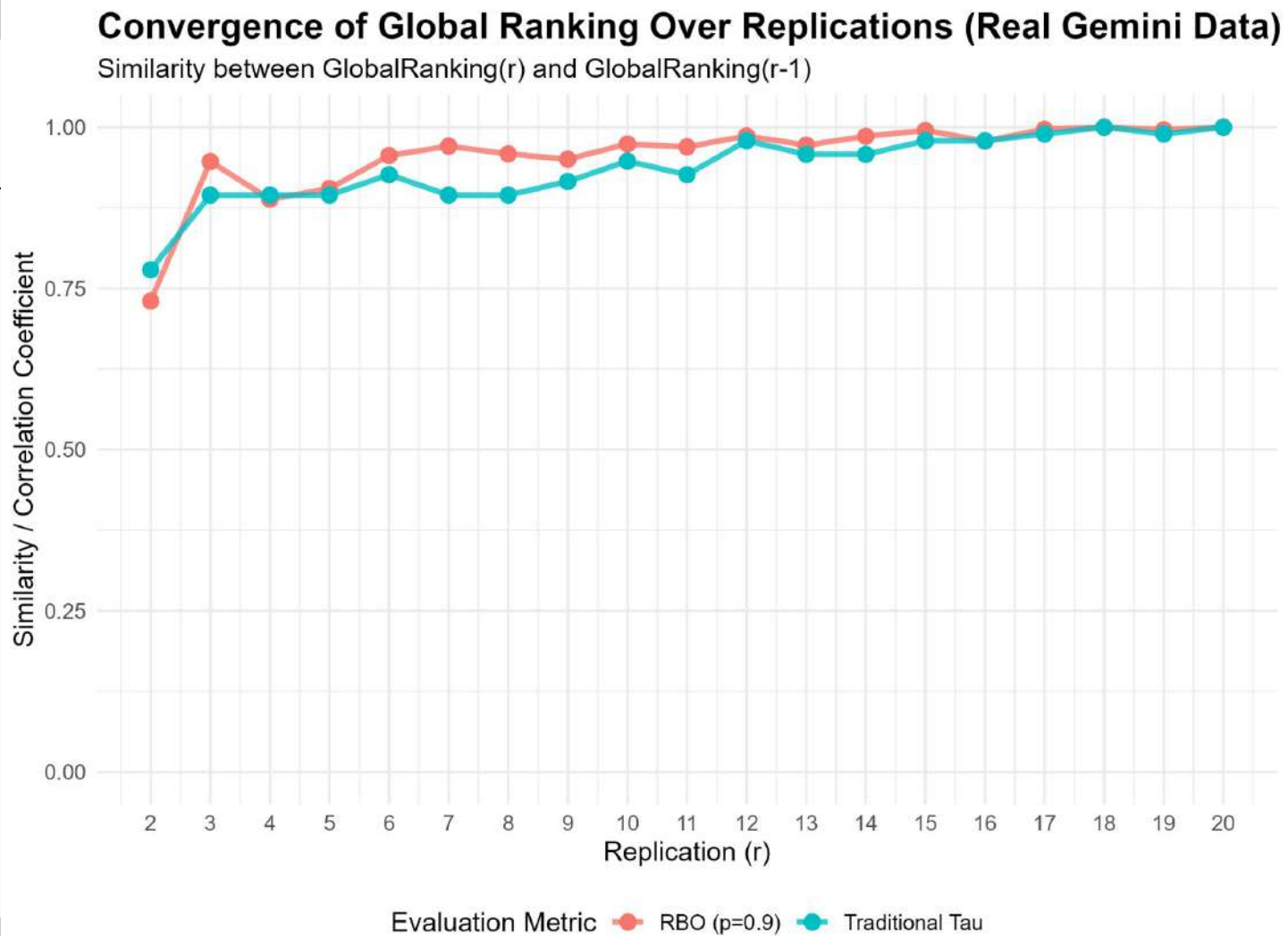
Convergence of Global Ranking Over Replications (Real Gemini Data)

Similarity between GlobalRanking(r) and GlobalRanking($r-1$)



Link

- $t = 20$



2026-06-16

Link

Local model result

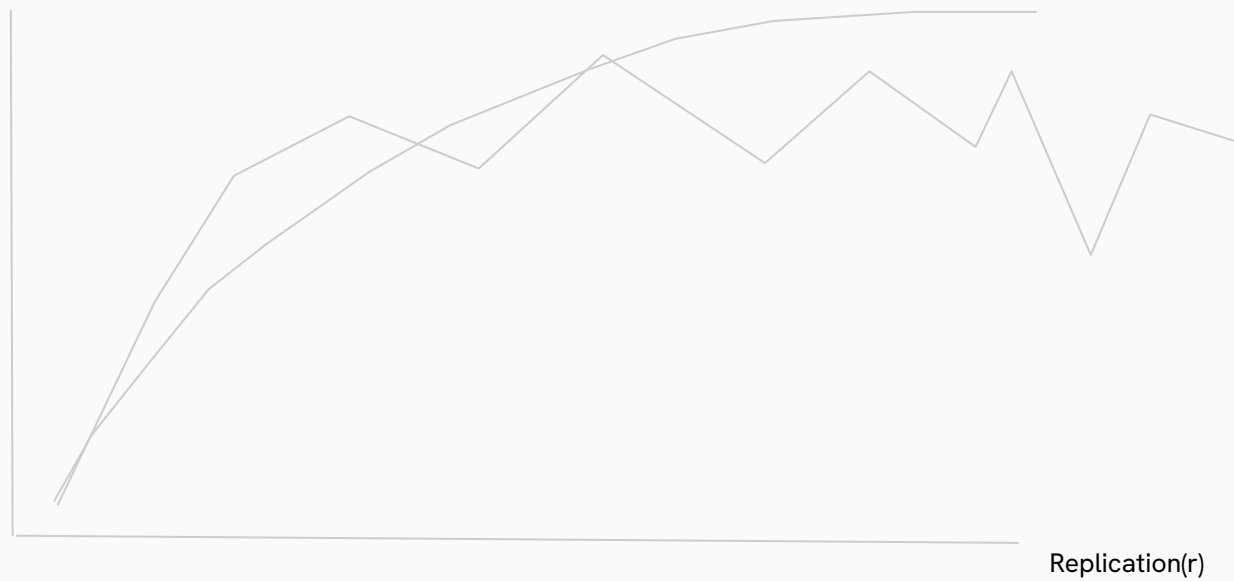
- 2 seeds result
 - [One](#)
 - [Two](#)

Link

Gemini result

- [Result](#)
- [Json](#)
- [PL result](#)
- [AI Rank Code](#)
- [PL code](#)

RBO (r vs r-1)



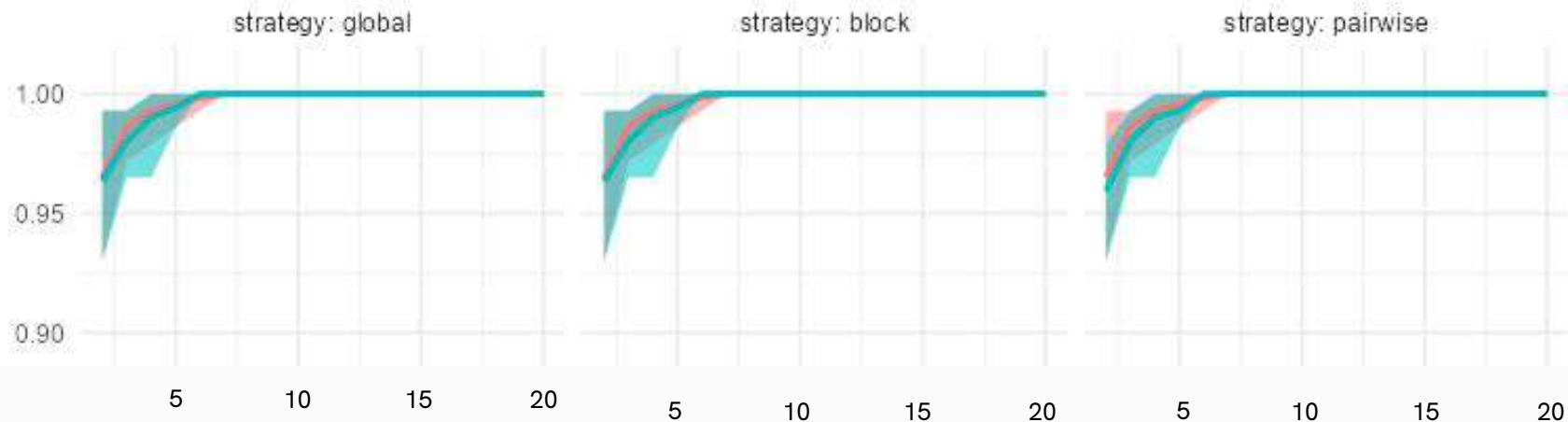
2026-06-02

Link

$$t = 12$$

Model Comparison (k=4): Spearman Rho

Ribbon represents the Min-Max range across seeds. Y-axis aligned.

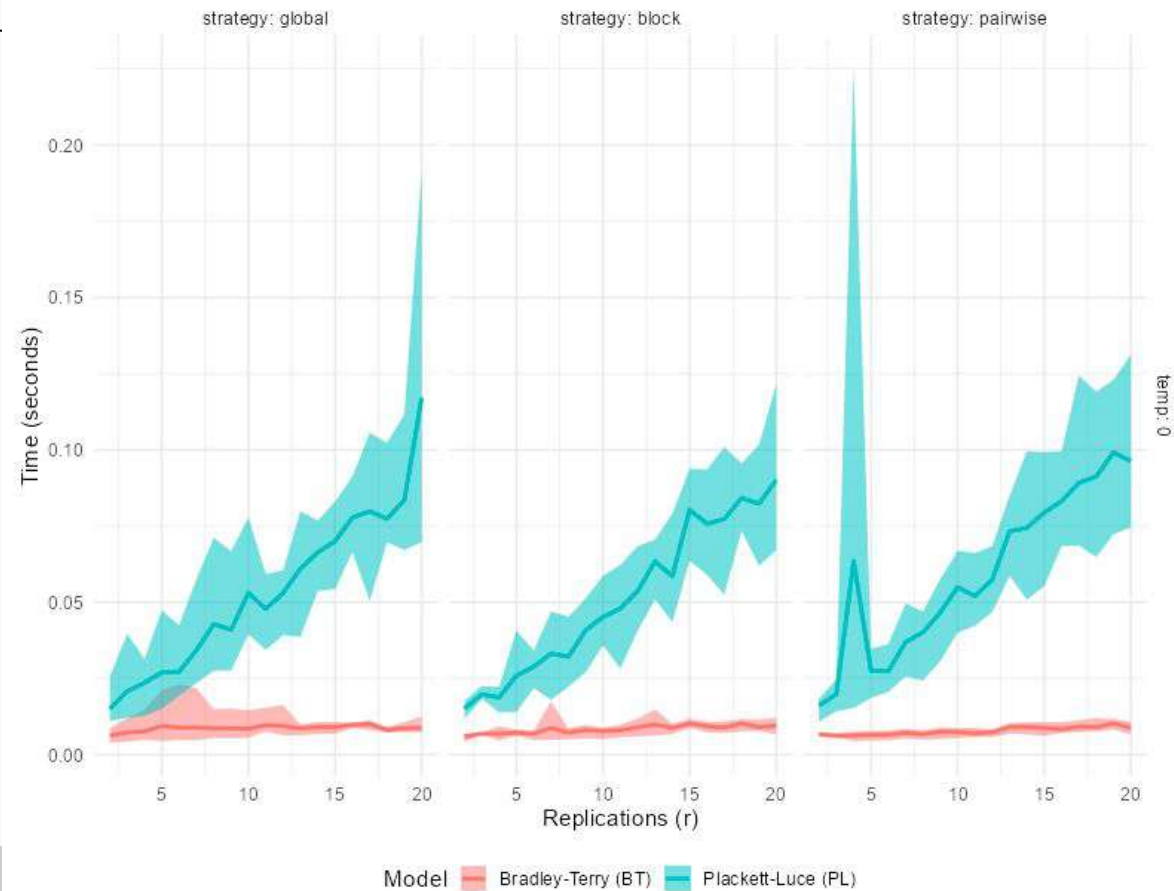


Select k for Time Comparison:

4

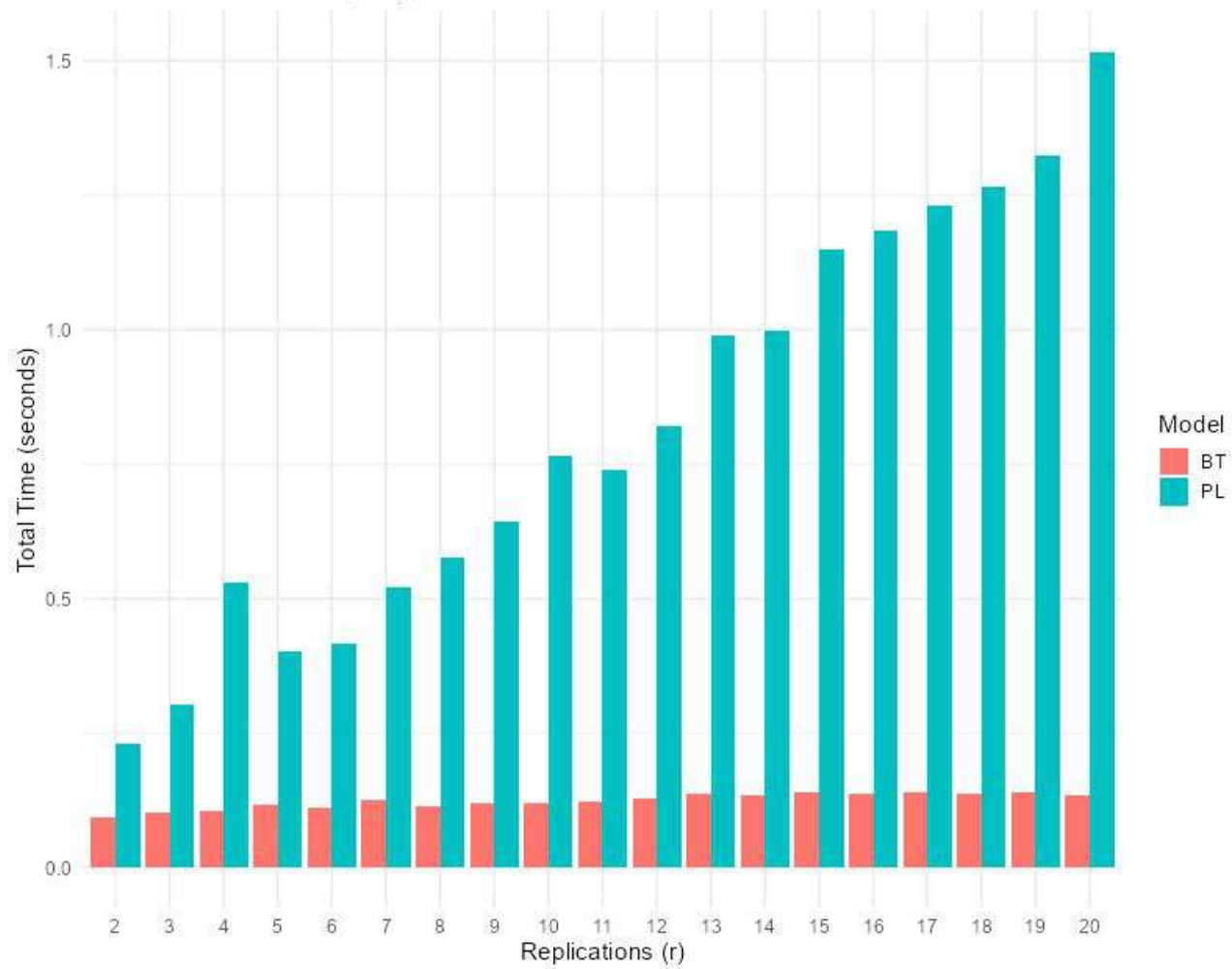
Computing Time Comparison (k=4)

Time in seconds. Ribbon represents variation across seeds.



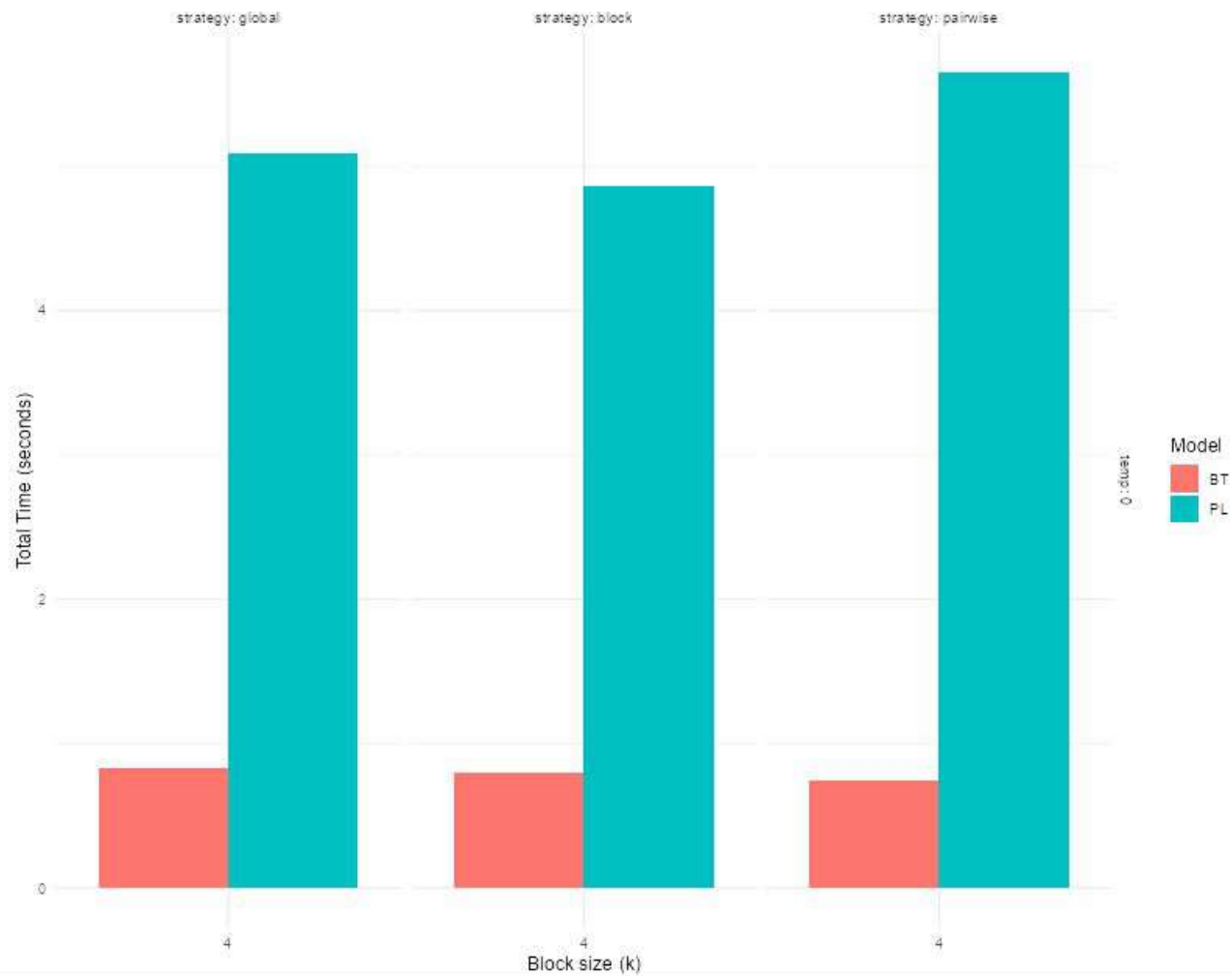
Total Computing Cost by Replication (r)

Sum of time across all k, temp, and seeds for each r



Total Computation Time per Scenario

Sum of all replications and seeds

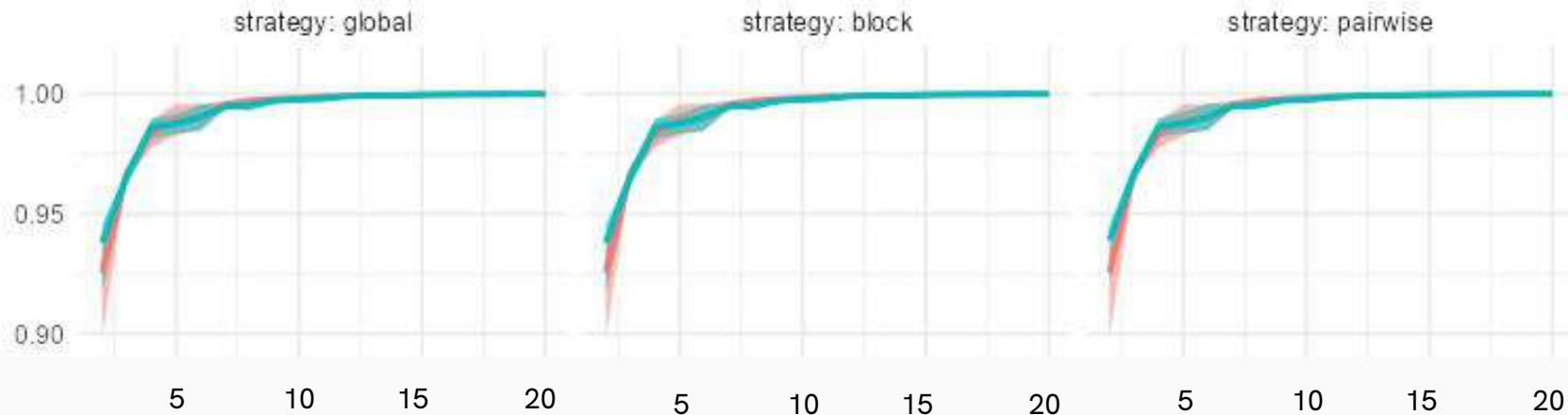


Link

$t = 48$

Model Comparison (k=4): Spearman Rho

Ribbon represents the Min-Max range across seeds. Y-axis aligned.

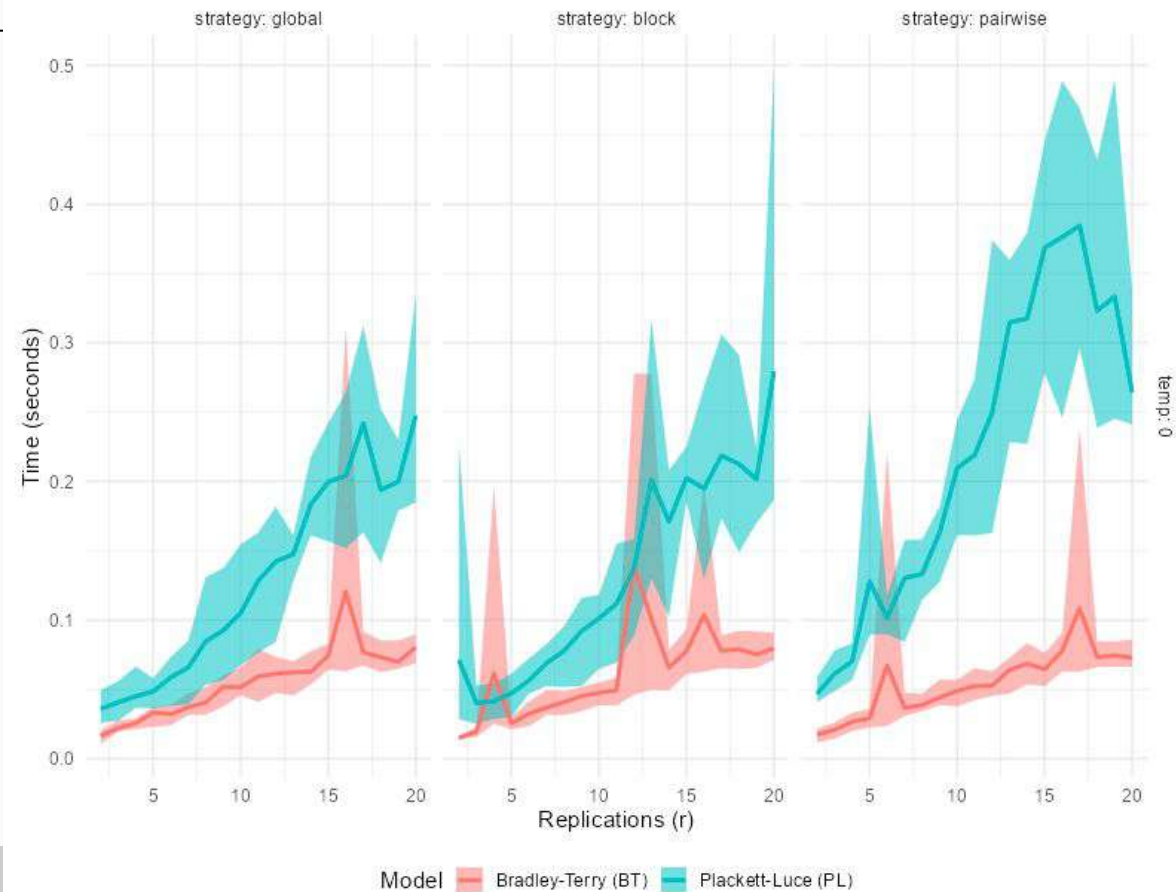


Select k for Time Comparison:

4

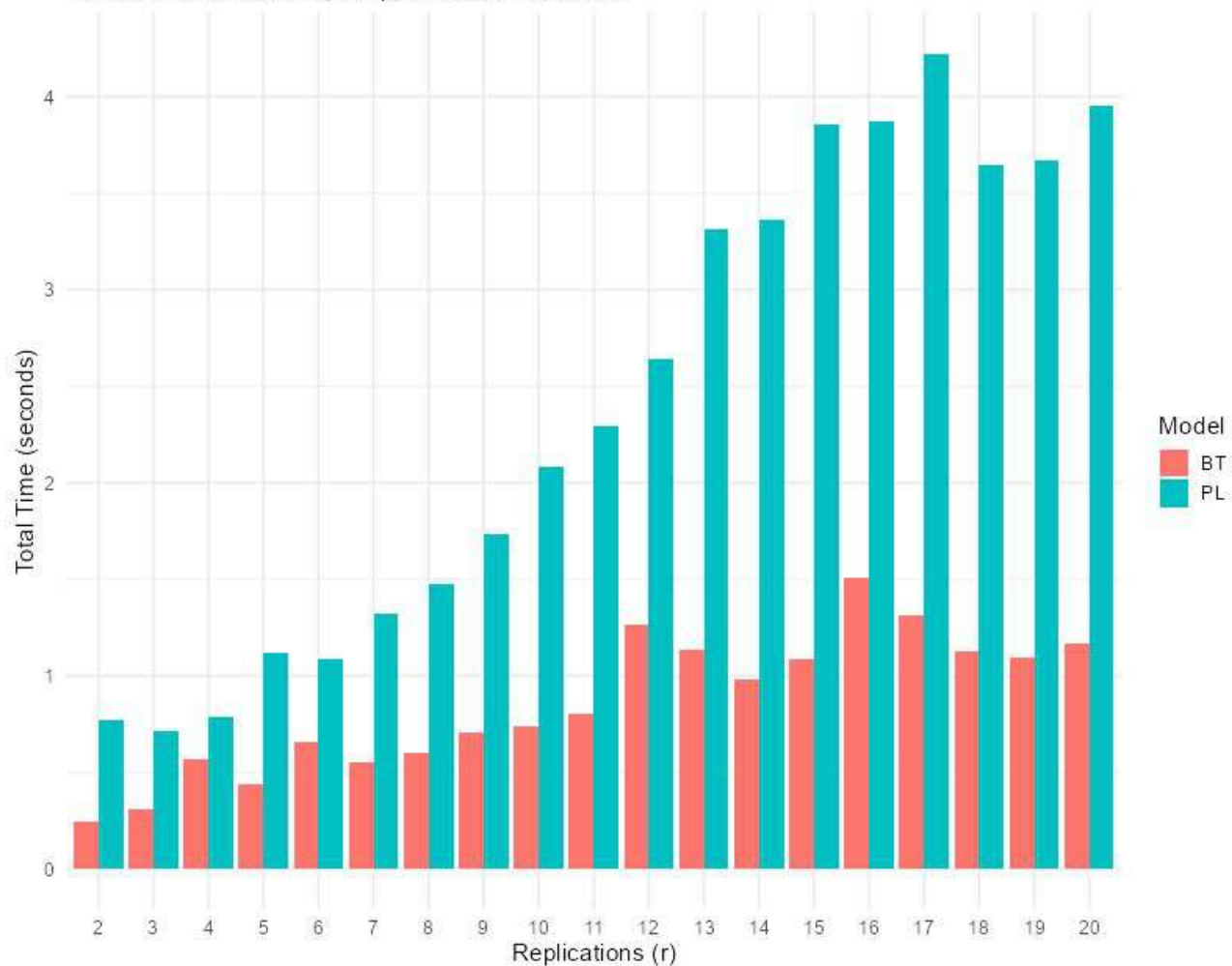
Computing Time Comparison (k=4)

Time in seconds. Ribbon represents variation across seeds.



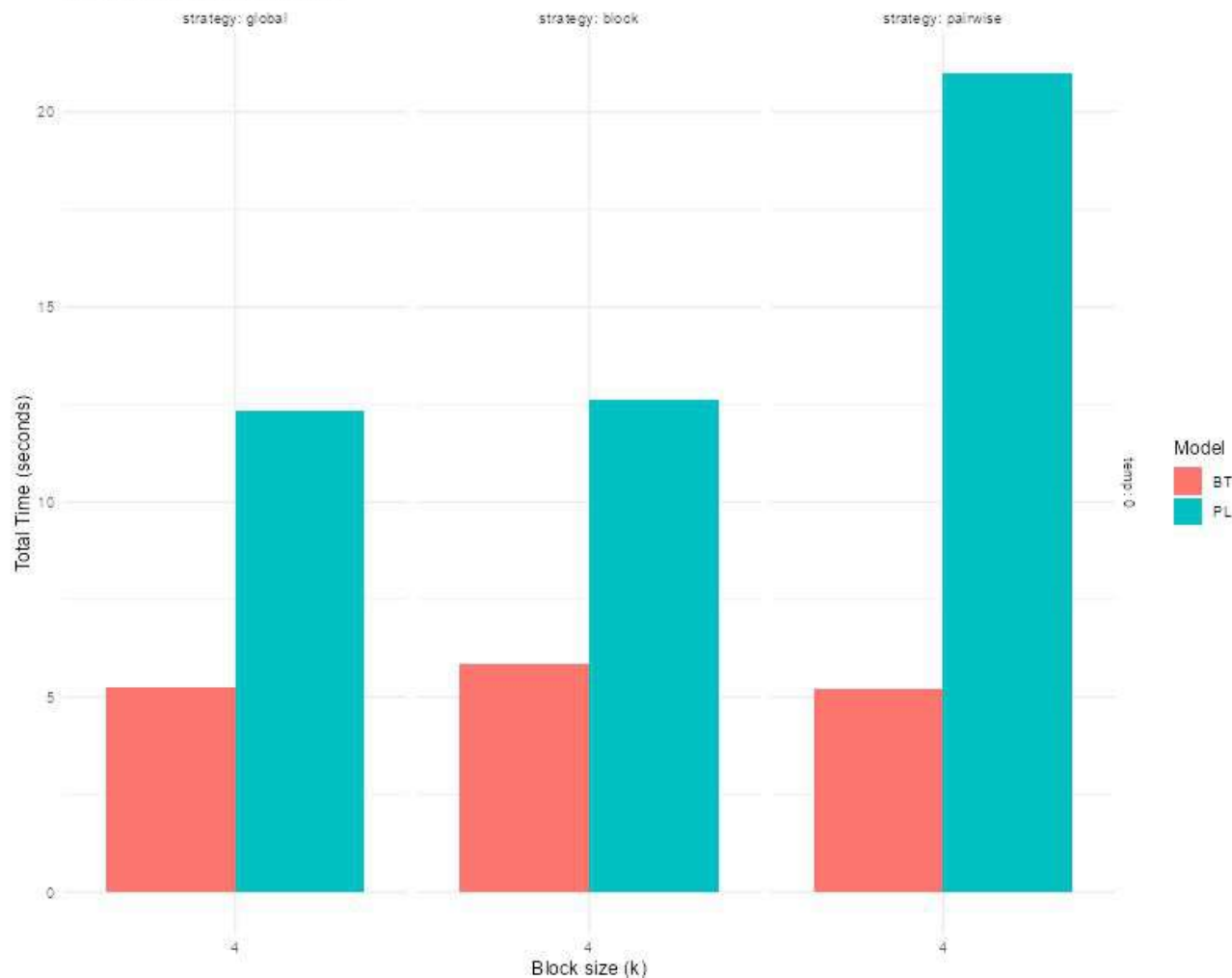
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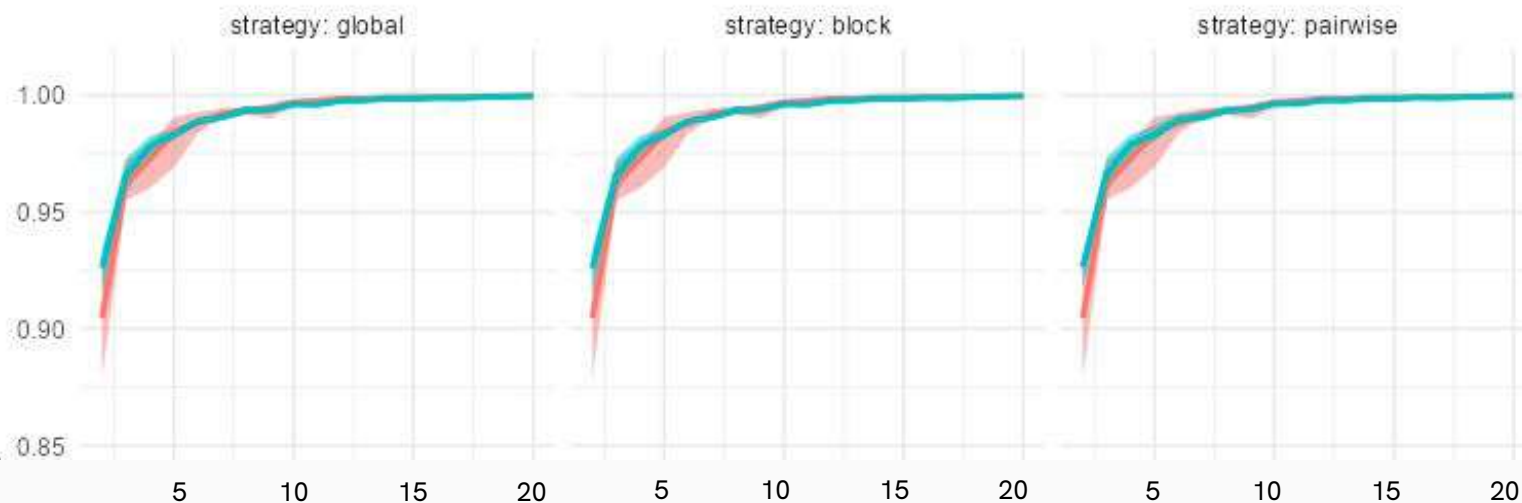


Link

$t = 80$

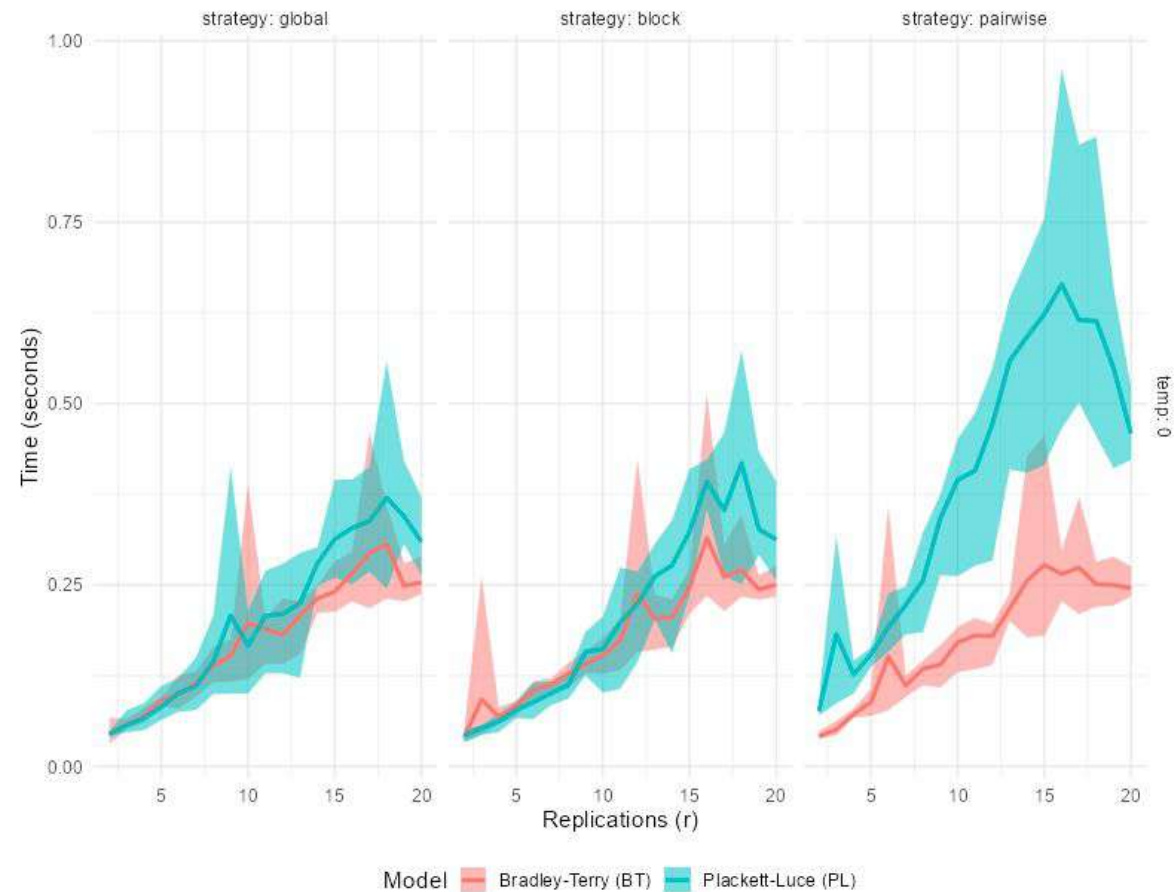
Model Comparison (k=4): Spearman Rho

Ribbon represents the Min-Max range across seeds. Y-axis aligned.



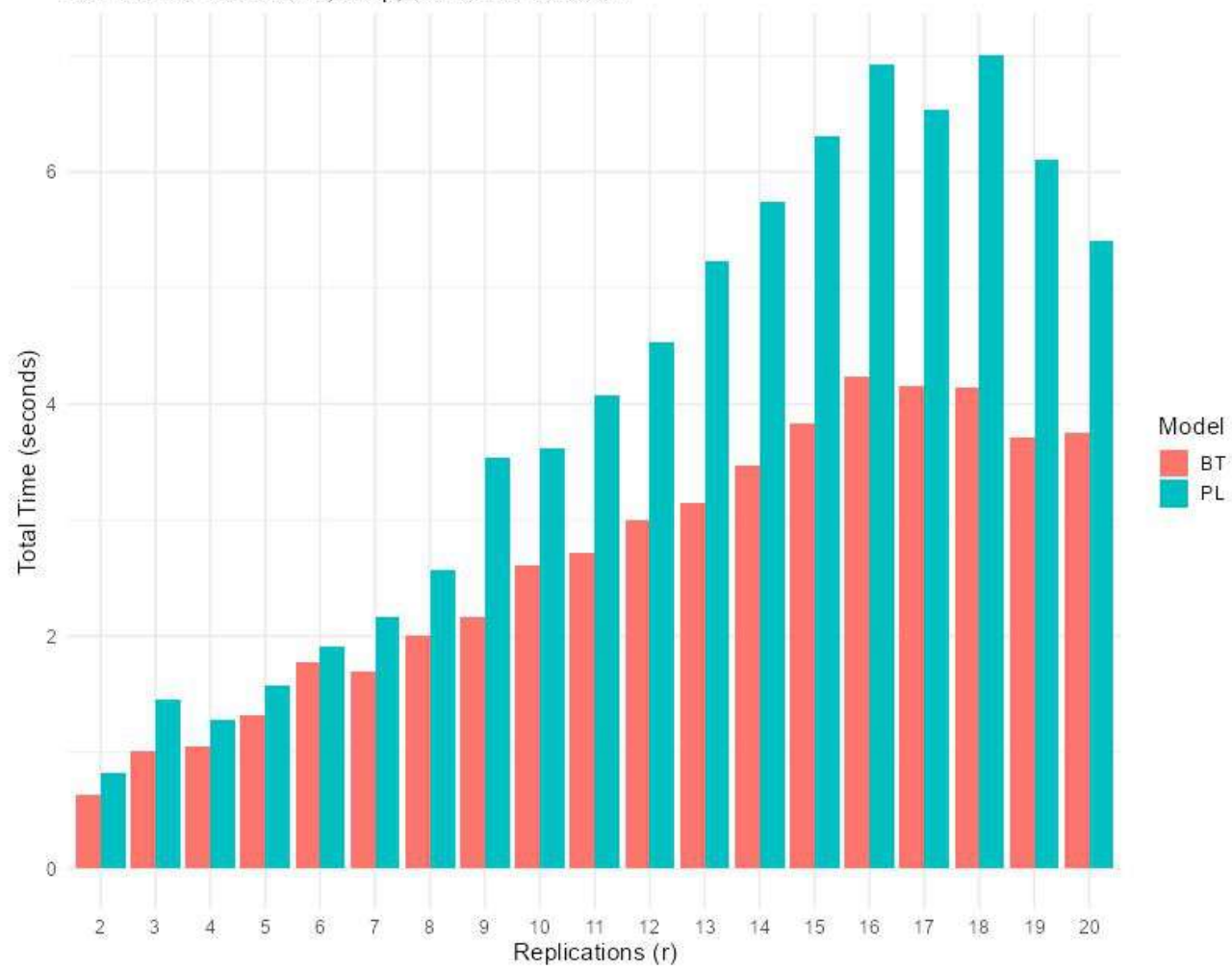
Computing Time Comparison (k=4)

Time in seconds. Ribbon represents variation across seeds.



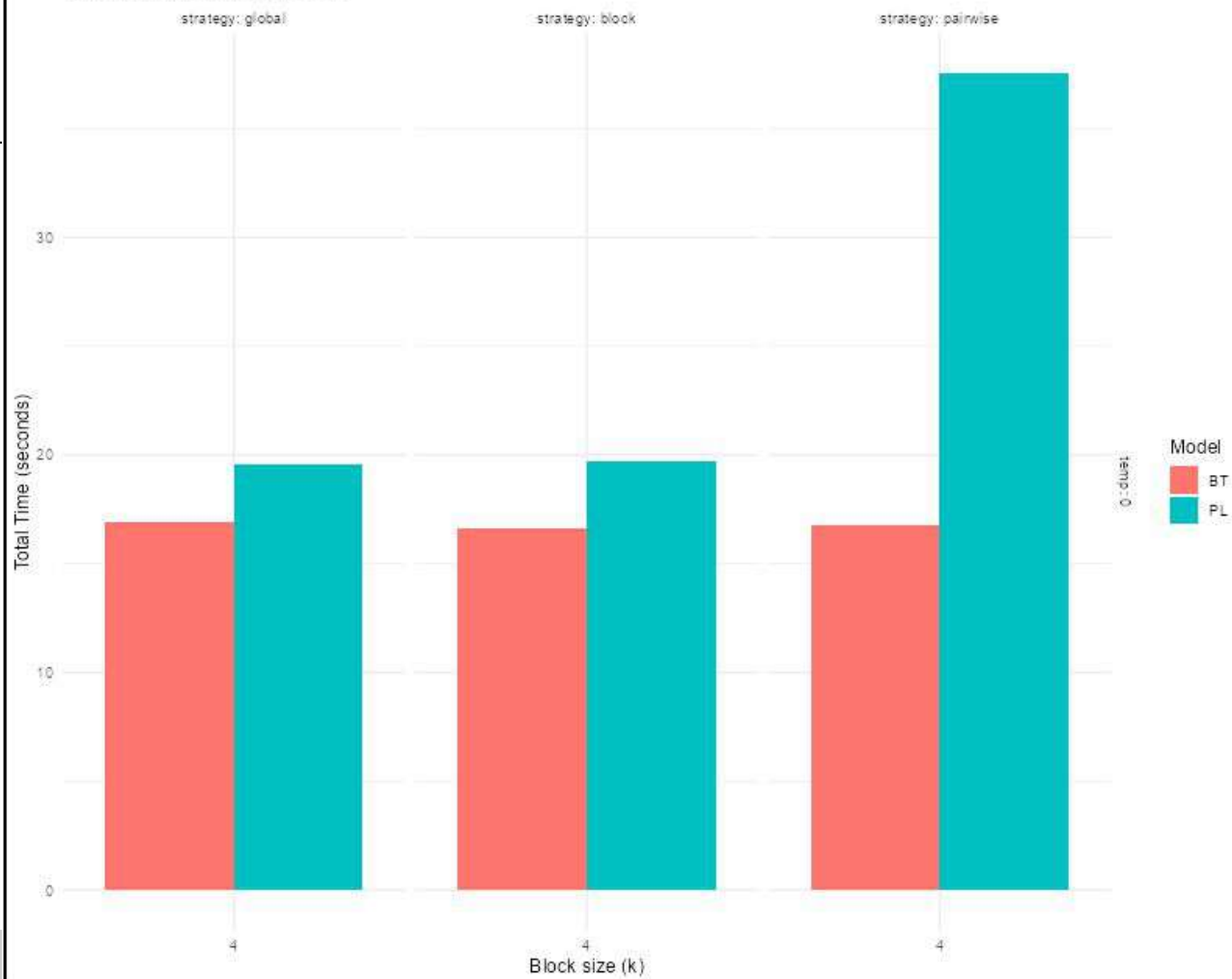
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Sum of time across all k, temp, and seeds for each r



Total Computation Time per Scenario

Sum of all replications and seeds

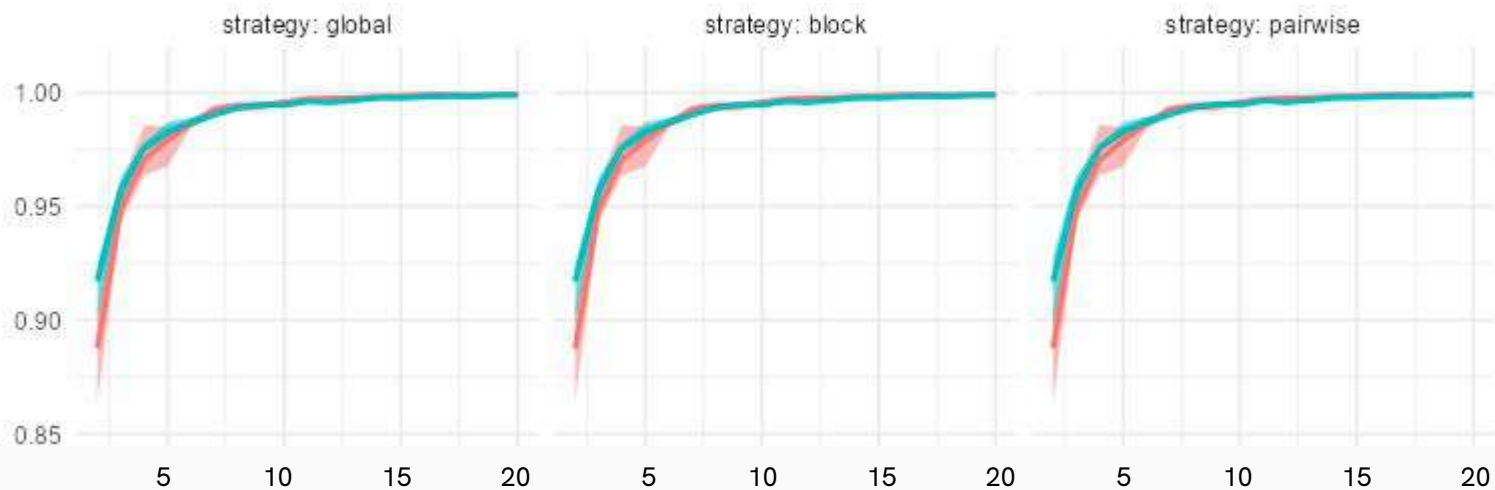


Link

$$t = 120$$

Model Comparison (k=4): Spearman Rho

Ribbon represents the Min-Max range across seeds. Y-axis aligned.

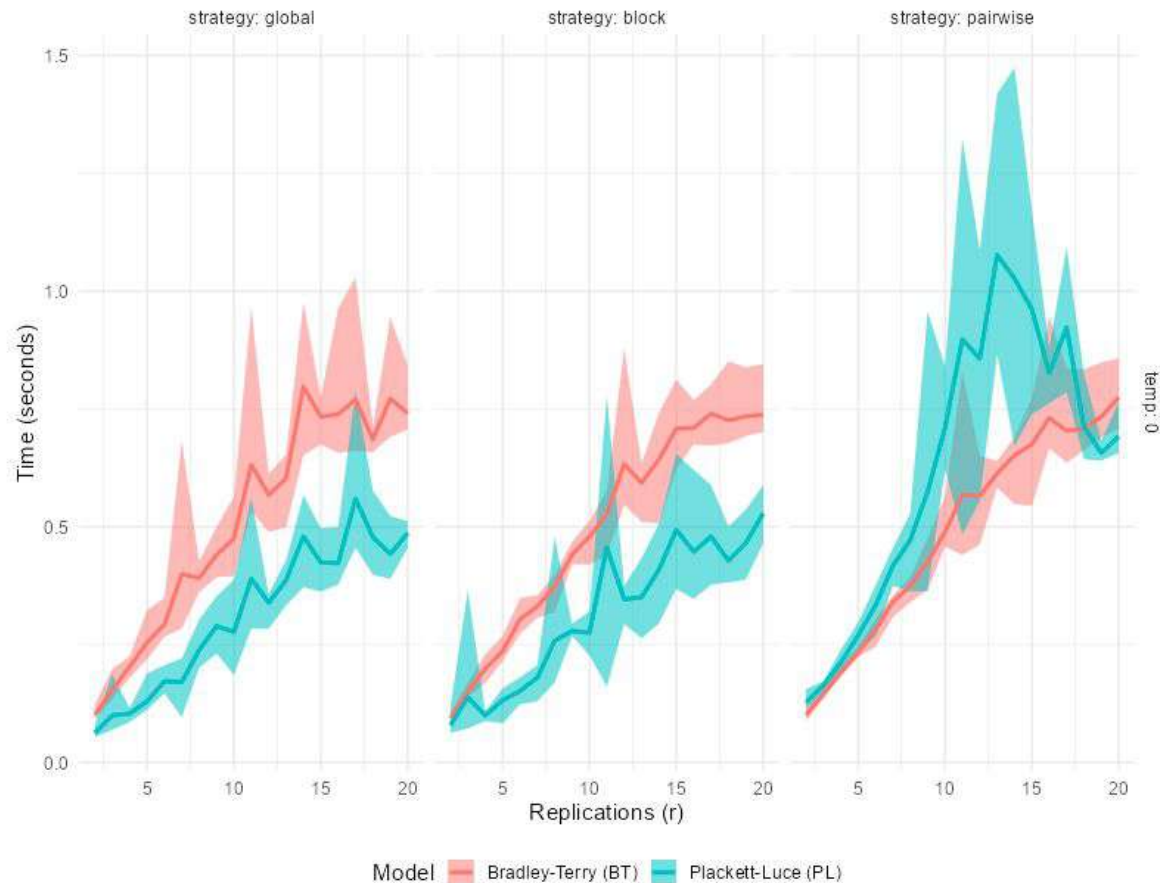


Select k for Time Comparison:

4

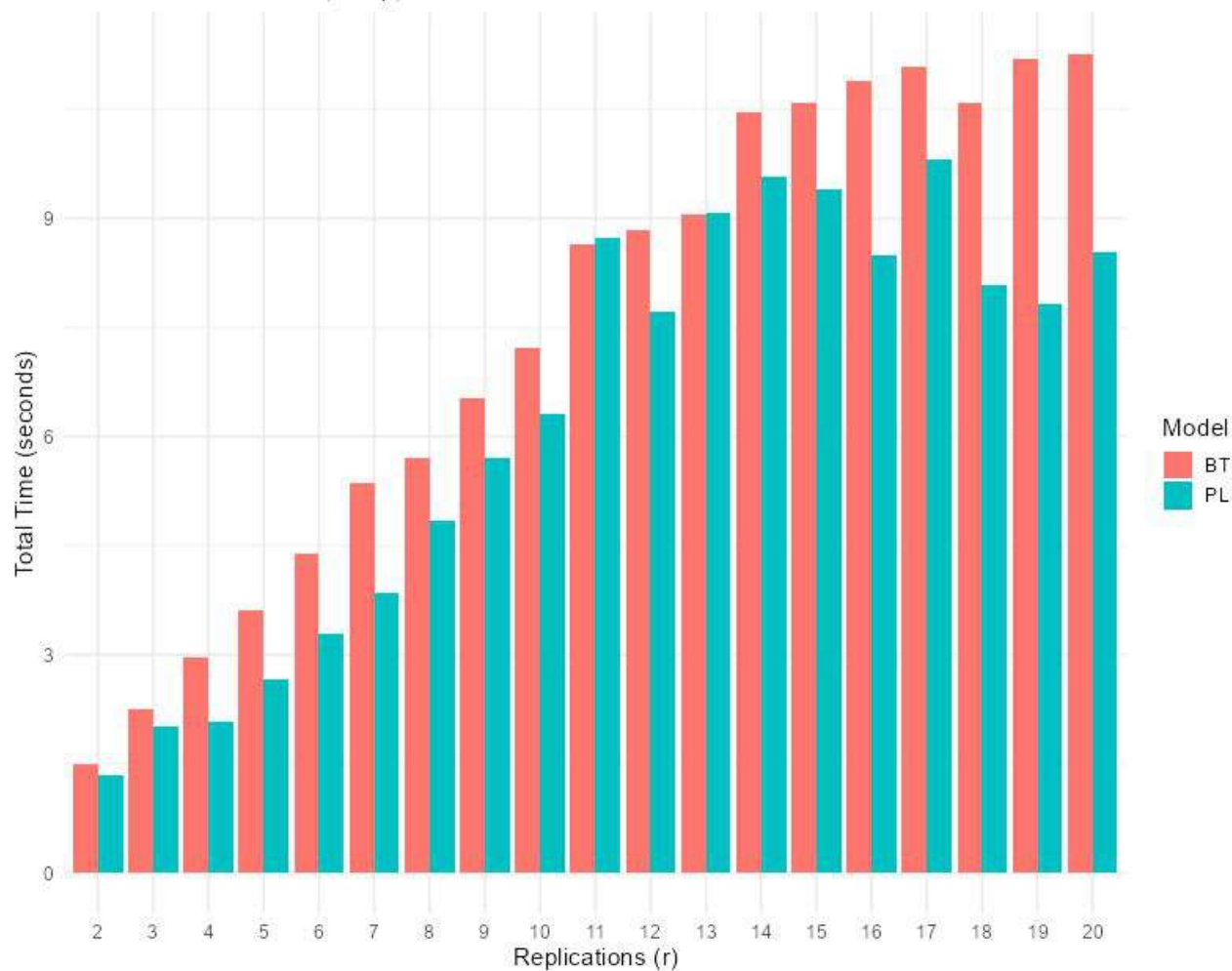
Computing Time Comparison (k=4)

Time in seconds. Ribbon represents variation across seeds.

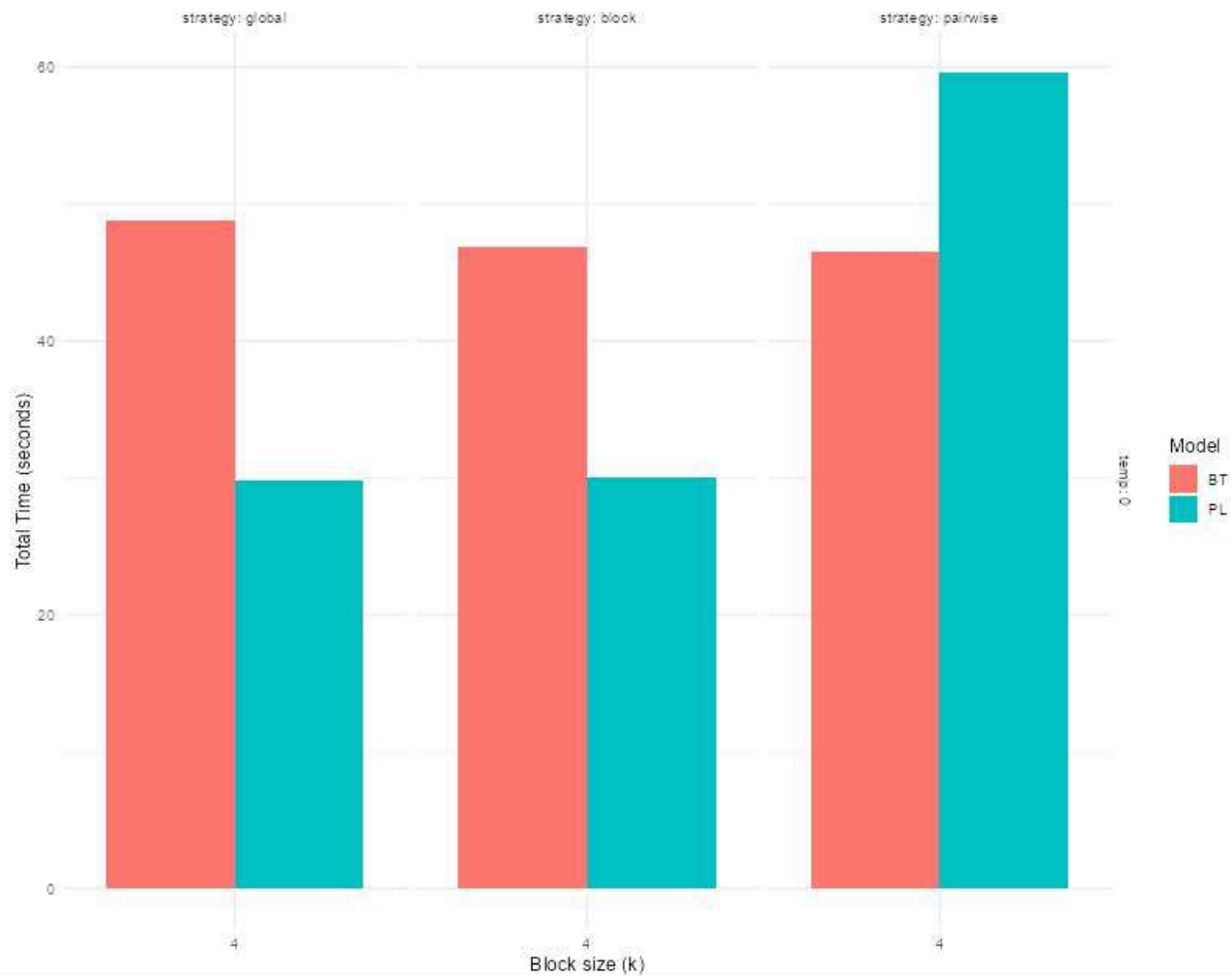


Total Computing Cost by Replication (r)

Sum of time across all k, temp, and seeds for each r



Total Computation Time per Scenario
Sum of all replications and seeds



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- https://github.com/megrez97-boop/Large-Choice-Sets-ranking/blob/main/studies/AI%20ranking%20test/app_llm_evaluator.R
 - <https://drive.google.com/file/d/1nBjG8Jxim82oTPfGMAzJm494JGzW4hIS/view?usp=sharing>
 -

2026-05-26

How does the Plackett-Luce (PL) Model Interpret Rankings?

- Assume we have $t = 12$, $r = 3$, $k = 3$. After using alpha design, we will have $12 \cdot 3 / 3 = 12$ blocks.
- Let's say we randomly pick a block, which with treatment A, B, C inside it.
- So, when we try to define PL model with the block we have

A single ranking is given by

$$R = \{C_1, C_2, \dots, C_J\}$$

where the items in set C_1 are ranked higher than (better than) the items in C_2 , and so on. If there are multiple objects in set C_j these items are tied in the ranking.

- It will generate a R for this block (and also other block).
- So the result after the block is provided, we might have the following situation

How does the Plackett-Luce (PL) Model Interpret Rankings?

- No draw in the result , for example $A > B > C$

Then we will have $R = \{C_1, C_2, C_3\}$

$R = \{A, B, C\}$

- Draw in the result , for example $A = B > C$

Then we will have $R = \{C_1, C_2\}$

$R = \{(A, B), C\}$

- Or $A = B = C$

$R = \{C_1\}$

$R = \{(A, B, C)\}$

How does the Plackett-Luce (PL) Model Interpret Rankings?

- Over all, we will have $R_1, R_2, \dots, R_{12}$

How does the Plackett-Luce (PL) Model Interpret Rankings?

- **Step 1: Ranking Definition (R)**

Converts group outcomes into a sequence of "winning clusters":

$$R_i = \{C_1, C_2, \dots, C_J\}$$

$$R_1 = \{A, (B, C), D\}$$

$$R_2 = \{H, G, (E, F)\}$$

- Note: Ties are grouped into the same cluster

How does the Plackett-Luce (PL) Model Interpret Rankings?

- **Step 2:** Dynamic Pruning (Exclusion Rule)

Data Scan: Determines the *maximum tie order* (D) observed in the dataset.

Restrict: If 3-way ties never occur, it forces $\delta_3 = 0$.

Impact: Trims down the denominator formula (e.g., from 15 combinations to 10 for $k=4$), preventing combinatorial explosion.

How does the Plackett-Luce (PL) Model Interpret Rankings?

For a set of items S , let

$$f(S) = \delta_{|S|} \left(\prod_{i \in S} \alpha_i \right)^{\frac{1}{|S|}}$$

where $|S|$ is the cardinality (size) of the set, δ_n is a parameter related to the prevalence of ties of order n (with $\delta_1 \equiv 1$), and α_i is a parameter representing the worth of item i . Then under an extension of the Plackett-Luce model allowing ties up to order D , the probability of the ranking R is given by

$$\prod_{j=1}^J \frac{f(C_j)}{\sum_{k=1}^{\min(D_j, D)} \sum_{S \in \binom{A_j}{k}} f(S)}$$

where D_j is the cardinality of A_j , the set of alternatives from which C_j is chosen, and $\binom{A_j}{k}$ is all the possible choices of k items from A_j . The value of D can be set to the maximum number of tied items observed in the data, so that $\delta_n = 0$ for $n > D$.

When the worth parameters are constrained to sum to one, they represent the probability that the corresponding item comes first in a ranking of all items, given that first place is not tied.

The 2-way tie prevalence parameter δ_2 is related to the probability that two items of equal worth tie for first place, given that the first place is not a 3-way or higher tie. Specifically, that probability is $\delta_2 / (2 + \delta_2)$.

How does the Plackett-Luce (PL) Model Interpret Rankings?

-

The 3-way and higher tie-prevalence parameters are similarly interpretable, in terms of tie probabilities among equal-worth items.

When intermediate tie orders are not observed (e.g. ties of order 2 and order 4 are observed, but no ties of order 3), the maximum likelihood estimate of the corresponding tie prevalence parameters is zero, so these parameters are excluded from the model.

How does the Plackett-Luce (PL) Model Interpret Rankings?

- **Step 3: Worth Function $f(S)$**

Single Item: $f(i) = \alpha_i$

Tied Cluster: $f(\{i, j\}) = \delta_2 \cdot (\alpha_i \cdot \alpha_j)^{0.5}$.

* δ is estimated by the package

Advantage: Perfectly absorbs tie information without data loss.

extension of the Plackett-Luce model allowing ties up to order D , the probability of the ranking R is given by

$$\prod_{j=1}^J \frac{f(C_j)}{\sum_{k=1}^{\min(D_j, D)} \sum_{S \in \binom{A_j}{k}} f(S)}$$

where D_j is the cardinality of A_j , the set of alternatives from which C_j is chosen, and $\binom{A_j}{k}$ is all the possible choices of k items from A_j . The value of D can be set to the maximum number of tied items observed in the data, so that $\delta_n = 0$ for $n > D$.

How does the Plackett-Luce (PL) Model Interpret Rankings?

- **Step 4: *Iterative Estimation*** (Stage-wise Selection)

Probability: $P(R) = \prod (\text{Worth of the selected cluster}) / (\text{Sum of worths for all possible clusters (Denominator)})$

MLE Process: Iterates thousands of times to refine α and δ values until the joint probability is maximized.

How does the Plackett-Luce (PL) Model Interpret Rankings?

- **Step 5:** Interpretation of Parameters

Worth Parameter (α) = Winning Probability

If $\alpha_{t_5} = 0.4$, t_5 has a 40% probability of coming in first.

The α values provide the final aggregated ranking for all 12 treatments.

Link

$$t = 12$$

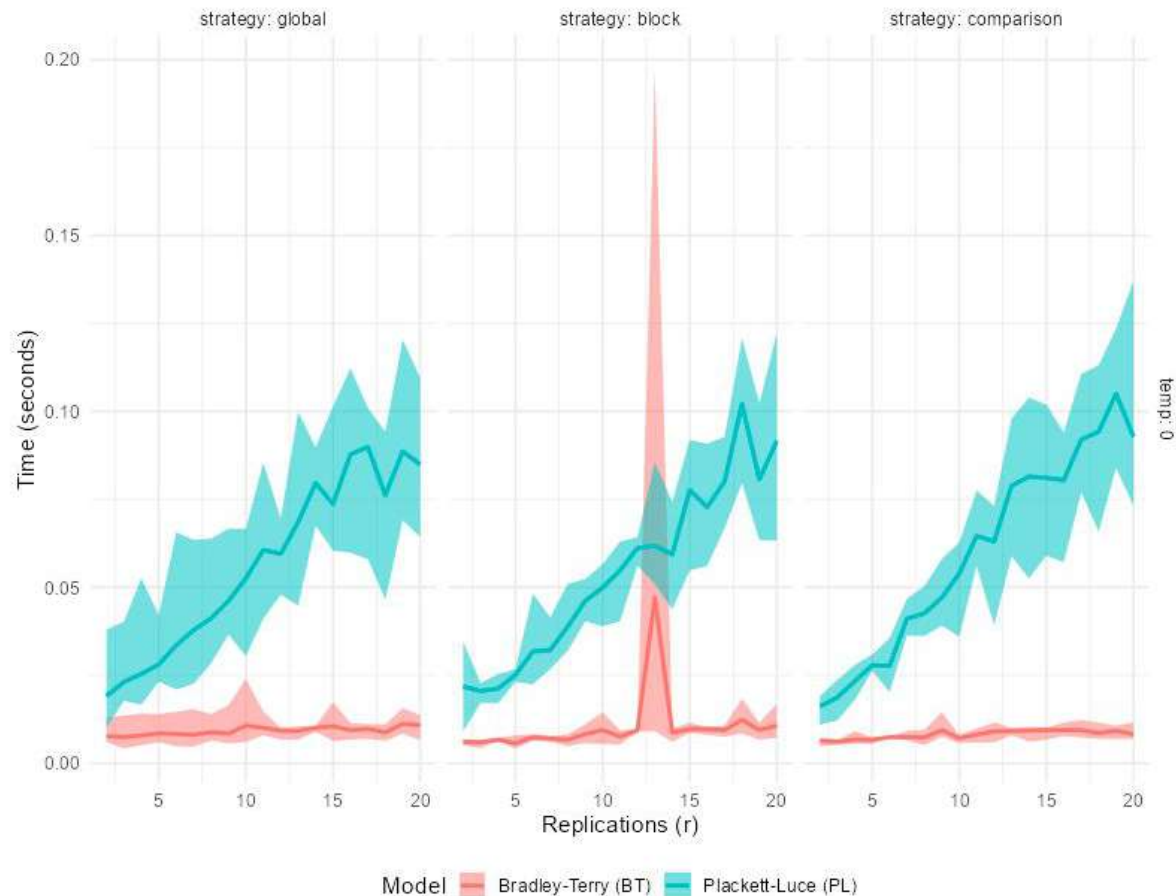
Model Comparison (k=4): Spearman Rho

Ribbon represents the Min-Max range across seeds. Y-axis aligned.



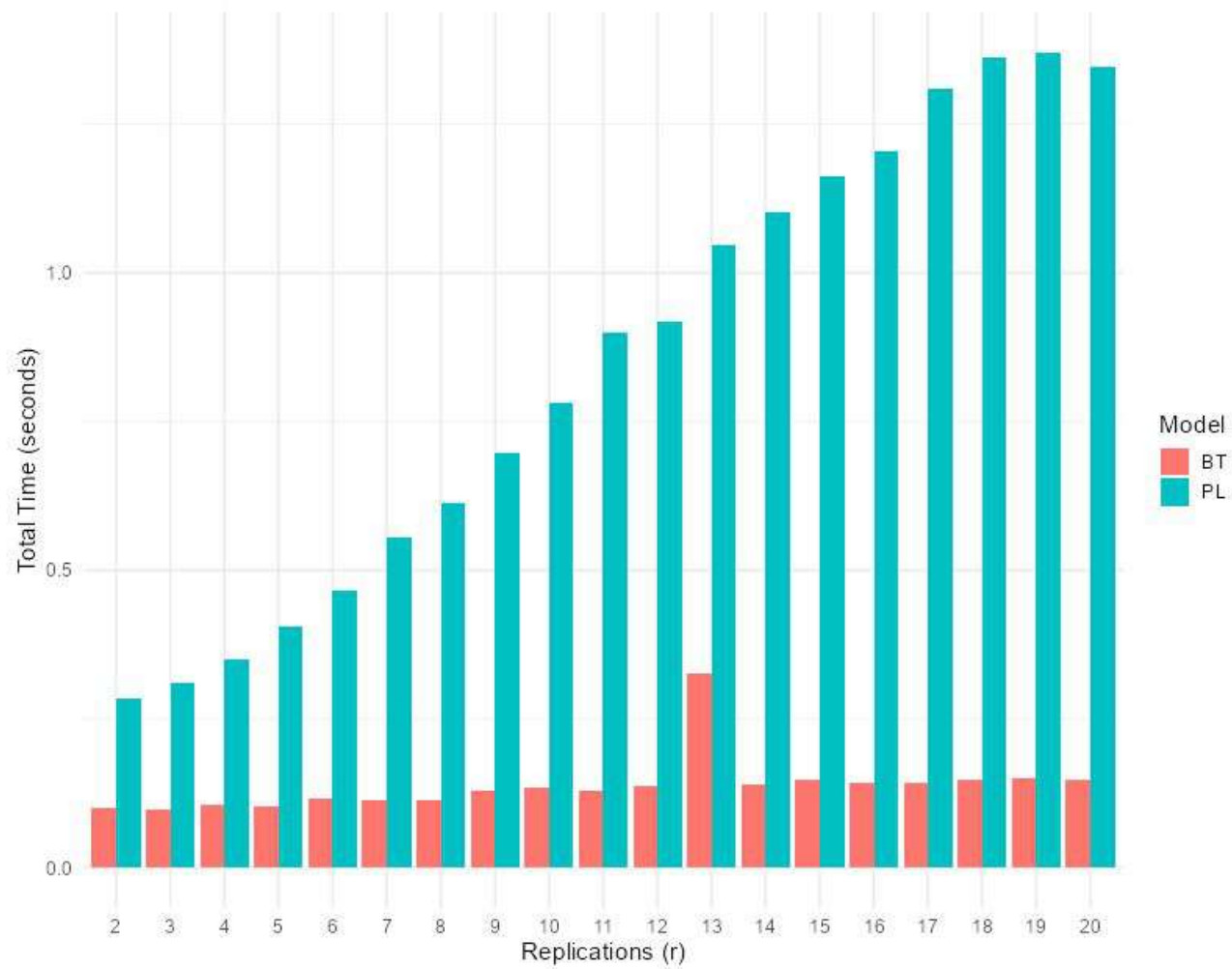
Computing Time Comparison (k=4)

Time in seconds. Ribbon represents variation across seeds.

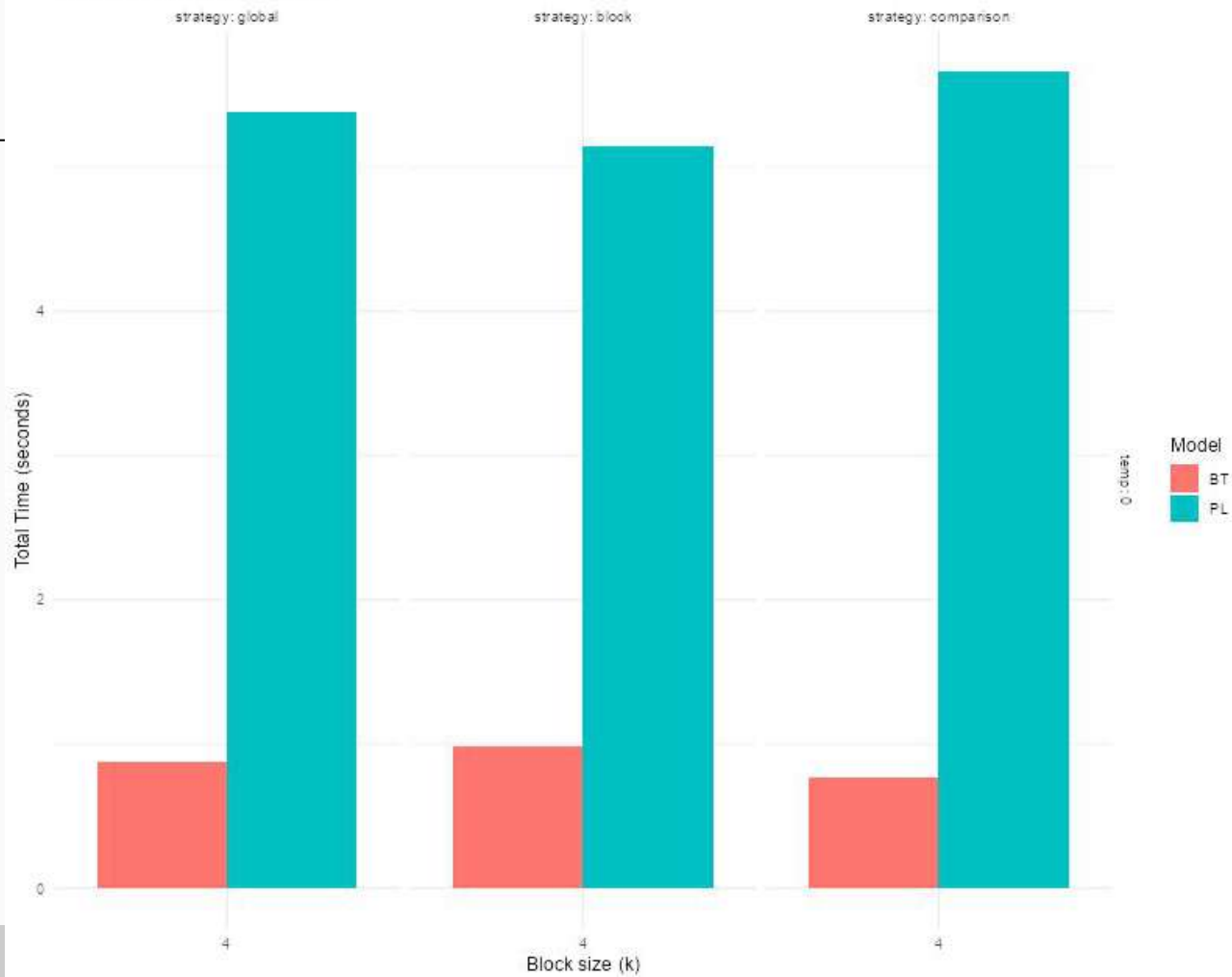


Total Computing Cost by Replication (r)

Sum of time across all k, temp, and seeds for each r



Total Computation Time per Scenario
Sum of all replications and seeds

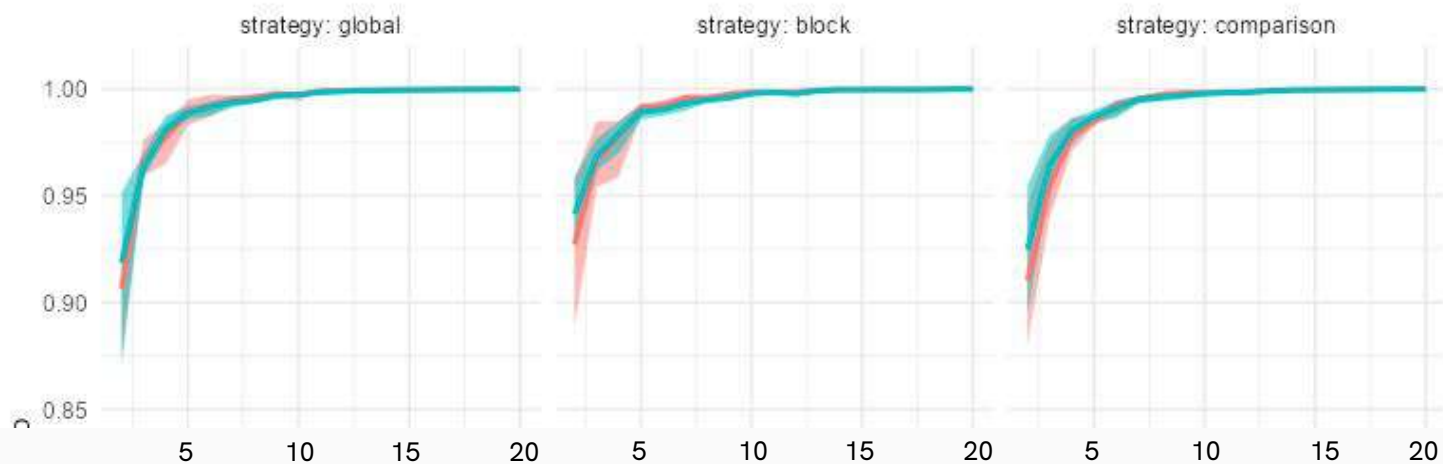


Link

$t = 48$

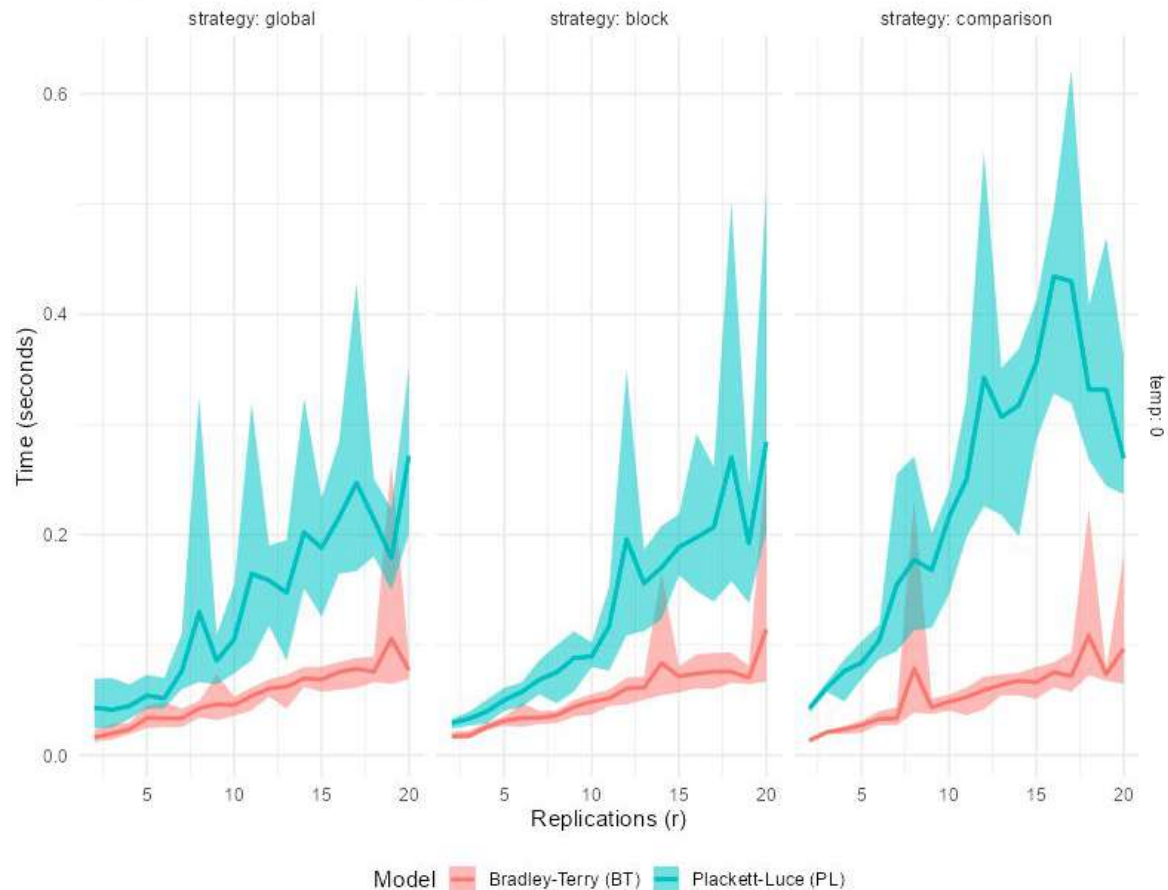
Model Comparison (k=4): Spearman Rho

Ribbon represents the Min-Max range across seeds. Y-axis aligned.



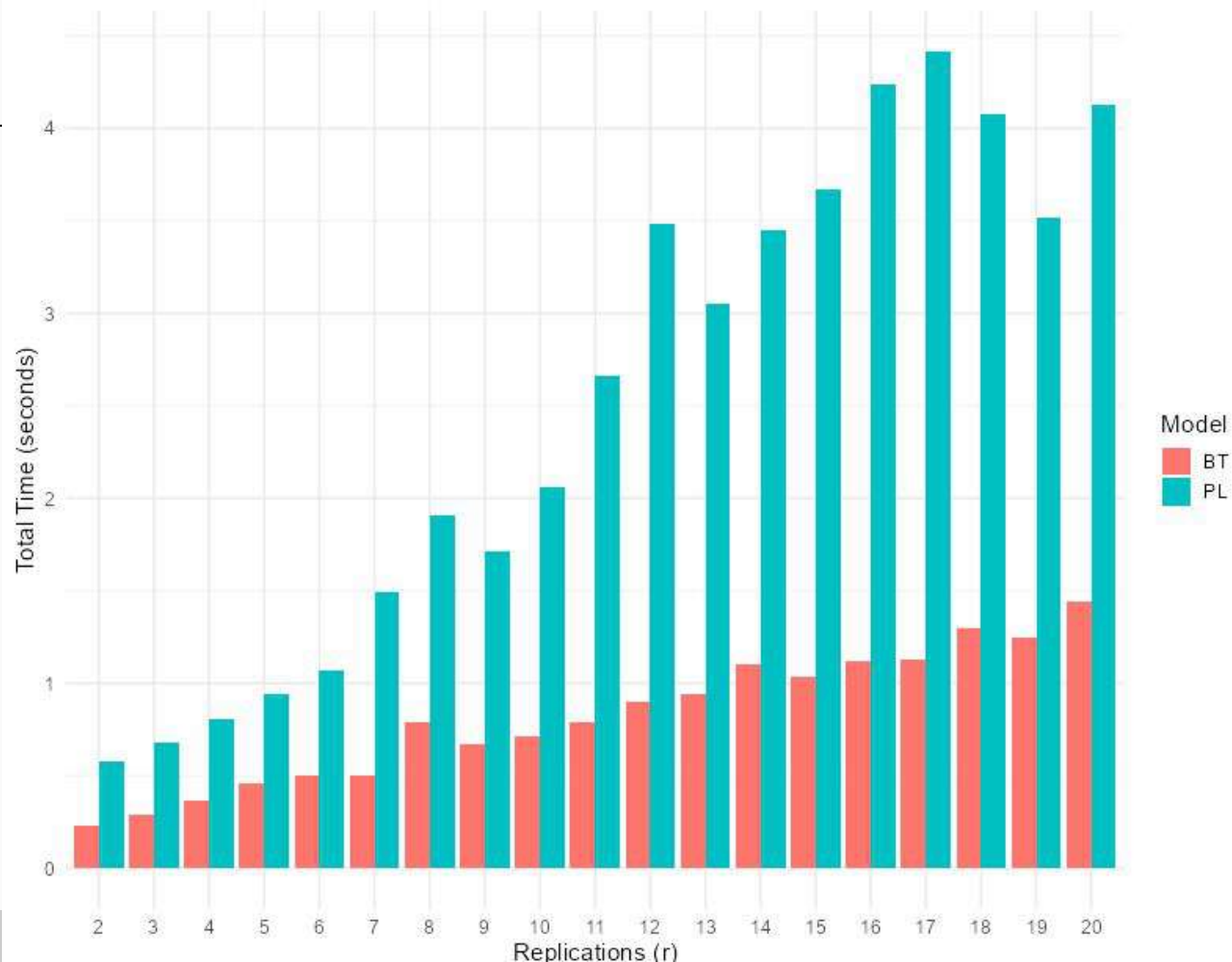
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Time in seconds. Ribbon represents variation across seeds.

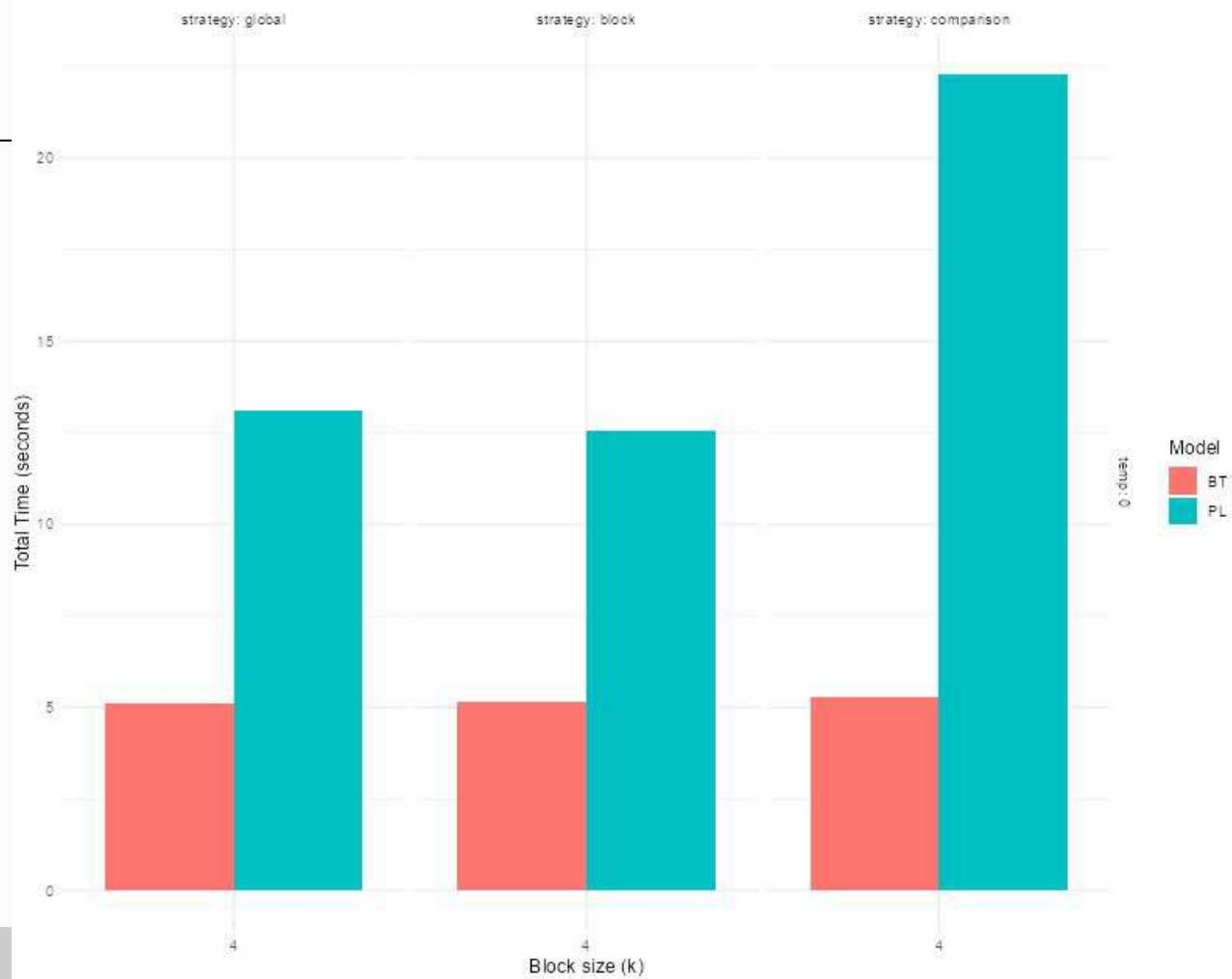


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Total Computation Time per Scenario
Sum of all replications and seeds

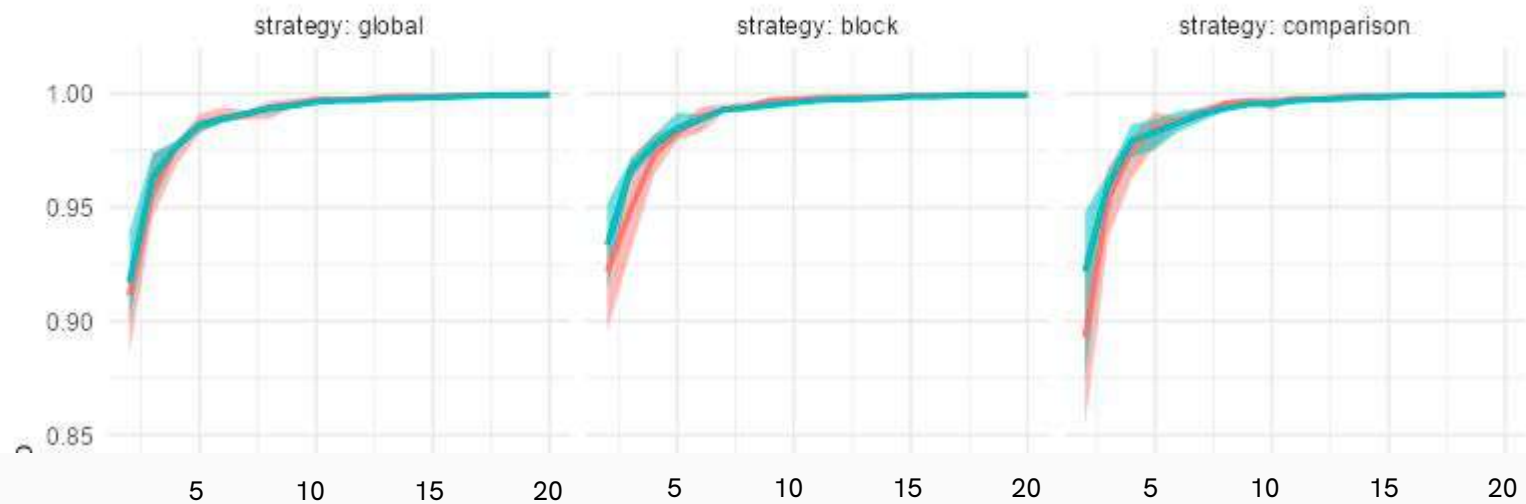


Link

$t = 80$

Model Comparison (k=4): Spearman Rho

Ribbon represents the Min-Max range across seeds. Y-axis aligned.

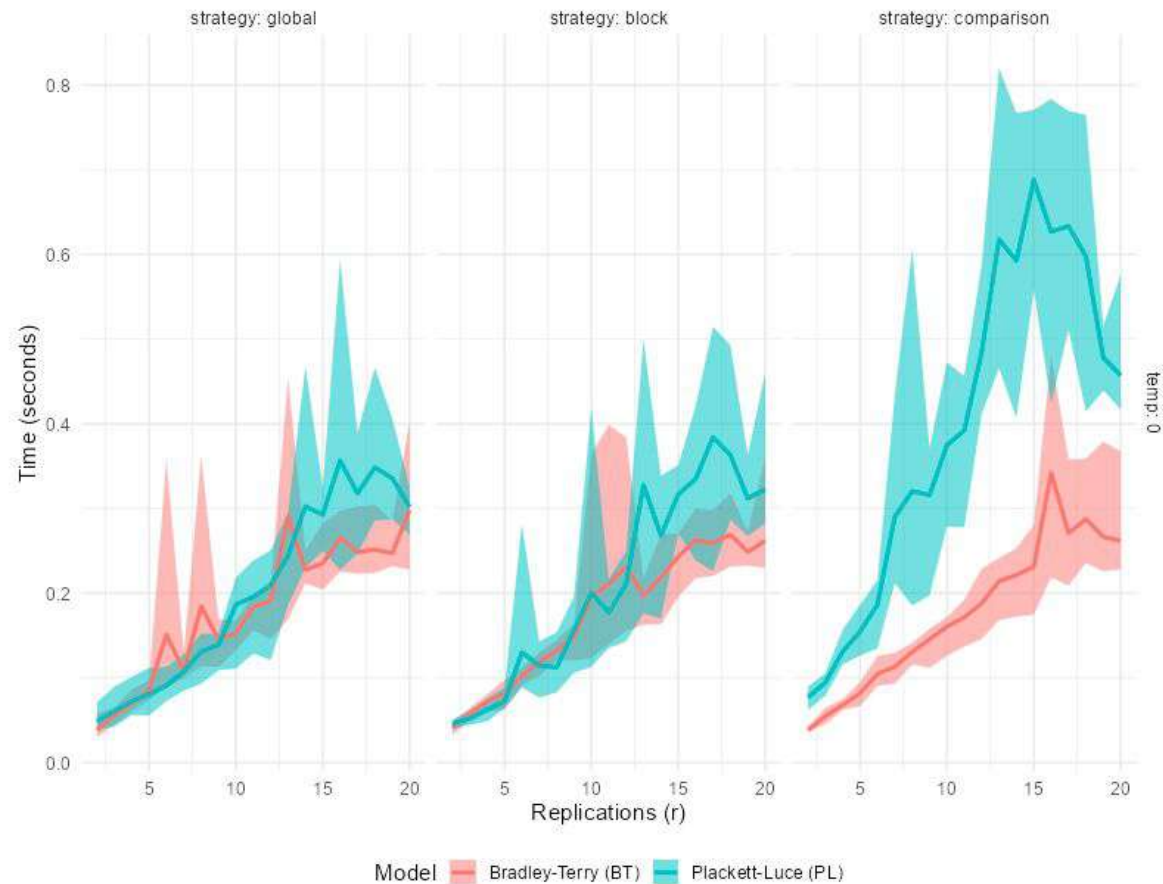


Select k for Time Comparison:

4

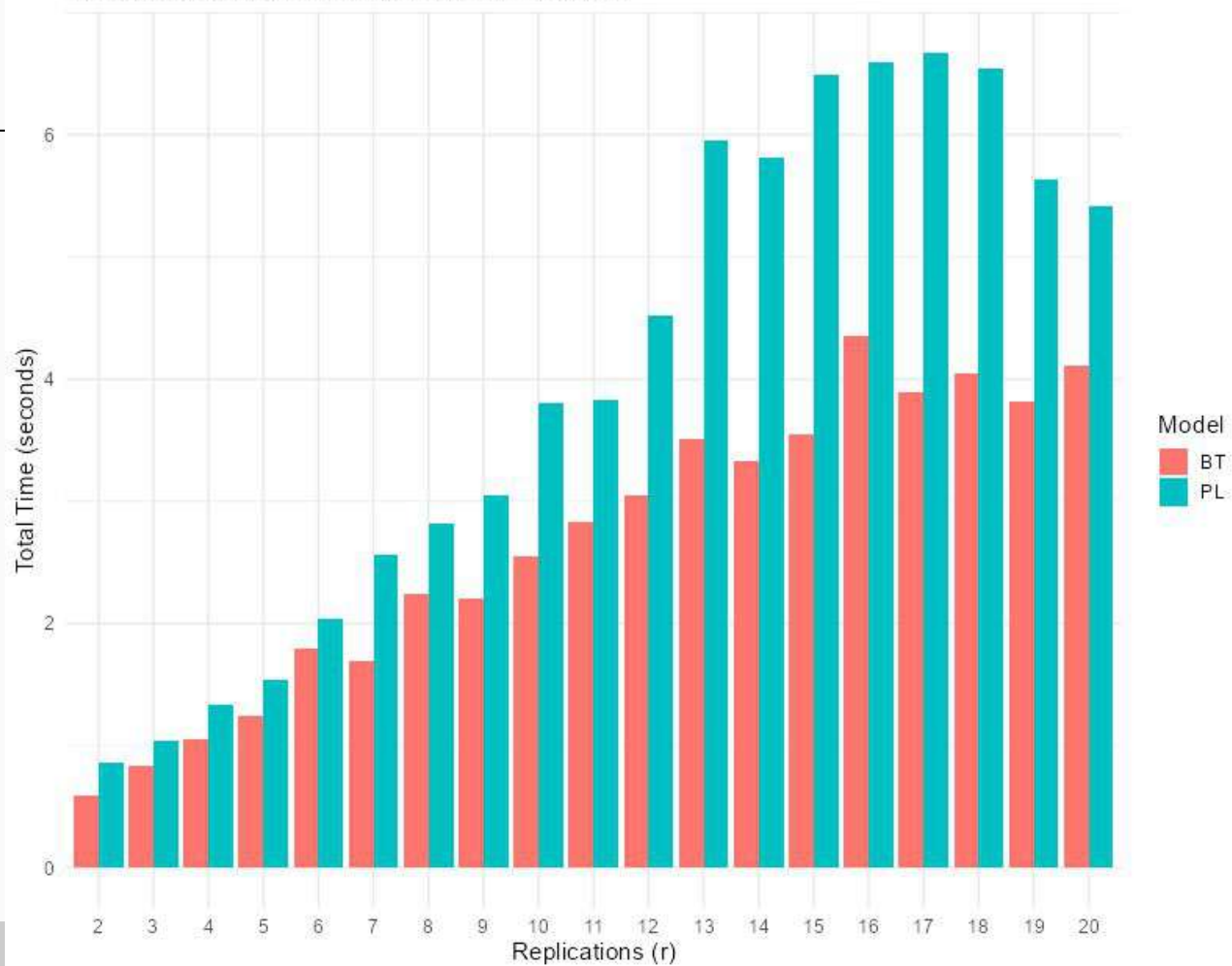
Computing Time Comparison (k=4)

Time in seconds. Ribbon represents variation across seeds.



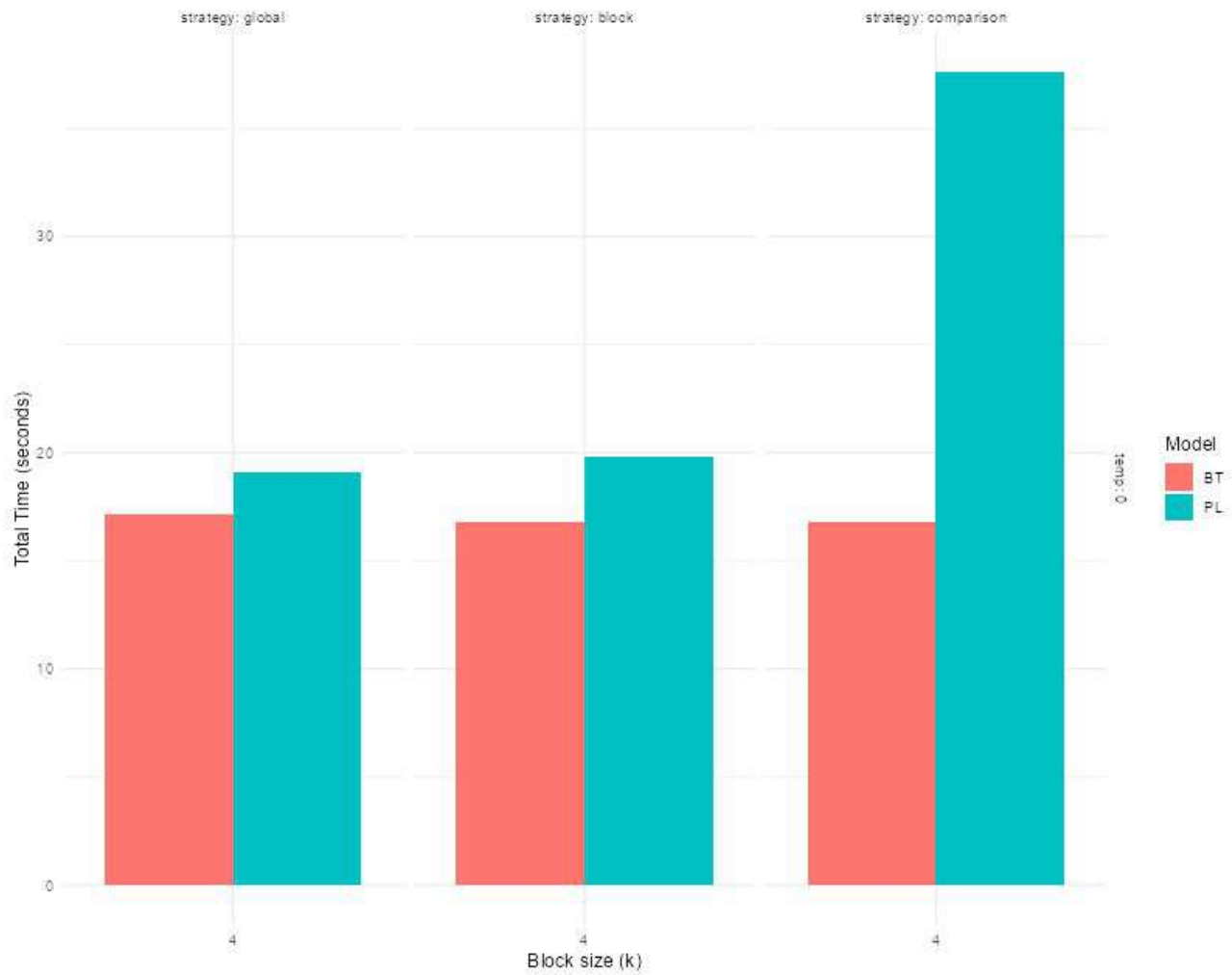
Total Computing Cost by Replication (r)

Sum of time across all k, temp, and seeds for each r



Total Computation Time per Scenario

Sum of all replications and seeds

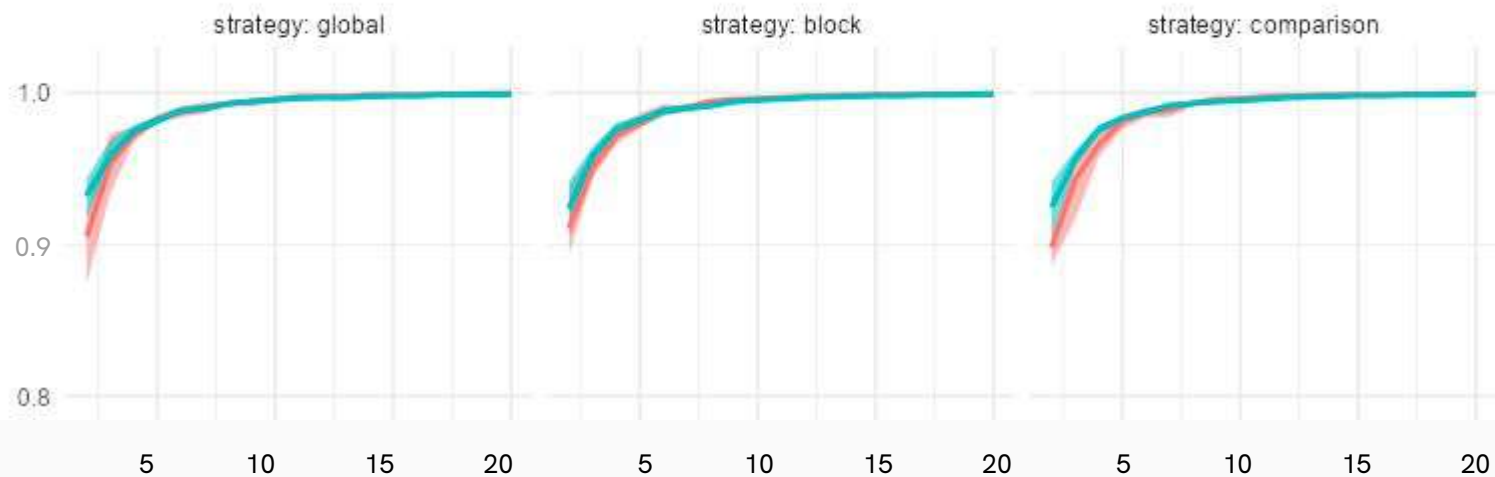


Link

$$t = 120$$

- Model Comparison (k=4): Spearman Rho

Ribbon represents the Min-Max range across seeds. Y-axis aligned.

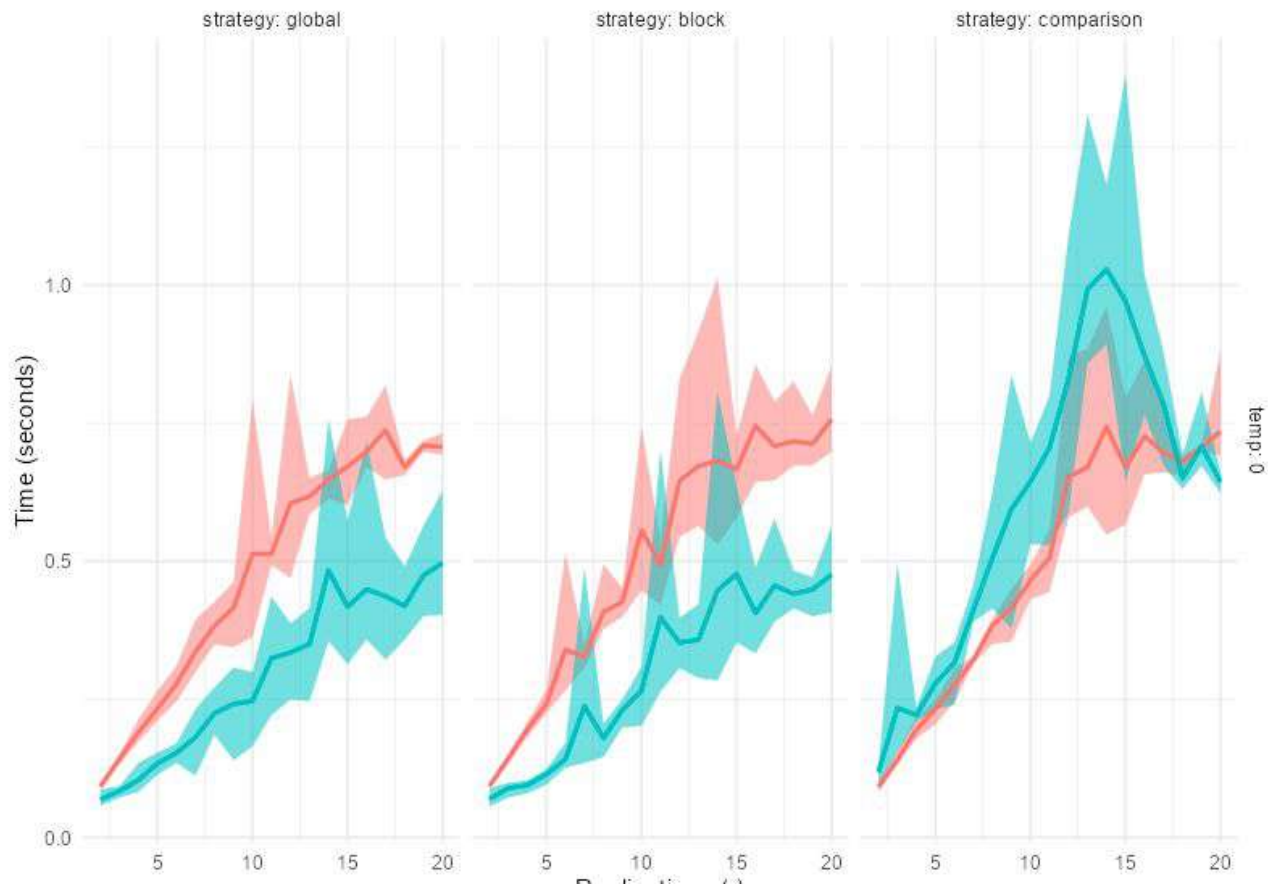


Select k for Time Comparison:

4

Computing Time Comparison (k=4)

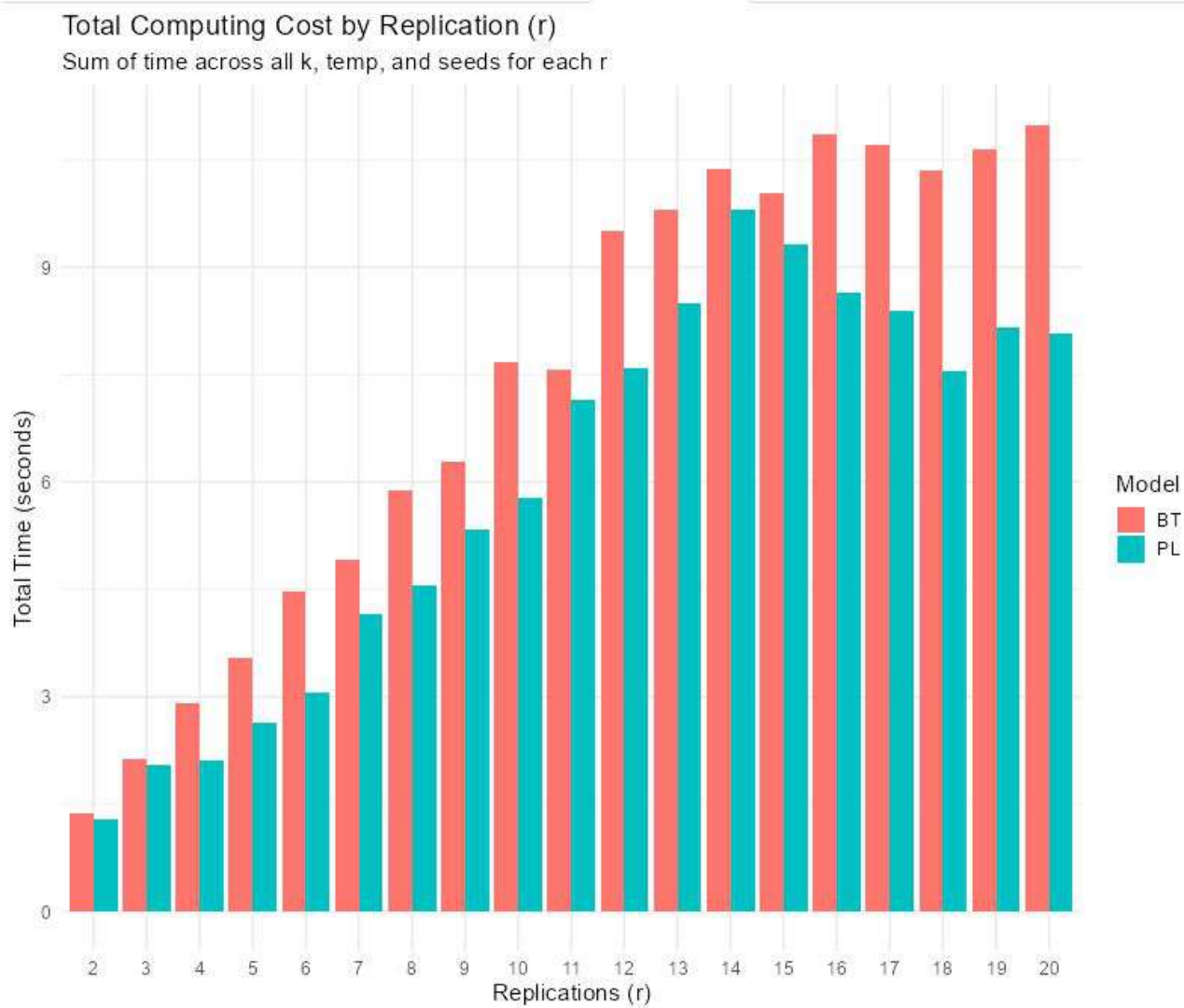
Time in seconds. Ribbon represents variation across seeds.



Under this replication
Average time for calculation
for per seed

How

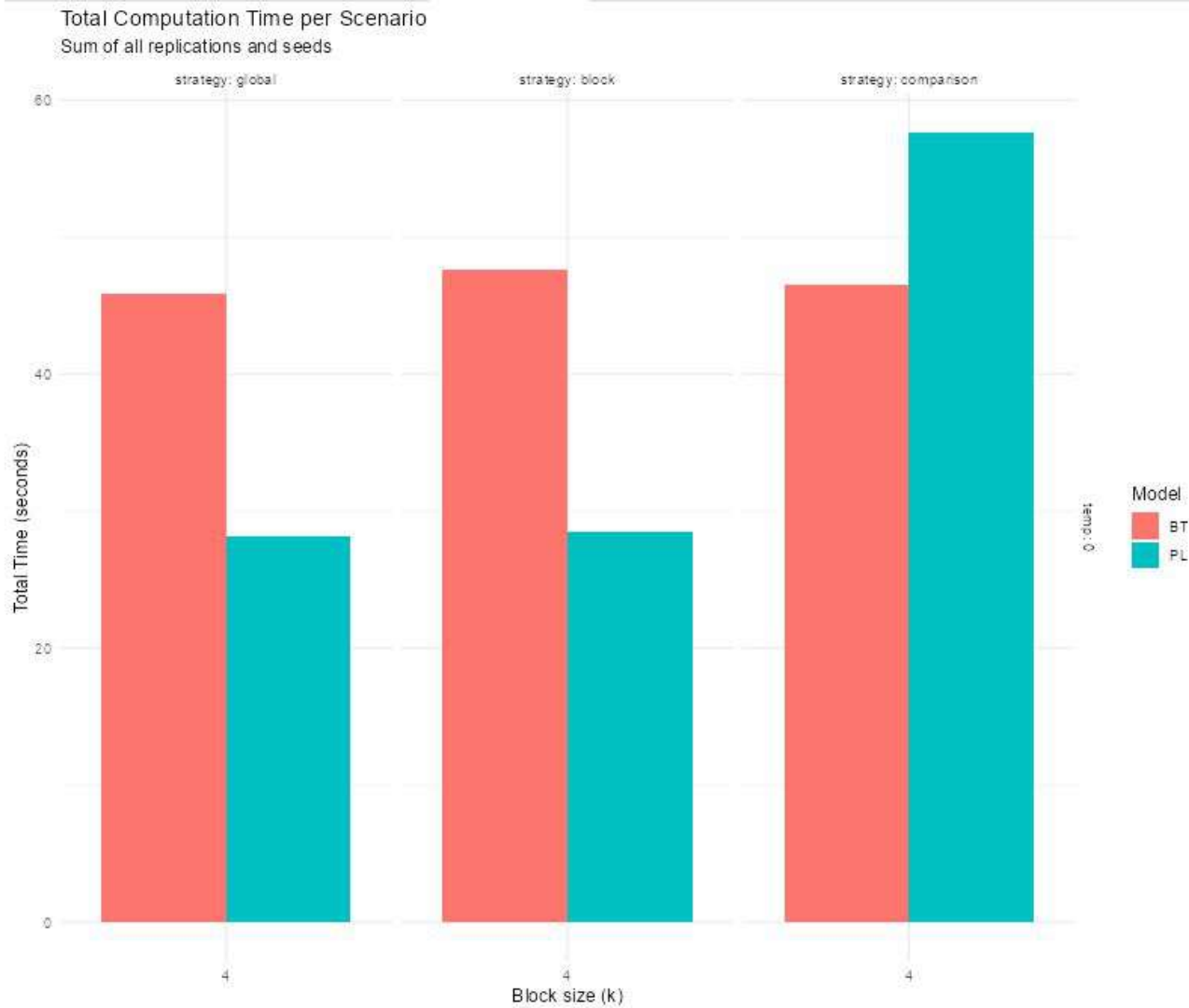
-



ings?

How

ings?



2026-05-19

Link

Behind the implementation of BT
(bradleyterry2 package)

How does the Bradley-Terry (BT) Model Interpret Rankings?

- Original Ranking: $1 > 2 > 3 > 4$
- BT Model's Perspective (Dimensionality Reduction & Splitting):
 - $1 > 2, 1 > 3, 1 > 4$
 - $2 > 3, 2 > 4$
 - $3 > 4$
- Subsequent Processing: Feed these 6 independent binomial outcomes (pairwise win/loss data) into **Maximum Likelihood Estimation (MLE)** to calculate the final abilities

How does the Bradley-Terry (BT) Model Interpret Rankings?

- Generate all **pairwise combinations within a block** using `combn()`

```
pm <- combn(as.character(block$TREATMENT), 2)
```

- Forcibly create a **binomial Winner/Loser record table**

```
bt_matches_list[[j]] <- data.frame(  
  Winner = ifelse(u1 > u2, pm[1,], pm[2,]),  
  Loser = ifelse(u1 > u2, pm[2,], pm[1,])
```

- Finally, **feed the pairwise data into BTm()** for MLE calculation

```
fit_bt <- BTm(1, Winner, Loser, data = bt_matches)
```

How does the Bradley-Terry (BT) Model Interpret Rankings?

- *In order to fit a Bradley-Terry model using BTm... the data must first be converted to binomial frequencies. That is, the data need to be organised into pairs (player1, player2) and corresponding frequencies of wins and losses for player1 against player2.——Official BradleyTerry2 package documentation (Turner & Firth)*

Link

Behind PL implementation
(PlackettLuce package)

How does the Plackett-Luce (PL) Model Interpret Rankings?

- Original Ranking: $1 > 2 > 3 > 4$
- PL Model's Perspective (A Sequence of Choices):

Stage 1: From $\{1, 2, 3, 4\}$, item 1 is chosen.

Stage 2: From the remaining $\{2, 3, 4\}$, item 2 is chosen.

Stage 3: From the remaining $\{3, 4\}$, item 3 is chosen.

- Probability Calculation: $\text{Prob}(\text{Stage 1}) * \text{Prob}(\text{Stage 2}) * \text{Prob}(\text{Stage 3})$

How does the Plackett-Luce (PL) Model Interpret Rankings?

- Directly **store the sorted items of the entire block as a single row** in a matrix

```
pl_rankings_matrix[j, ] <-  
  as.character(block$TREATMENT[order(-u_scores)])
```

- **Convert the matrix into a dedicated "rankings" object**
(preserving the full sequence)

```
R <- PlackettLuce::as.rankings(pl_rankings_matrix, input = "orderings")
```

- Finally, feed the object directly into `PlackettLuce()` for MLE calculation

```
fit_pl <- PlackettLuce::PlackettLuce(R, maxit = c(500, 100), epsilon = 1e-6)
```

How does the Plackett-Luce (PL) Model Interpret Rankings?

- Behind the scenes, the function uses algorithms like Iterative Scaling or BFGS to find the Maximum Likelihood Estimate.
- This approach of not splitting the data prevents the artificial inflation of the sample size and perfectly preserves the original transitive information of the rankings.
- *“A single ranking is given by $R = \{C1, C2, \dots, CJ\}$, where the items in set $C1$ are ranked higher than (better than) the items in $C2$, and so on... the probability of the ranking R is given by the product of the probability of choosing Cj from the set of alternatives remaining.”*——**PlackettLuce package documentation (Turner, Kosmidis, & Firth)**

Link

BWS SIM question

Plackett-Luce with "Ties" (The Efficient Way)

- How: Treat BWS as a simple ranking: Best = 1, Middle items = 2, Worst = 3.
- Pros:
 - Very Fast: Perfect for our large dataset ($T=120$).
 - Consistent: It uses the same model as our other simulations.
- Cons: Not the "standard" software for marketing surveys, but mathematically the same.

Standard BWS Packages (The Traditional Way)

- How: Use support.BWS and mlogit packages.
- Pros:
 - Most marketing researchers use it.
- Cons:
 - Very Slow: It expands the data into thousands of rows, which might crash the simulation for $T=120$.
 - Instead of having one row per choice task, standard packages create a separate row for every possible alternative a respondent could have chosen in that task. When dealing with a large parameter like $T=120$ (whether that represents tasks, items, or respondents), this combinatorial expansion multiplies the dataset size exponentially.

Link

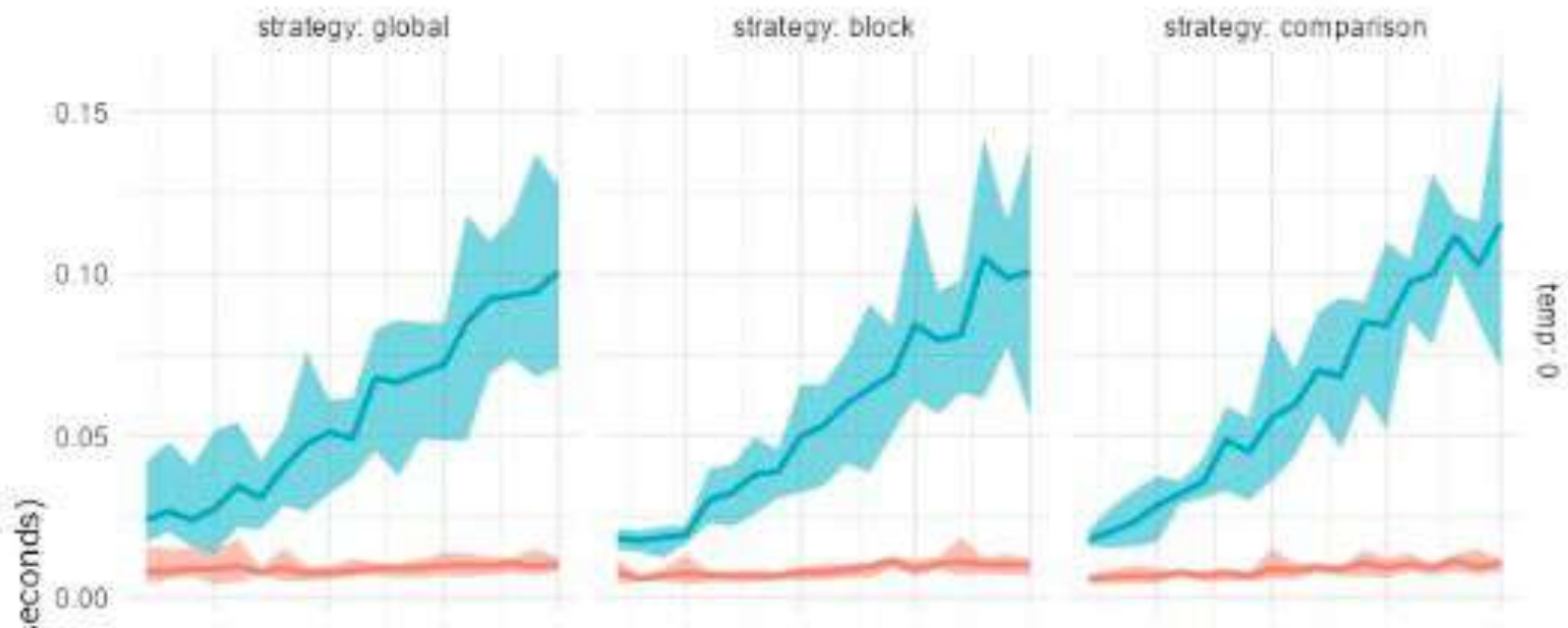
BT vs PL

[Link](#)

When t is small...(t=12)

Computing Time Comparison (k=4)

Time in seconds. Ribbon represents variation across seeds.

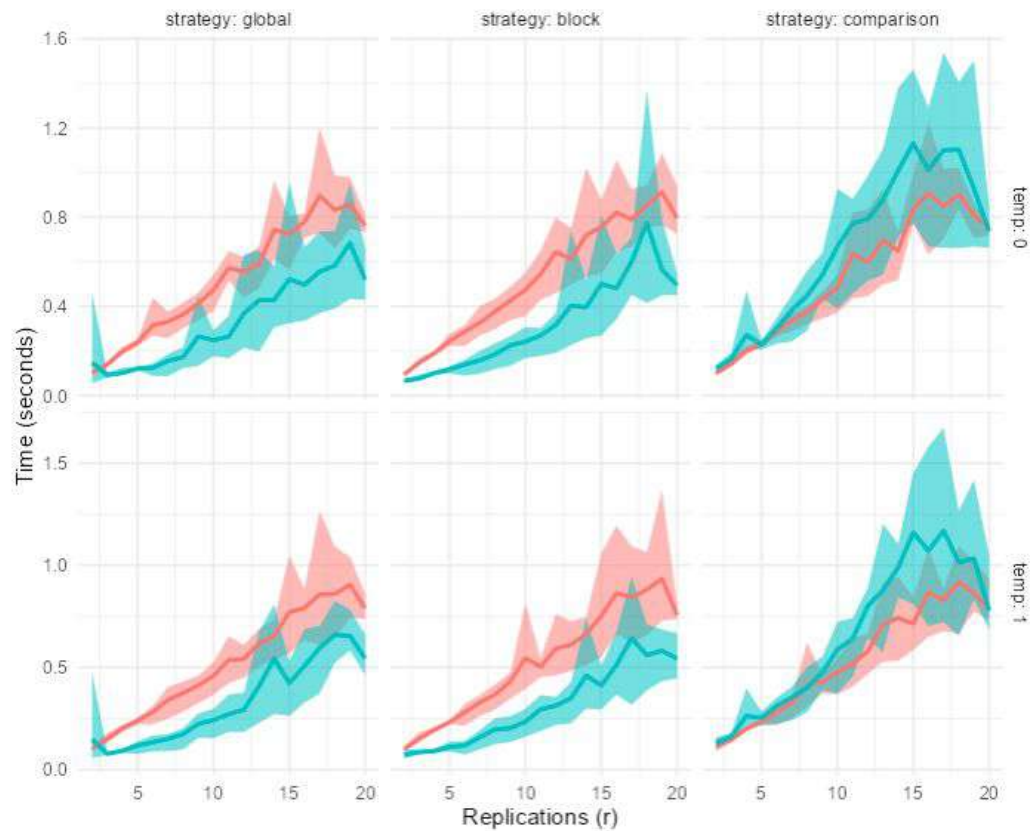


Select k for Time Comparison:

☐ 3 ☒ 4 ☐ 8 ☐ 10

Computing Time Comparison (k=4)

Time in seconds. Ribbon represents variation across seeds.



Model — Bradley-Terry (BT) — Plackett-Luce (PL)

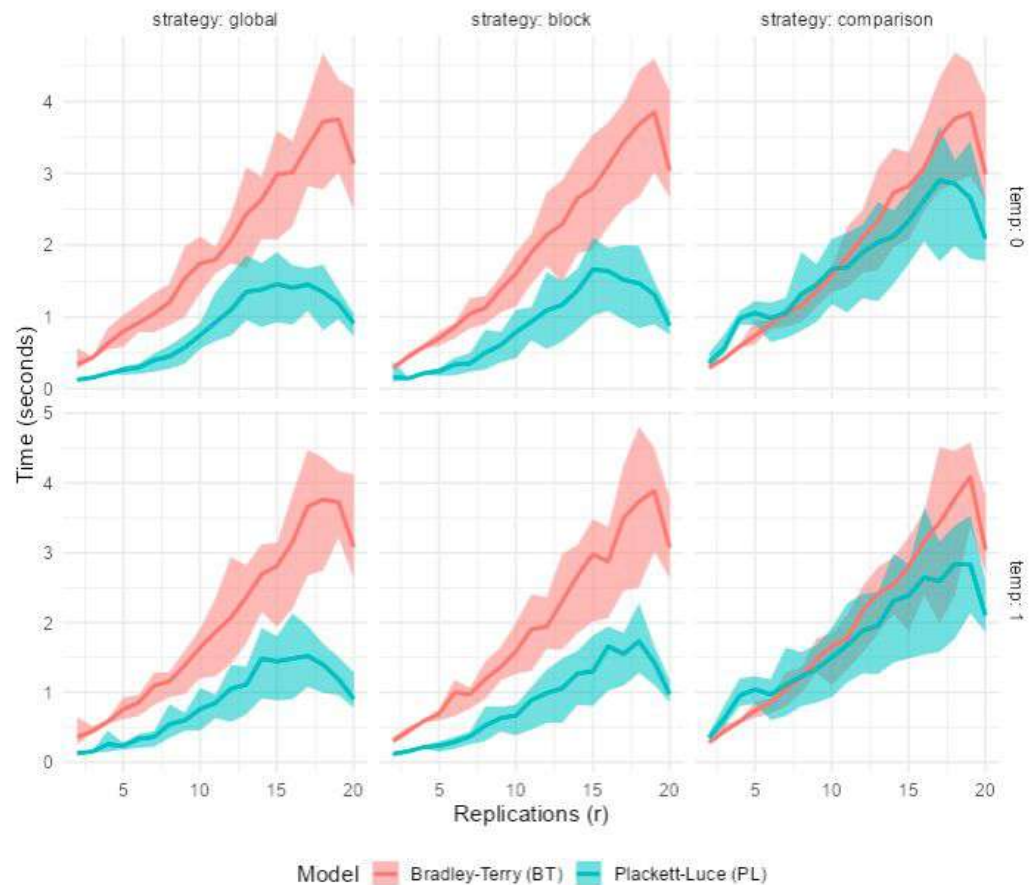
Select k for Time Comparison:

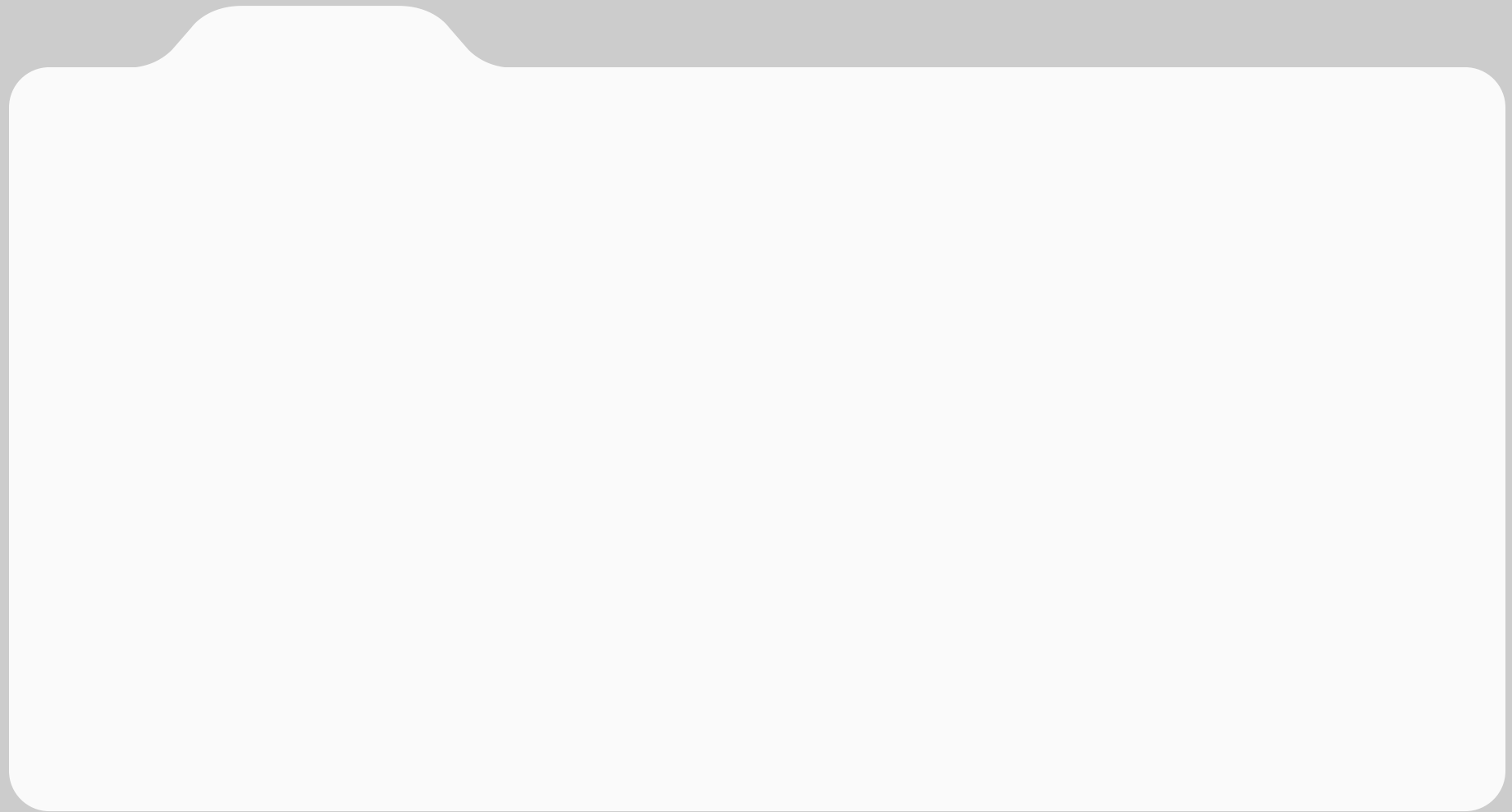
☐ 3 ☐ 4 ☐ 8 ☒ 10

Li

Computing Time Comparison (k=10)

Time in seconds. Ribbon represents variation across seeds.





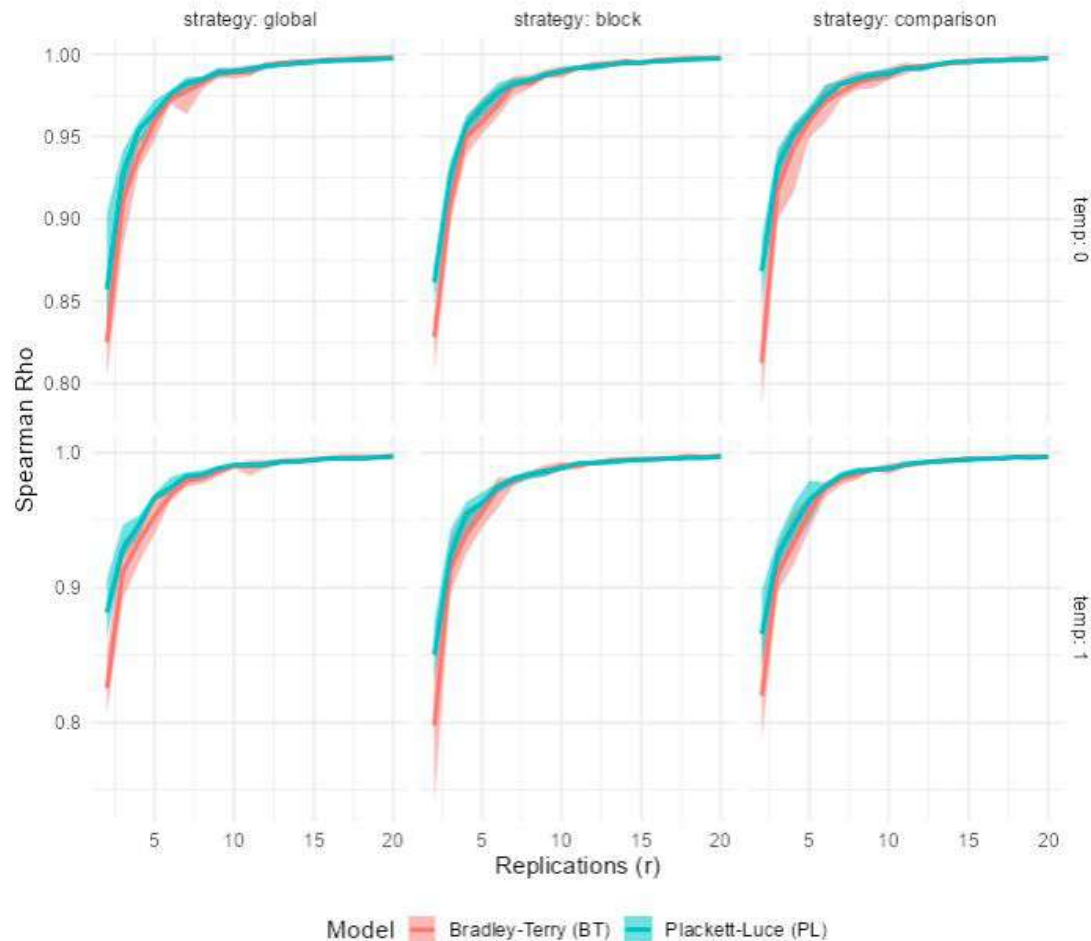
2026-05-12

Link

Comparison result

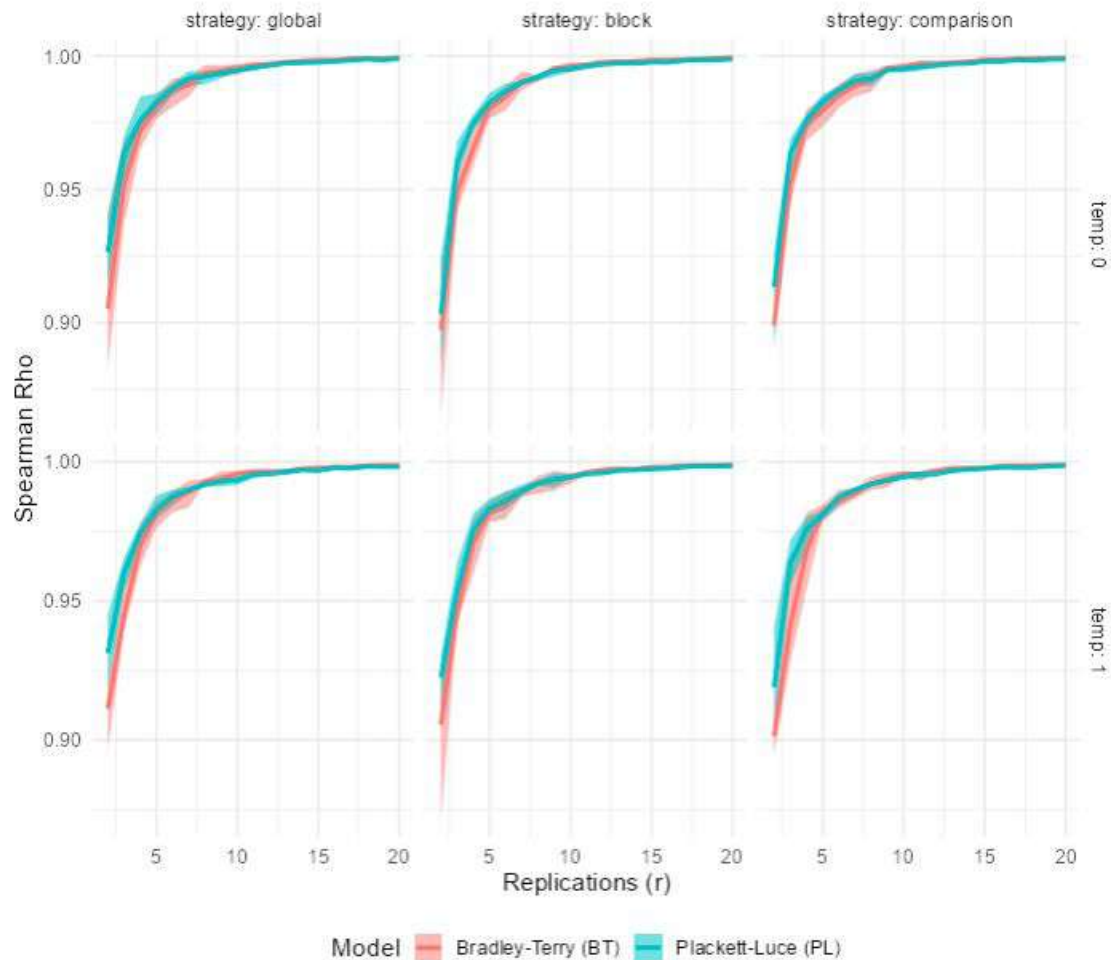
Model Comparison (k=3): Spearman Rho

Ribbon represents the Min-Max range across seeds



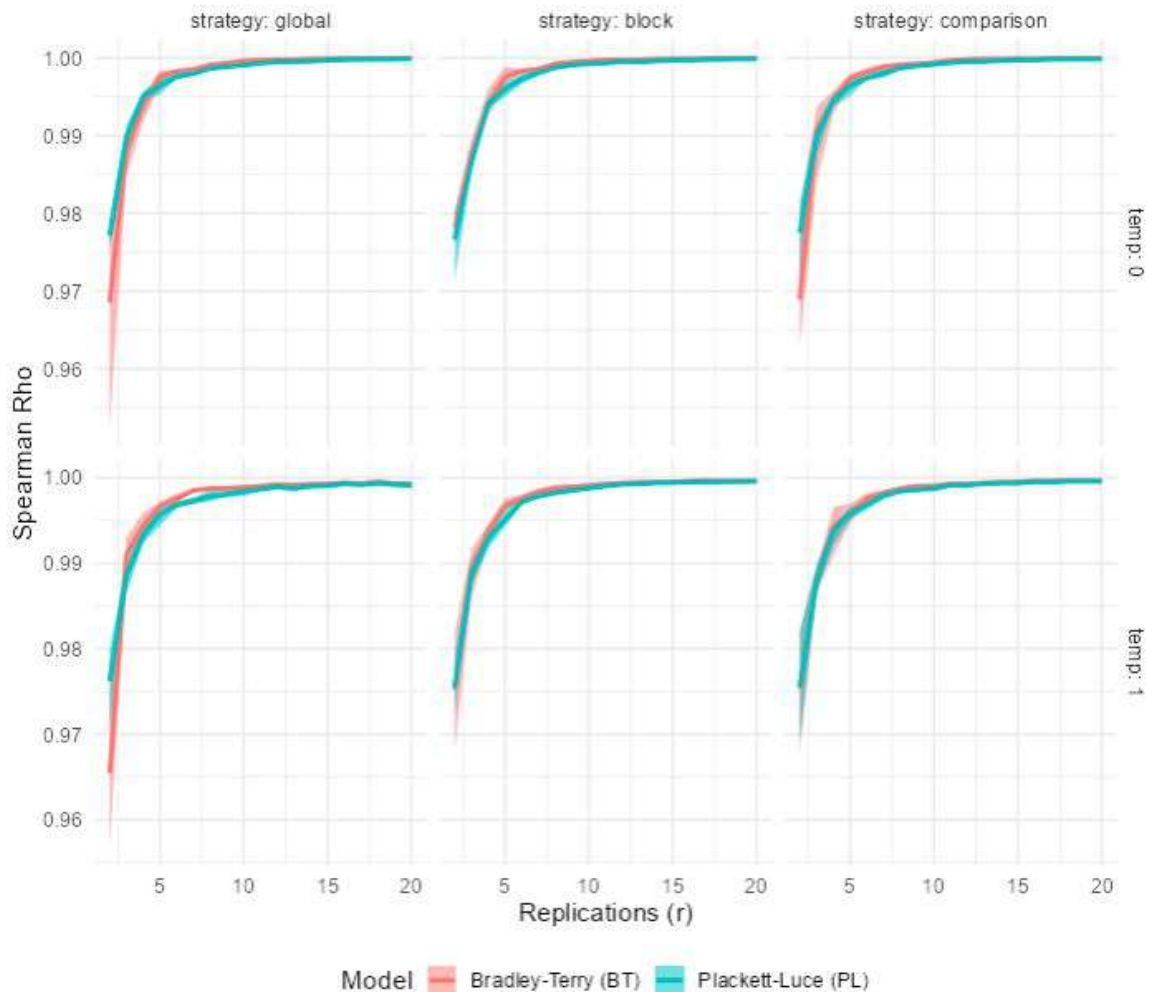
Model Comparison (k=4): Spearman Rho

Ribbon represents the Min-Max range across seeds



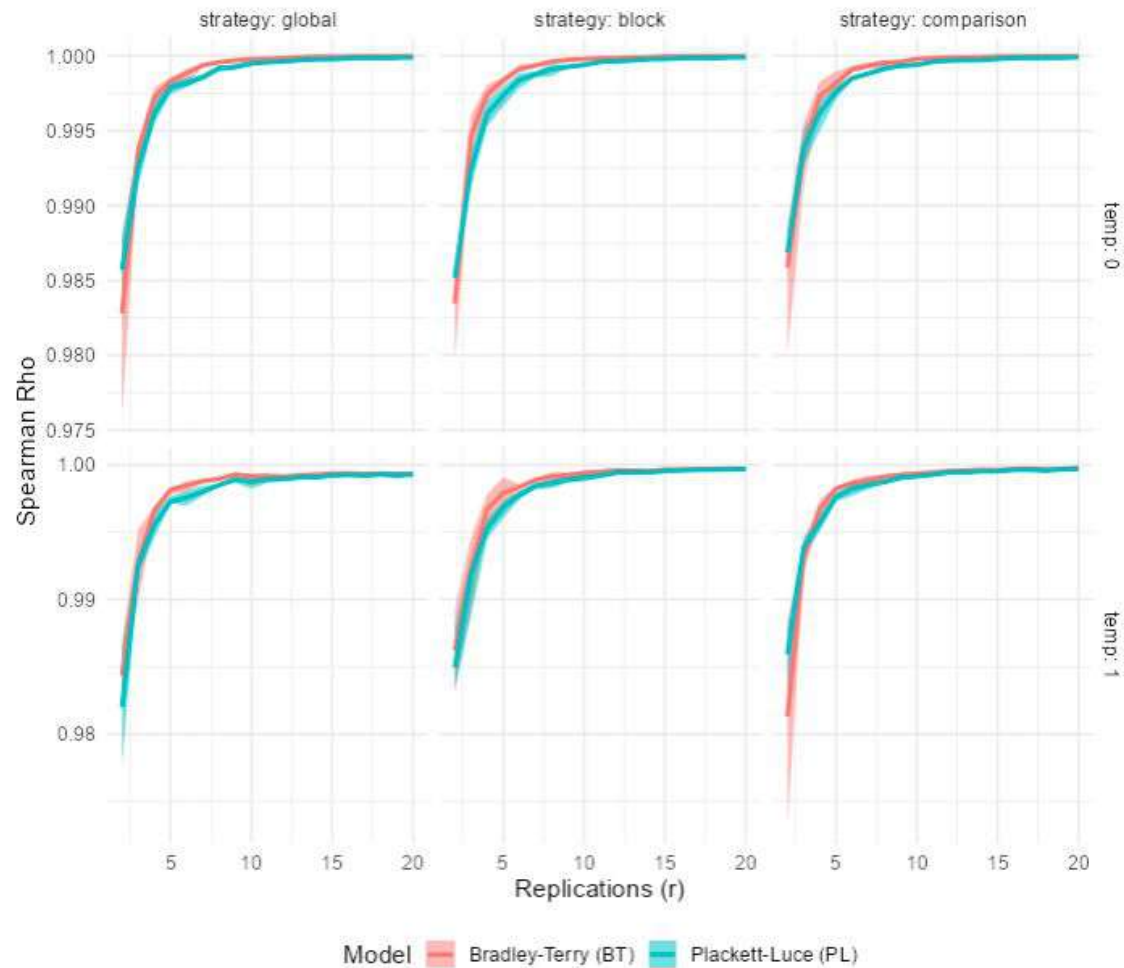
Model Comparison (k=8): Spearman Rho

Ribbon represents the Min-Max range across seeds



Model Comparison (k=10): Spearman Rho

Ribbon represents the Min-Max range across seeds



Link

Efficiency Comparison

Link

[Building upon Bradley-Terry and Plackett-Luce: some methods for modeling paired comparison and rank order data.](#)

Zhang, Sanqian. 2021.

Doctoral dissertation, Harvard University Graduate School of Arts and Sciences.

Conclusion

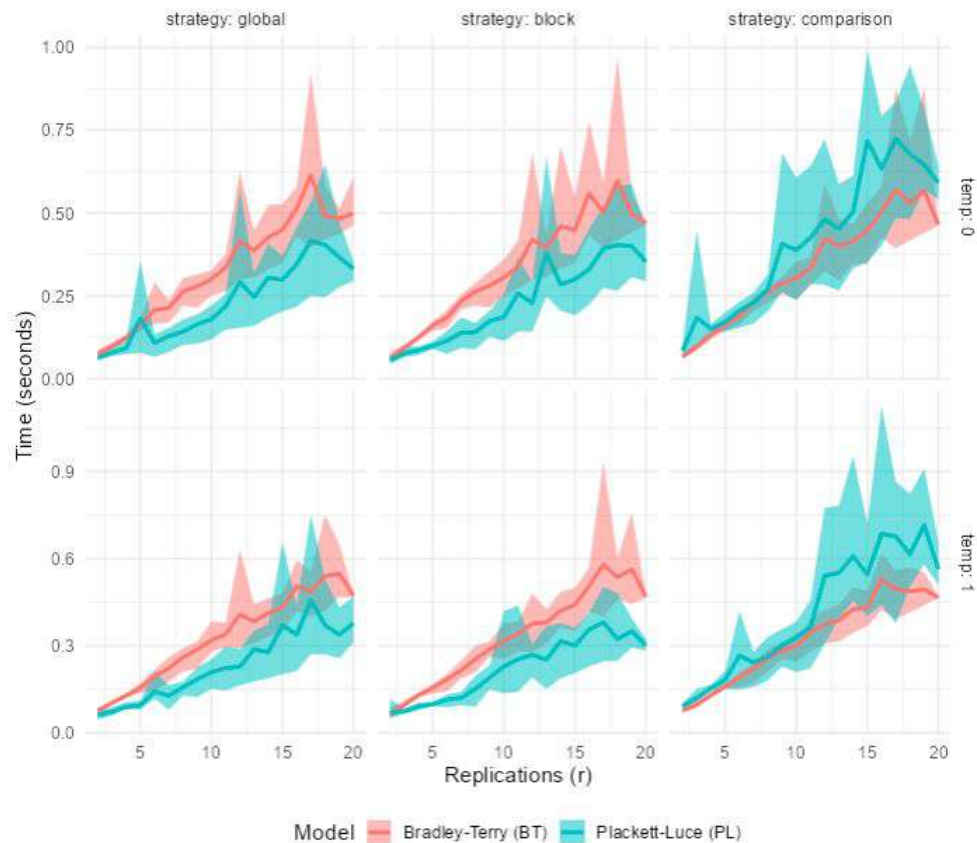
- Positive Correlation with k : The maximum possible loss of efficiency increases as the **number of ranked items (k)** increases.
- Quantifiable Loss: For a ranking of 10 items, the maximum possible loss of efficiency in using BT can reach up to 30%.
- Conclusion: For Alpha Lattice designs where **k (number of ranked items) > 2** , the Plackett-Luce model natively preserves the hierarchical structure of the data, ensuring optimal statistical efficiency without information loss.

Select k for Time Comparison:

Link ☒ 3 ☐ 4 ☐ 8 ☐ 10

Computing Time Comparison (k=3)

Time in seconds. Ribbon represents variation across seeds.

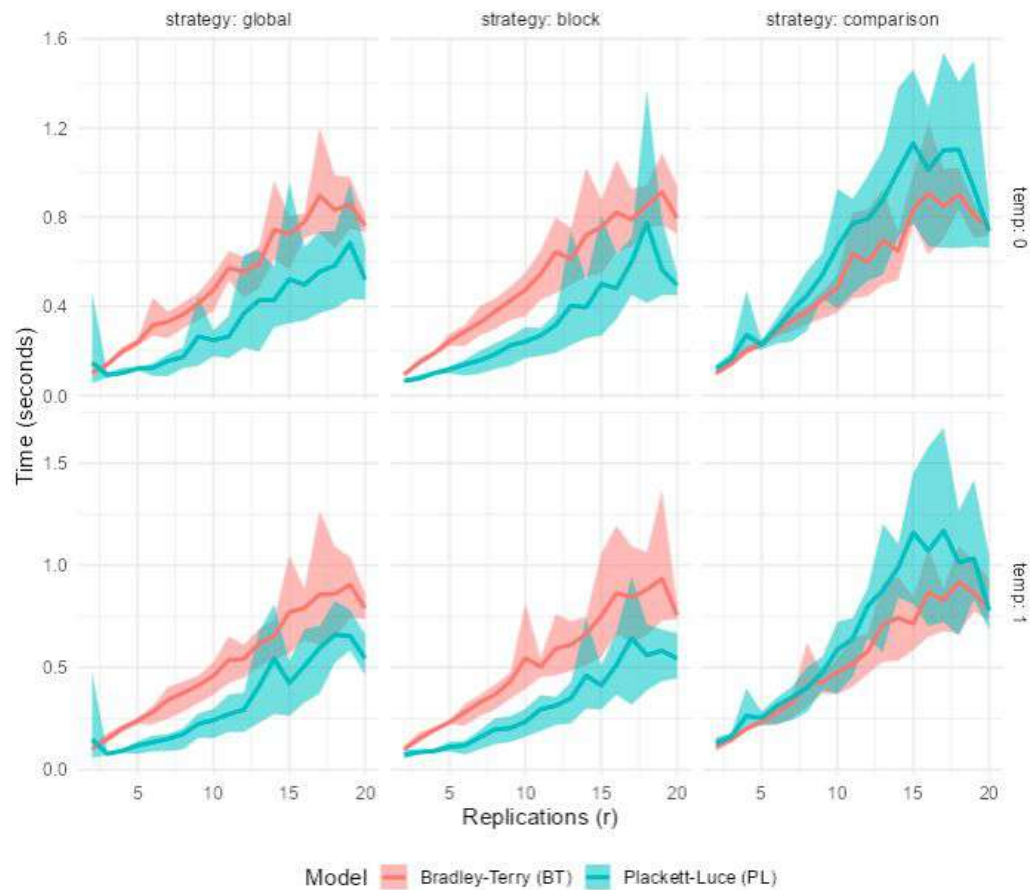


Select k for Time Comparison:

☐ 3 ☒ 4 ☐ 8 ☐ 10

Computing Time Comparison (k=4)

Time in seconds. Ribbon represents variation across seeds.

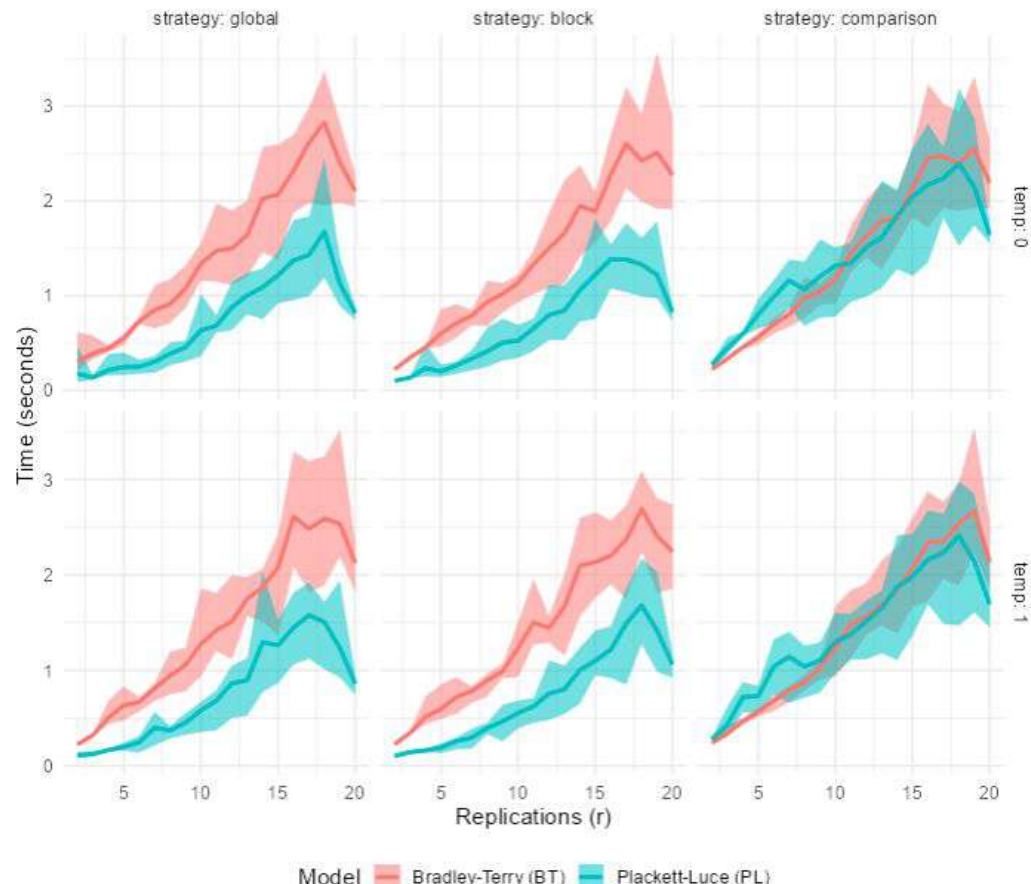


Select k for Time Comparison:

☐ 3 ☐ 4 ☒ 8 ☐ 10

Computing Time Comparison (k=8)

Time in seconds. Ribbon represents variation across seeds.



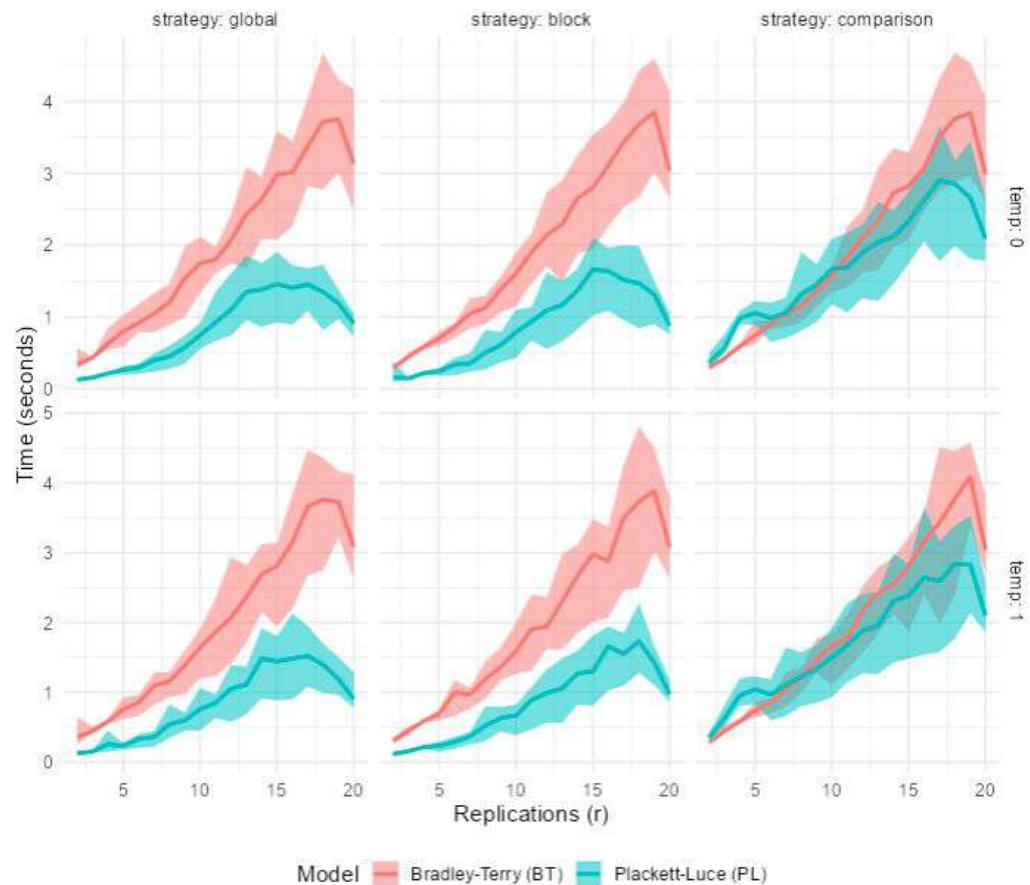
Select k for Time Comparison:

☐ 3 ☐ 4 ☐ 8 ☒ 10

Li

Computing Time Comparison (k=10)

Time in seconds. Ribbon represents variation across seeds.



Link

AI testing result

1.

Many of us dream of one day travelling to as beautiful a country as Japan, yet we often get so caught up in our fantasies of what we would like to do once we're there that we neglect to consider just how we will get there. With this short guide, you will be able to navigate through any of Japan's airport! First, make sure that you have your passport in order, as this will be your most important means of identification, so be sure to always carry it with you and to keep it safe. Once you arrive at the airport, follow signs for check-in and go through security. Japan's airports have various public transport links to help you find and navigate your way, shuttling you through the vast complex directly to your desired destination. There are also maps to ensure that you never lose your way. Listen for boarding announcements and follow staff instructions. You will reach board the plane in no time!

2.

a good travel idea would be getting the proper deal on a website like Groupon to combine different deals into one such as hotel , rides , and food. It would be good to go to a popular destination such as Dominican republic. having a nice beach around you with friendly tourists who are social and nice weather is optimal for the best experience in a travel idea. I'd also include different places like tourist attractions. something that would cause adrenaline. I would like to also learn more about the culture and history of the place I'm visiting and the price should be affordable.

3.

1. Arrive at the airport 3 hours before your flight.
2. Locate your airline's check-in counter; have your passport and flight details ready.
3. Check in and drop off your luggage.
4. Proceed to the security check; remove any metal objects and electronics from your bag.
5. Clear security and find your gate on the airport's display boards.
6. Use any available time to relax, eat, or shop in the terminal.
7. Listen for boarding announcements and proceed to your gate when called.
8. At immigration, undergo iris scan and fingerprinting.

Top 3 Bits of Advice:

Always double-check your gate number as it can change.

Keep your boarding pass and passport accessible.

Follow liquids and electronics rules strictly to avoid delays at security.

4.

The first thing that you should do is go over to immigration in the airport and have your passport, visa, and arrival card ready. You should collect your luggage afterwards and then pass through customs. You then might want to exchange some of your money into yen before leaving the airport area. You then want to purchase a sim card or pocket wifi if you want to have the ability to make calls and texts in Japan. Outside of the airport you will be able to use public transportation or a taxi to travel to your destination. The three most important tips that I have for you are to have your visa and paper work ready when you get off the plane, convert some of your money to yen, and be prepared to find a taxi driver or a bus along with a map so that you know where to go and how to get there.

```
> source("C:/Users/User/Documents/R/Thesis/studies/AI ranking test/dev_slm_functions.R")
+ ai_result <- ask_slm_ranking(c(1, 2, 3, 4))
+ cat(ai_result)
```

New names:

- `` -> `...1`
- `` -> `...2`

Attempt 1 for block [1,2,3,4]...

⚠ AI skipped items in the RANKING line. Retrying...

Attempt 2 for block [1,2,3,4]...

Ranking: 4 > 1 > 2 > 3

Reason: Item 4 provides a more comprehensive and detailed guide on arriving in Japan, covering immigration, customs, and transportation options. Item 1 is a general guide suitable for navigating airports but lacks specific details. Items 2 and 3 focus on travel planning and pre-flight preparation rather than airport navigation.

```
> result_phi <- ask_slm_ranking(c(1, 2, 3, 4), model_name = "phi4-mini")
```

Attempt 1 for block [1,2,3,4]...

```
> cat(result_phi)
```

```
+ result_gemma <- ask_slm_ranking(c(1, 2, 3, 4), model_name = "gemma3")
```

```
+ cat(result_gemma)
```

Ranking: [3 > 1 > 4 > 2]

Reason: Item 3 offers a concise step-by-step guide with top advice for travelers in Japan; item 1 covers basics but lacks structured guidance and detailed tips compared to item 3, while items 4 and 2 are less focused on the airport experience.

Attempt 1 for block [1,2,3,4]...

⚠ AI skipped items in the RANKING line. Retrying...

Attempt 2 for block [1,2,3,4]...

⚠ AI skipped items in the RANKING line. Retrying...

Attempt 3 for block [1,2,3,4]...

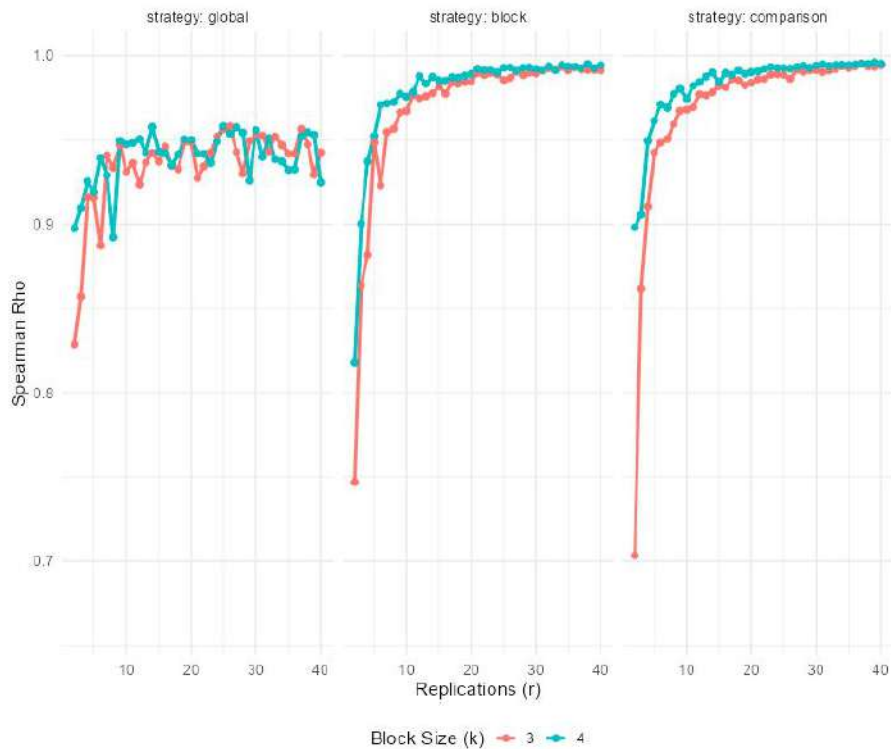
⚠ AI skipped items in the RANKING line. Retrying...

Error: AI repeatedly failed to include all items after max retries.

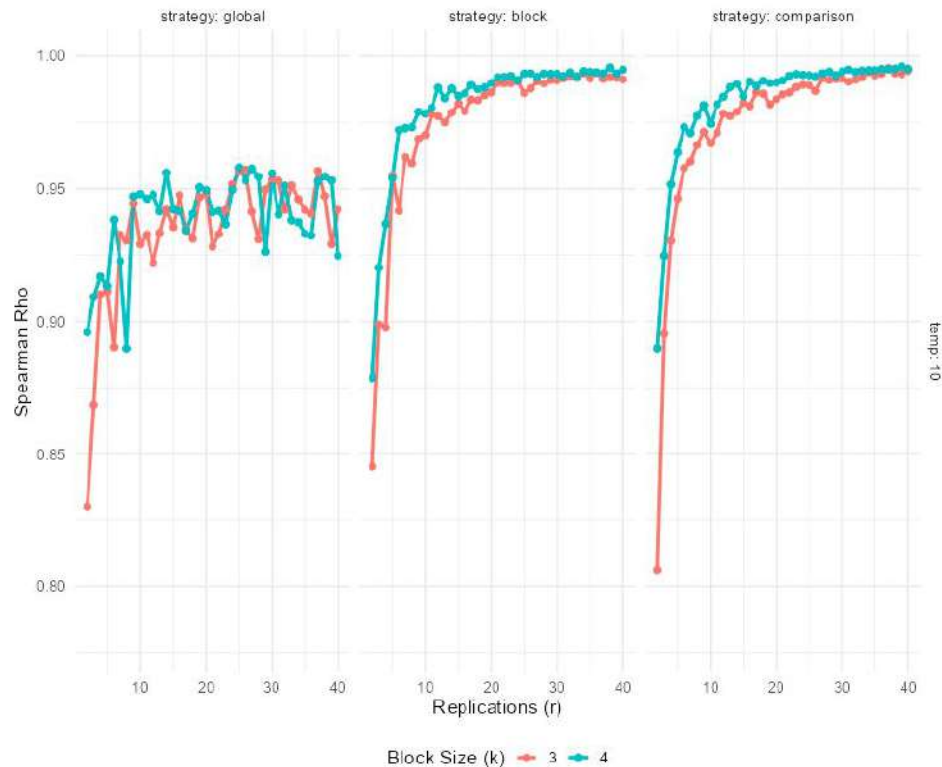
2026-05-05

PL VS BT, seed = 1000, temp = 10, r = 40, t = 120

Bradley-Terry: Spearman Rho



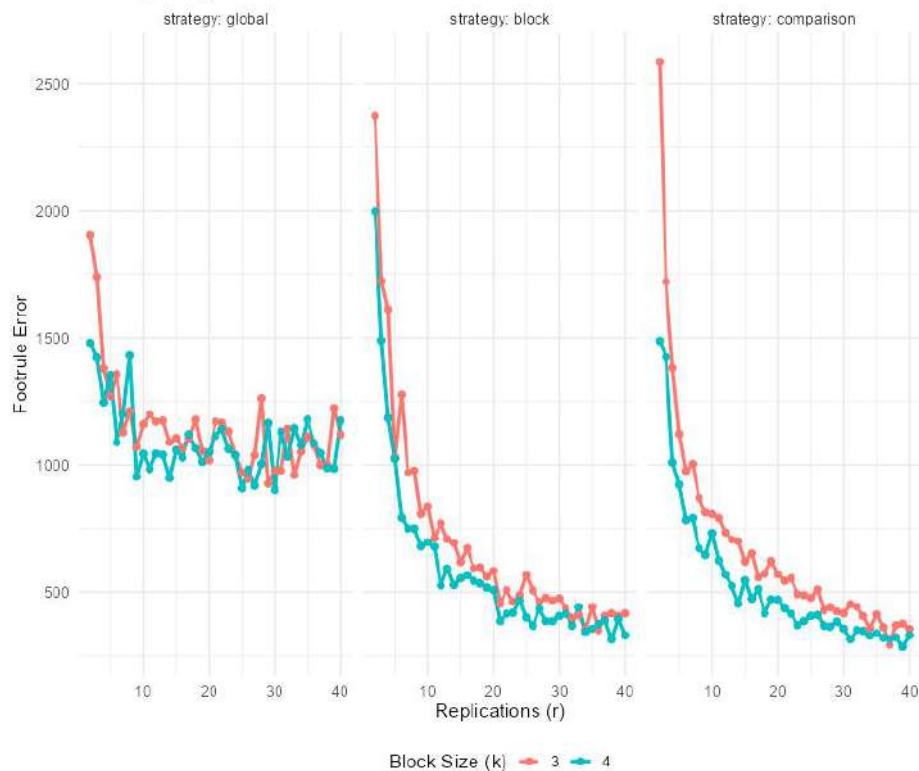
Plackett-Luce: Spearman Rho



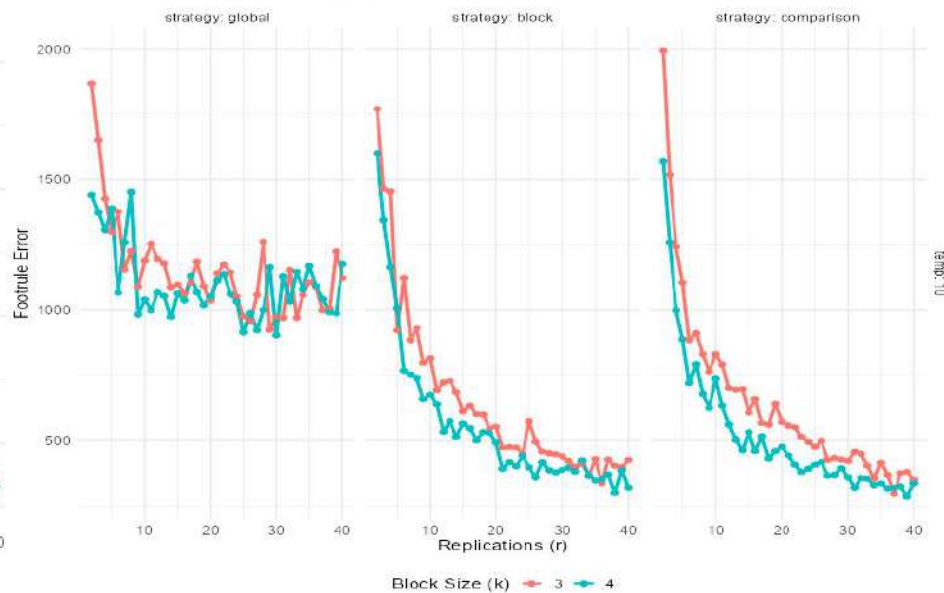
Link

PL VS BT, seed = 1000, temp = 10, r = 40, t = 120

Bradley-Terry: Footrule Error

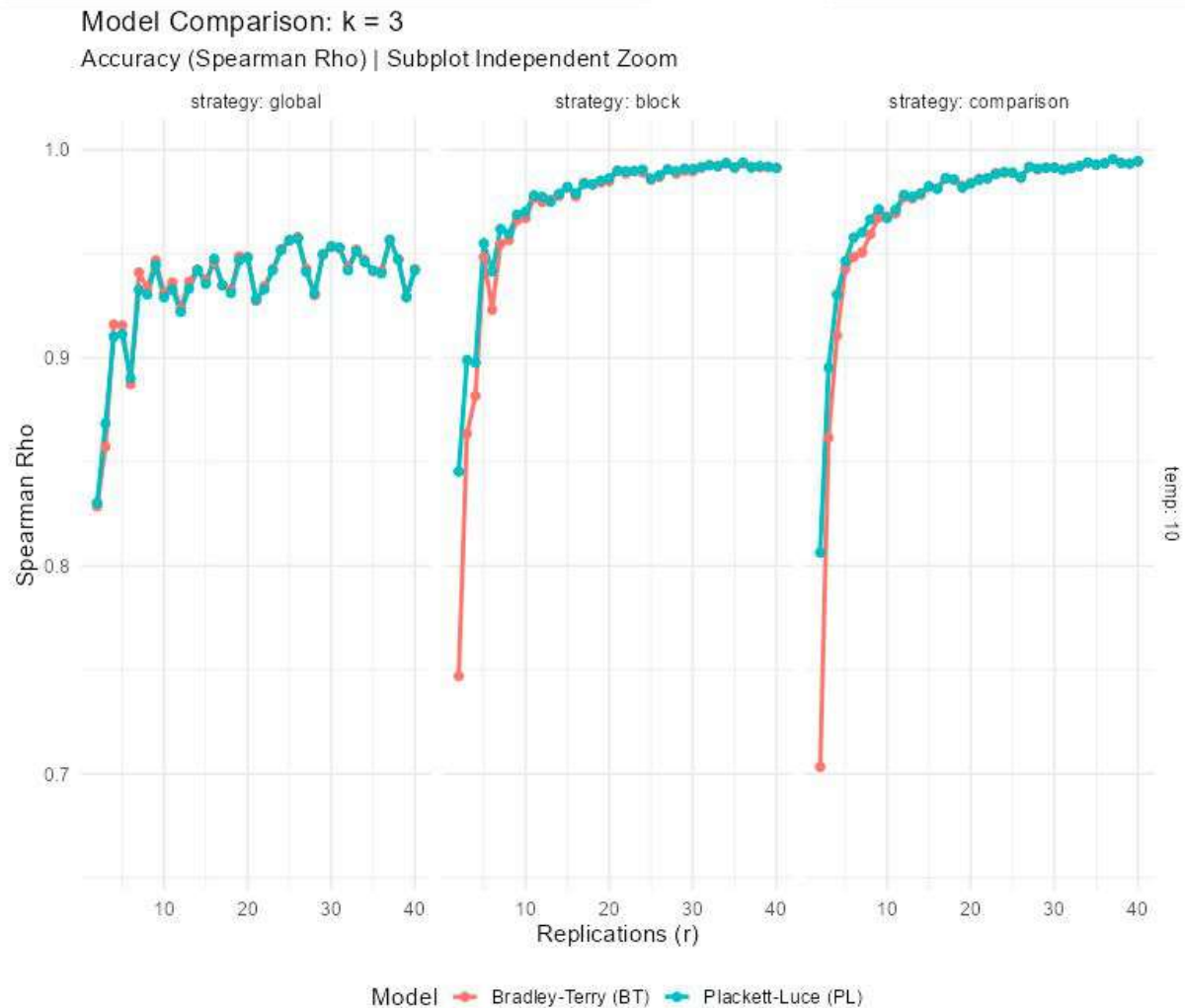


Plackett-Luce: Footrule Error



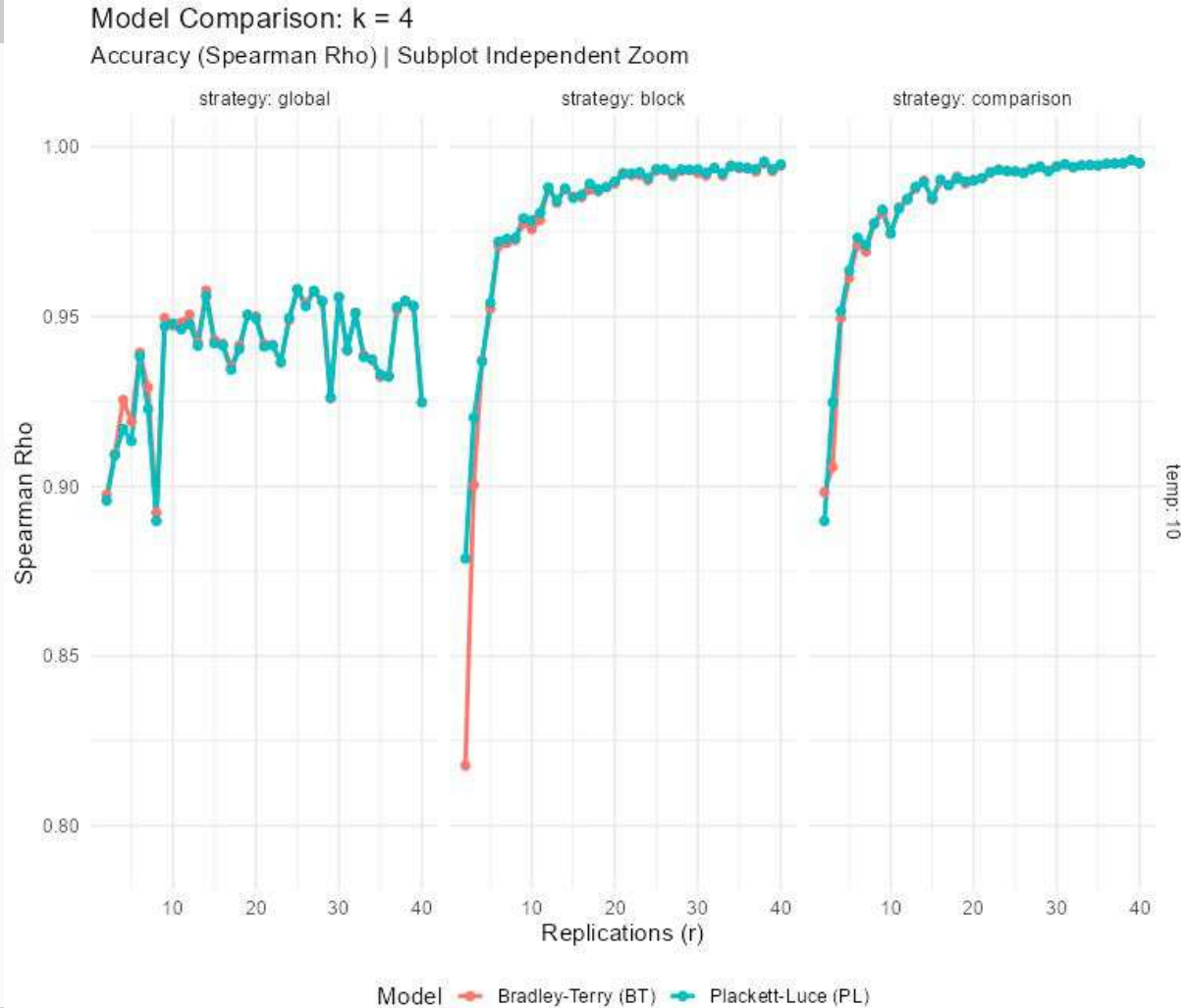
K = 3
models
compare

Seed =
1000



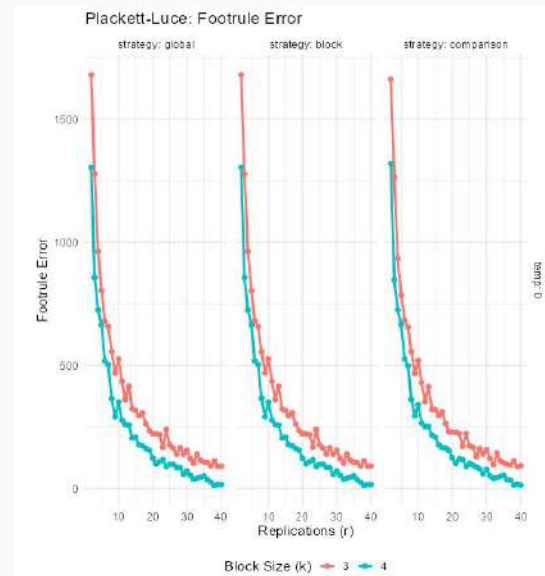
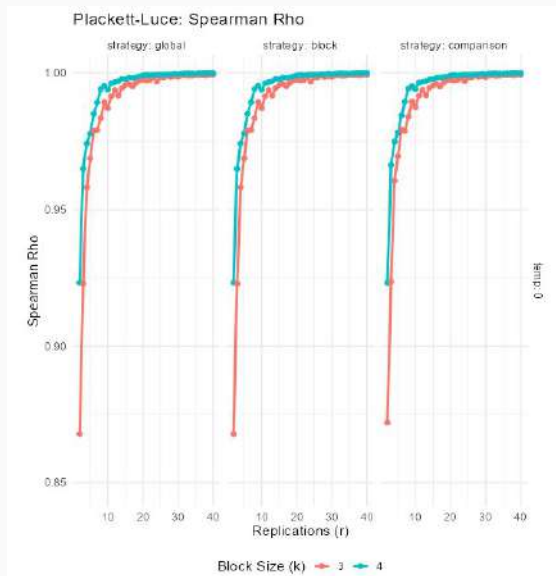
K = 4
models
compare

Seed =
1000



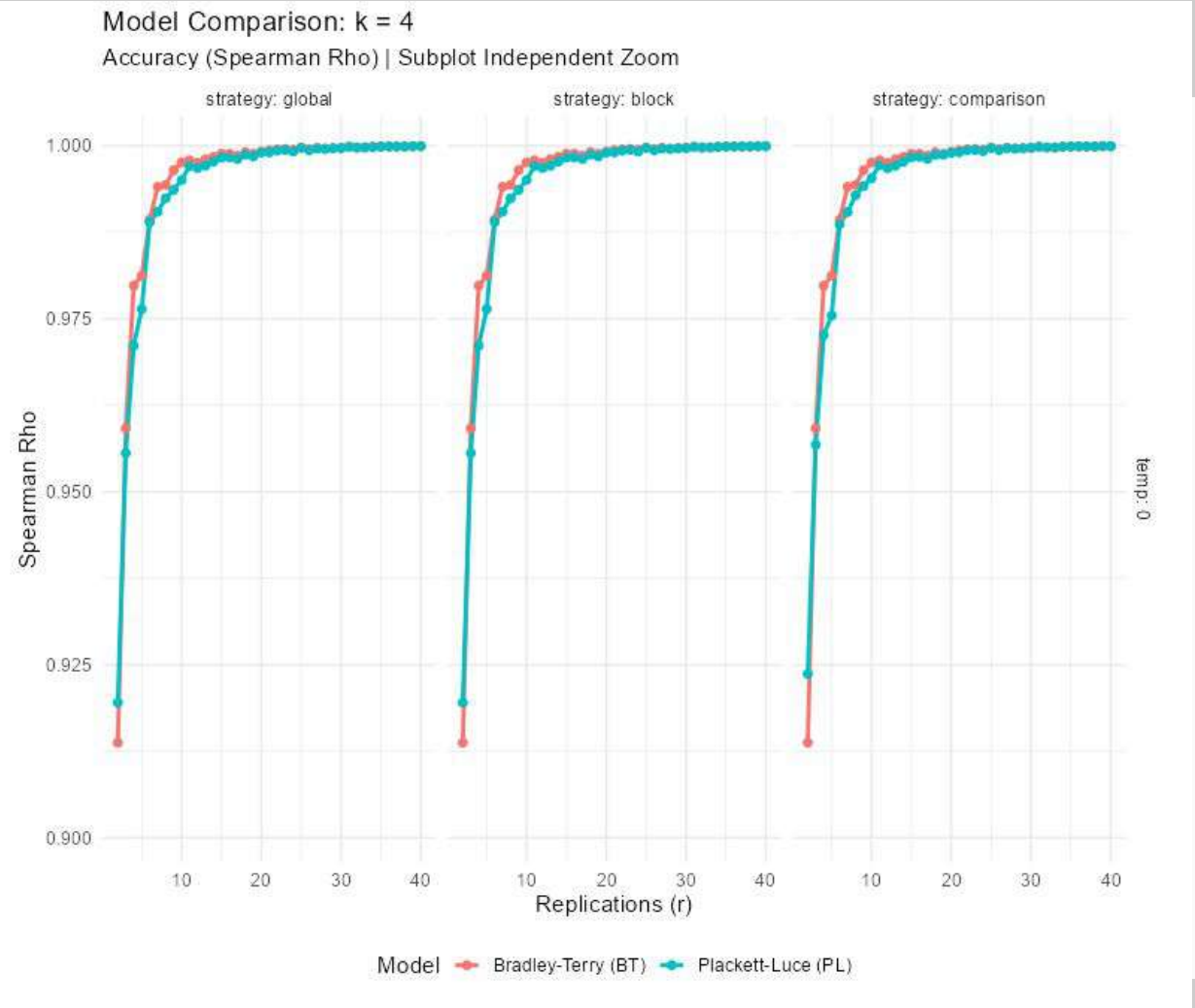
PL temp = 0, seed 1006, t = 120, r = 40

- *Same Graph with the previous slide
- PL works well with larger k



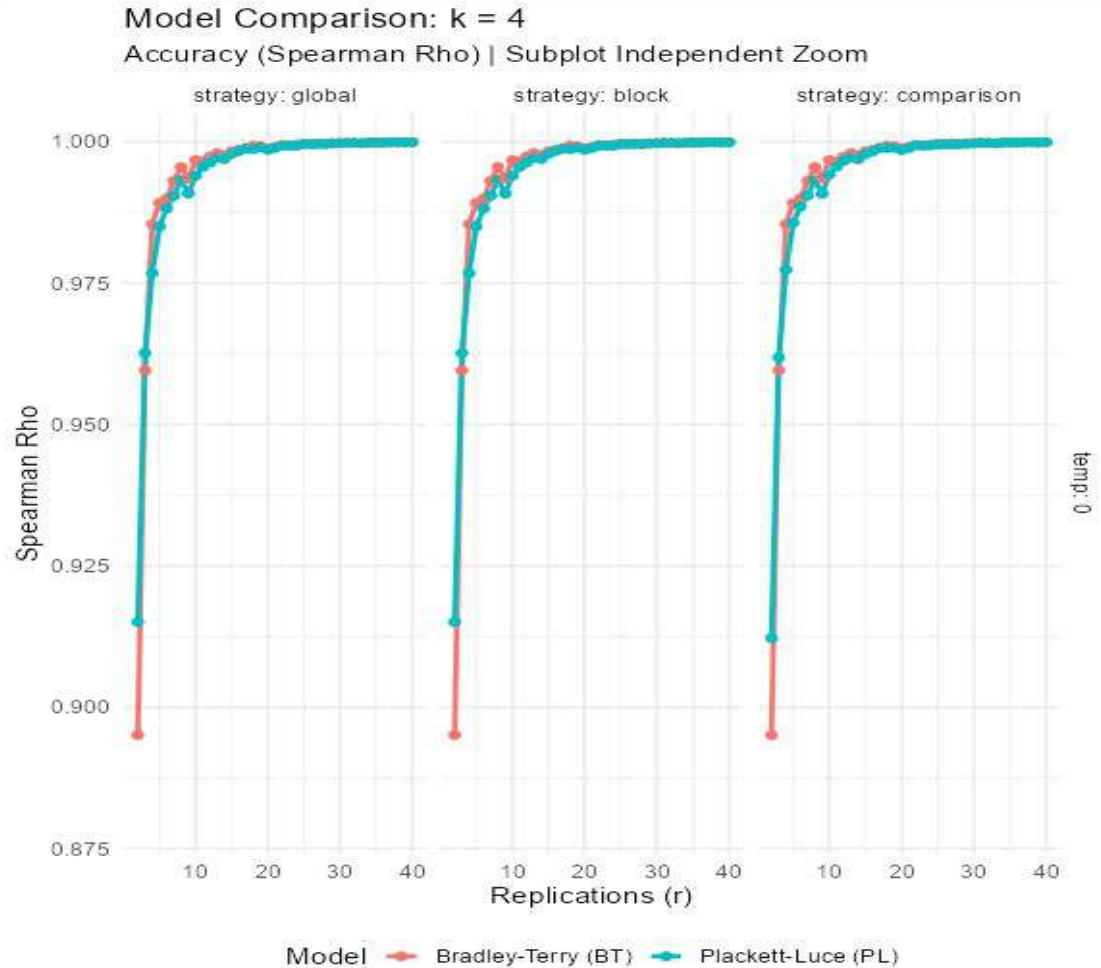
K = 4
models
Compare

Seed
=20260430



K = 4
models
Compare

Seed
=20260505



AI results - failed, still trying with other prompt and method

- [link](#)

Best-Worst Scaling

Theory, Methods and Applications

**Jordan J. Louviere, University of South Australia,
Terry N. Flynn, University of Western Sydney,
A. A. J. Marley, University of Victoria, British Columbia**

Publisher: Cambridge University Press

Publication date: 05 October 2015

BWS - how it make pack everything into rank

- A simple way to summarize the best and worst choice data is to count the best and worst choices for each choice object, as shown in Table 2.6. A simple scale and weak order of the objects can be obtained by subtracting the worst count for each object from the best count for that object, and ordering the objects by those scores.

Table 2.6 *Summary choices of public policy issues*

Issue no.	Issues (spend more on)	Best	Worst	B minus W
1	Streets and roads	2	1	1
2	K-12 education	2	0	2
3	Tertiary education	4	0	4
4	Parks and recreation	0	1	-1
5	Sports facilities	0	4	-4
6	Housing developments	0	2	-2
7	Job creation	3	0	3
8	Broadband access/speed	1	1	0
9	Tourism facilities	0	3	-3

BWS - how it make pack everything into rank

- Also

More sophisticated estimation approaches

One can write special-purpose estimation code in Matlab, Gauss or other software to maximize the likelihood of the maxdiff model given the observed best-worst choice data; the likelihood is given by Marley and Louviere (2005, eq. 30) for Case 1 and by Marley and Pihlens (2012) for Case 3. One also can estimate the parameters of this model using conditional logistic regression in readily available software such as Stata. For example, Flynn *et al.* (2007) show how to estimate parameters using a pairwise expansion of the data that assumes simultaneous best and worst choices.

Interesting fact

The Rank-Ordered Logit (ROL) and Plackett-Luce (PL) models are **often used interchangeably to analyze ranked data**, with ROL typically referring to the regression framework (incorporating covariates) and PL focusing on estimating item "worth" parameters based solely on ranks. Both are based on Luce's axiom of choice, treating rankings as a sequence of choices.  Stata +4

Key Differences and Relationship:

- **Definition:** Plackett-Luce is a specific case of the ROL (or "exploded" logit) model where utility depends only on an item's worth.
- **Covariates:** ROL models allow, and are often used for, incorporating item-specific or respondent-specific covariates to explain rankings. PL typically focuses on the intrinsic worth of items themselves.
- **Implementation:** ROL is often implemented via survival analysis software (e.g., [Stata's rologit](#) or R's `ROlogit` package) using maximum likelihood. The PL model can be fitted as a log-linear model, but often requires specialized packages for efficiency.
- **Ties:** While standard models cannot handle ties, extensions of the Plackett-Luce model (e.g., [PlackettLuce R package](#)) allow for modeling tied rankings.  Stata +6

2026-4-28

PL model Plackett-Luce

- How it cal the ranking?

Like BT , estimating alpha (the true value) , after taking in all the comparison result.

- BT:

Probability(Item i beats Item j) = (Strength of i) / (Strength of i + Strength of j)

- PL:

Probability(Specific Ranking Sequence) = [(Strength of 1st place) / (Total strength of all items)] × [(Strength of 2nd place) / (Total strength ...)] × [(Strength of 3rd place) / (Total strength ...)] × ... (until the ranking is complete)

PL model Assumption:

- Independence of Irrelevant Alternatives (IIA) / Luce's Choice Axiom
Assumes that the **relative probability of choosing Item A over Item B remains exactly the same**, regardless of what other items are added to or removed from the competition.
- **Path Independence**
The "path" or grouping structure used to make a choice does not change the final probability of an item winning. Mathematically, the model assumes that the probability of Item A being selected from the entire pool in one step is exactly equal to the probability of first selecting a subgroup containing Item A, and then selecting Item A from that subgroup. The outcome is independent of the decision path.

PL model Assumption:

-

Consider a set X of possible outcomes, and consider a selection rule P , such that for any $a \in A \subset X$ with A a finite set, the selector selects a from A with probability $P(a \mid A)$.

☐ A

Luce proposed two choice axioms. The first one is usually called "[independence from irrelevant alternatives](#)" (IIA), and the second one is usually referred to as "Luce's choice axiom".^[2]

☒ L

☐ C

Luce's choice axiom 1 (IIA): if $P(a \mid A) = 0$, $P(b \mid A) > 0$, then for any $a, b \in B \subset A$, we still have $P(a \mid B) = 0$.

Luce's choice axiom 2 ("[path independence](#)"):

$$P(a \mid A) = P(a \mid B) \sum_{b \in B} P(b \mid A)$$

for any $a \in B \subset A$.^[3]

Luce's choice axiom 1 is implied by choice axiom 2.

Link

Questions

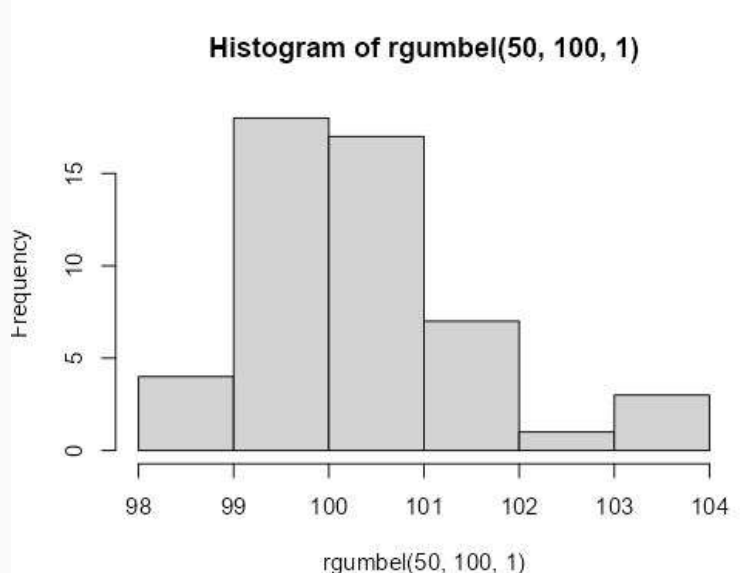
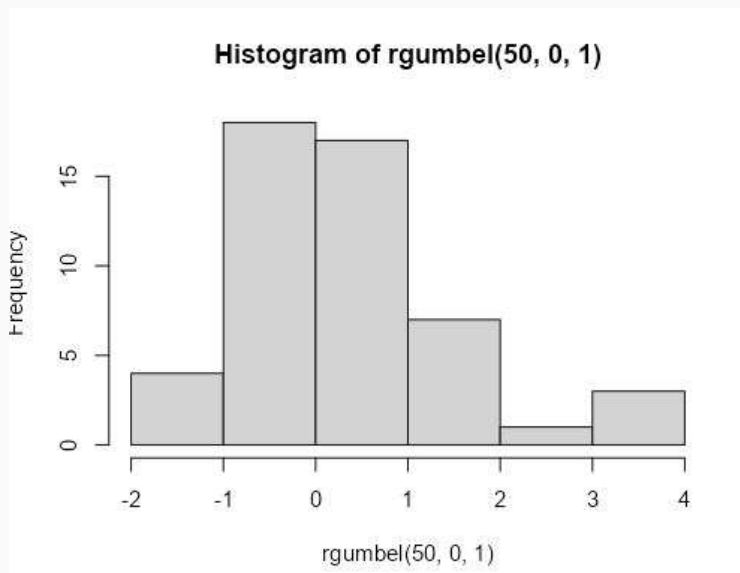
- Should I dive into efficiency comparison?
- CLI training

2026-04-21

amplify temperature or noise to create better variability

- Temperature
- `gumbel(0,1) => gumbel(location, scale)`

Location = where the most of the data gather , for 0, the peak will near 0



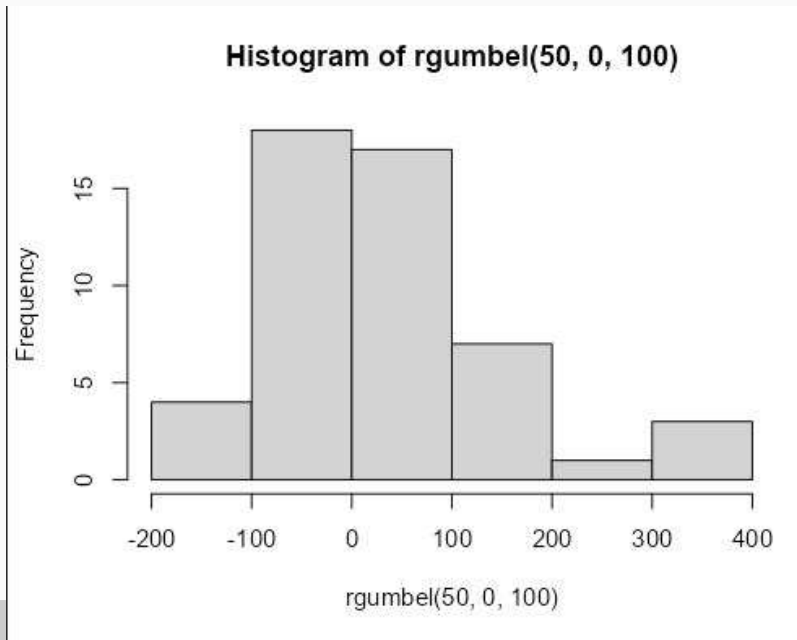
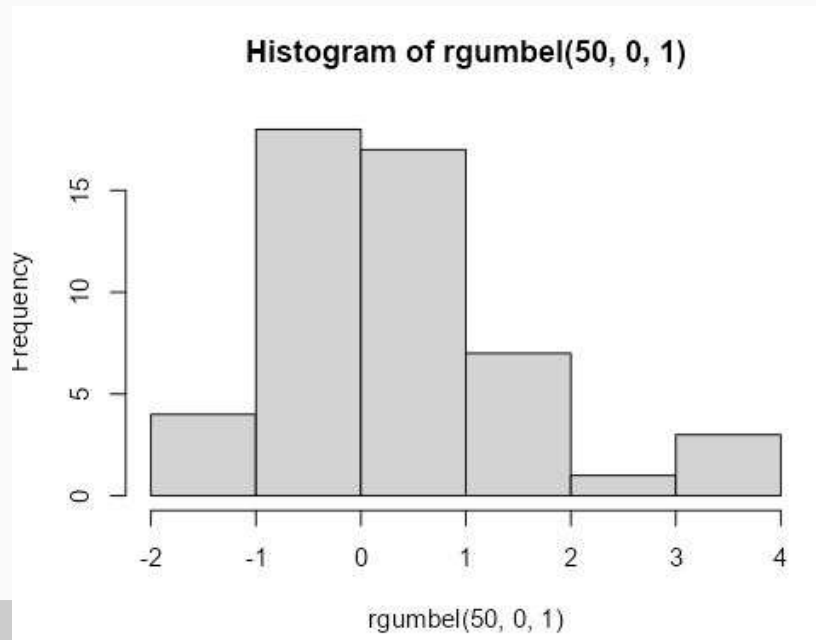
Link

amplify temperature or noise to create better variability

- `gumbel(0,1) => gumbel(location, scale)`

Scale = the x axis, how large the scale of x axis

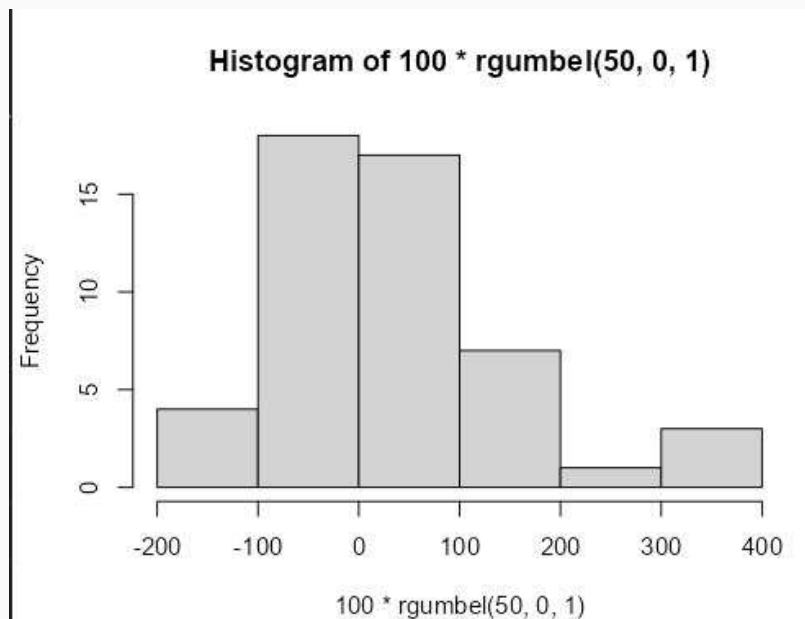
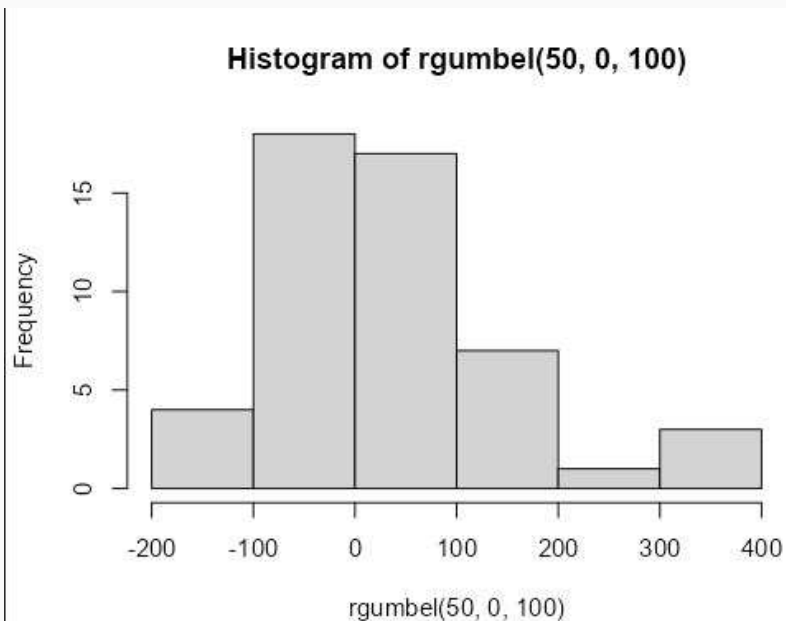
Seed = 123



Link

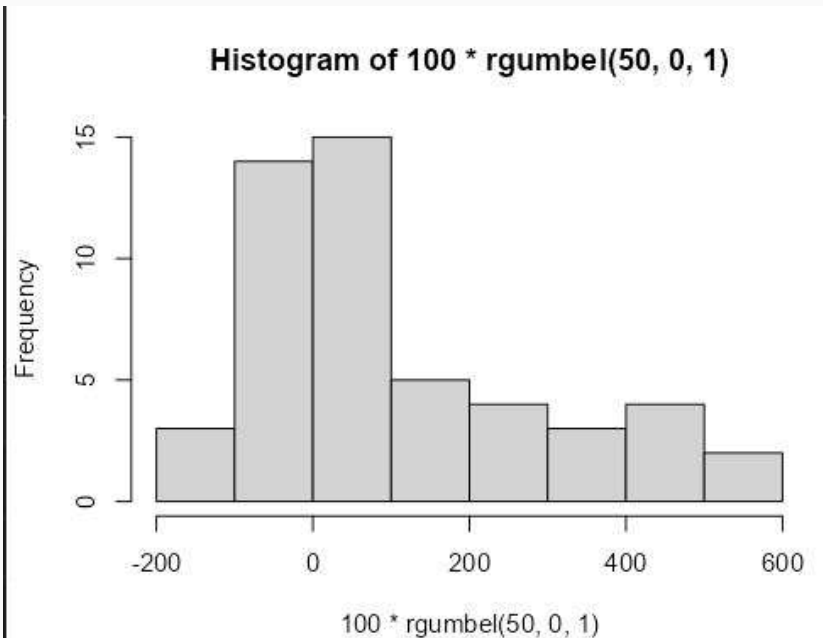
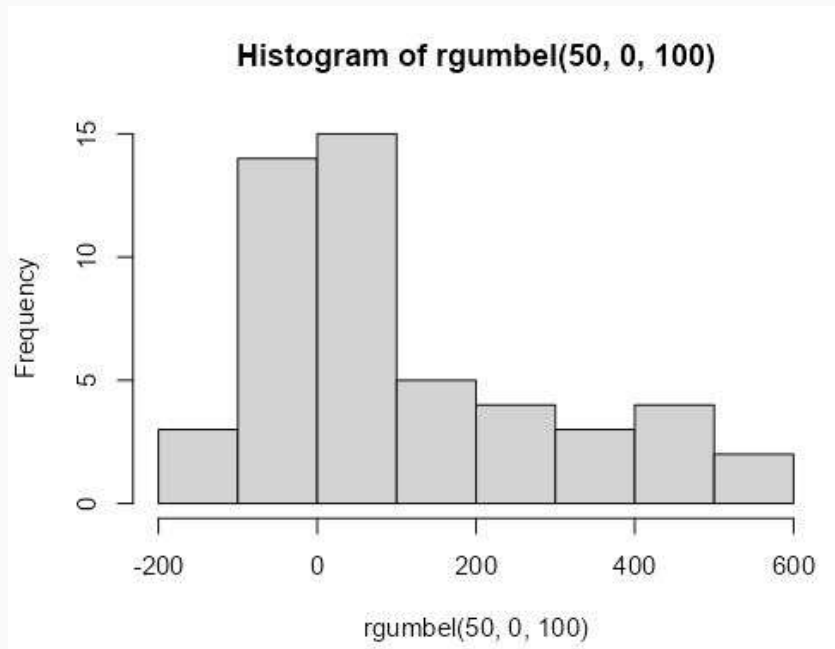
amplify temperature or noise to create better variability

- Surprisingly, mathematically $\text{temp} * \text{gumbel}(0,1) = \text{gumbel}(0,\text{temp})$
Seed = 123



amplify temperature or noise to create better variability

- Surprisingly, mathematically $\text{temp} * \text{gumbel}(0,1) = \text{gumbel}(0,\text{temp})$
Seed = 1234



Link

amplify temperature or noise to create better variability

- For us, changing the location doesn't affect our result (the ranking) at all
But why we have to change the temp, not the scale?

- **Identification Restriction**

Since utility has no absolute scale, we use $T \times \text{Gumbel}(0, 1)$ to provide an anchor for the unit of utility, treating $\text{Gumbel}(0, 1)$ as the fixed "Standard Meter" of noise.

- **Scale Parameter Logic**

Following Kenneth Train's Discrete Choice Theory, the model only estimates the ratio of signal to noise; separating T allows researchers to explicitly quantify the scale parameter (the inverse of variance) relative to a standardized benchmark.

- **Theoretical Consistency**

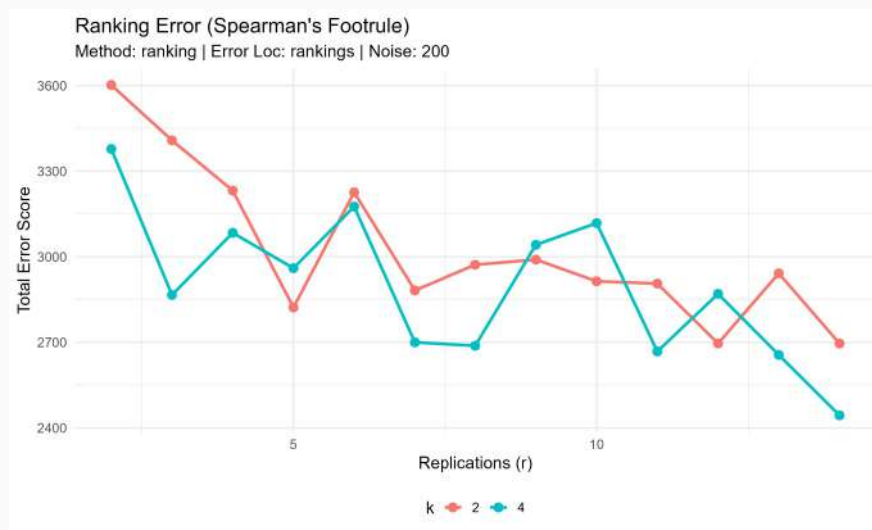
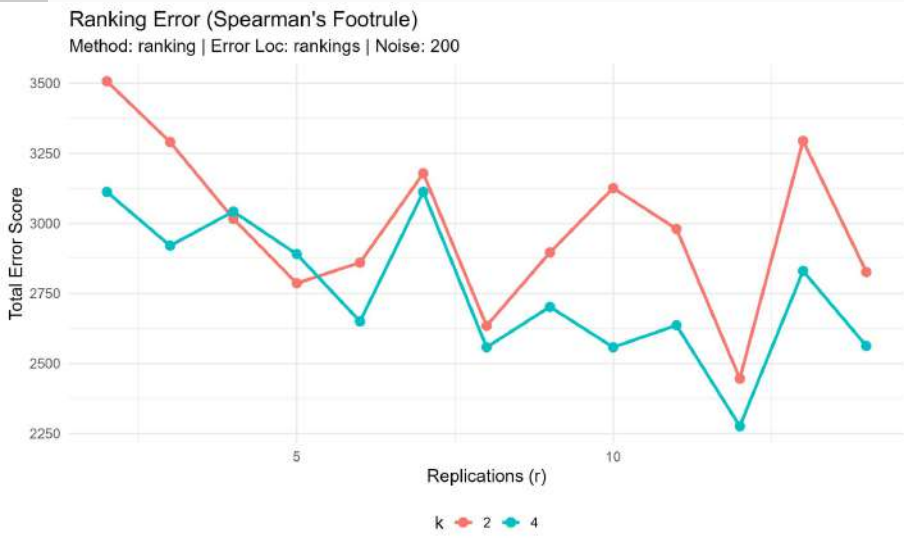
This structure maintains the structural integrity of Random Utility Models (RUM), ensuring the error term remains a "Standardized Residual," which is essential for comparing results across different experimental designs or datasets.

Link

Some simulation(although seems not so useful)

Temp = 200 , t100, k 2,4, r 14 , **error in ranking**, footrule

Note, same side(left or right), same seed

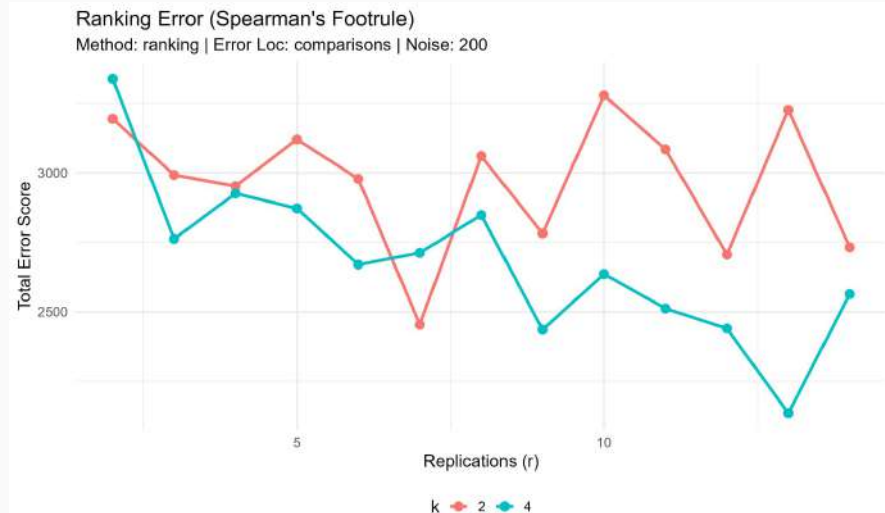
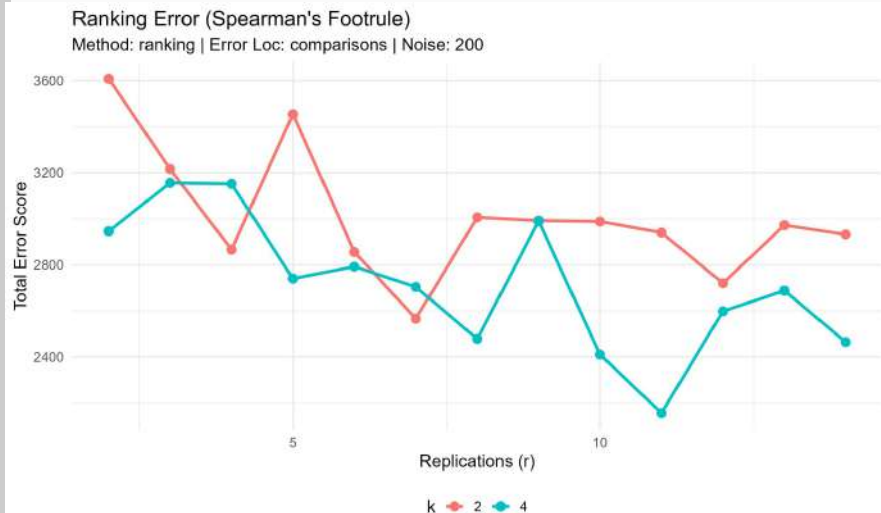


Link

Some simulation(although seems not so useful)

Temp = 200 , t100, k 2,4, r 14 , **ranking error in comparison**, footrule

Note, same side(left or right), same seed



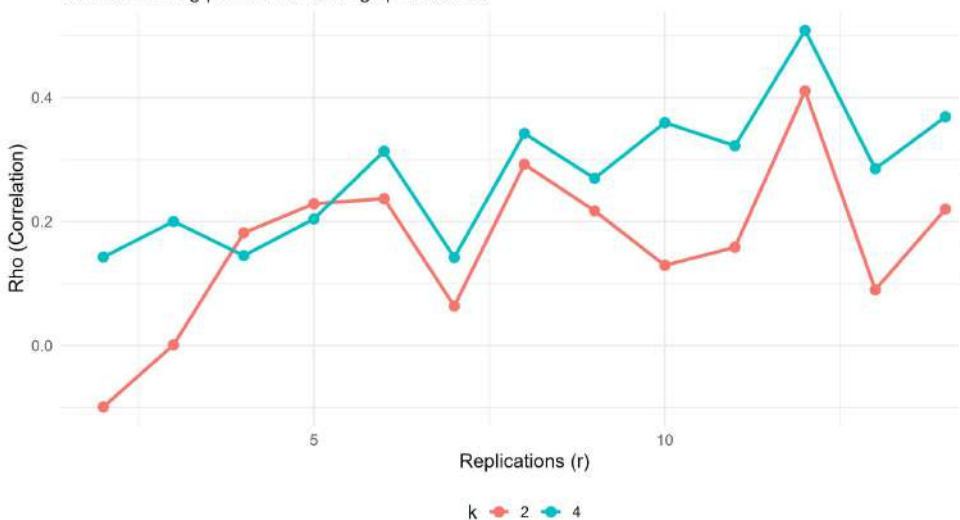
Link

Some simulation(although seems not so useful)

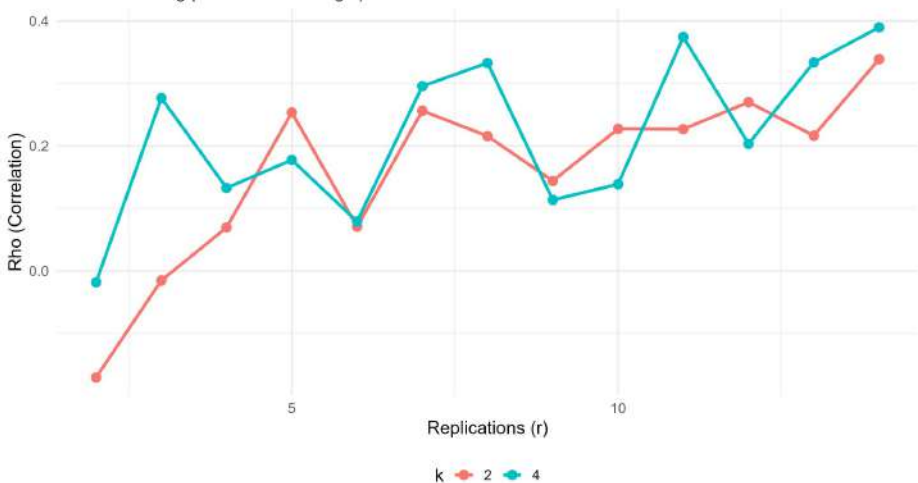
Temp = 200 , t100, k 2,4, r 14 , error in ranking, rho

Note, same side(left or right), same seed

Ranking Accuracy (Spearman's Rho)
Method: ranking | Error Loc: rankings | Noise: 200



Ranking Accuracy (Spearman's Rho)
Method: ranking | Error Loc: rankings | Noise: 200



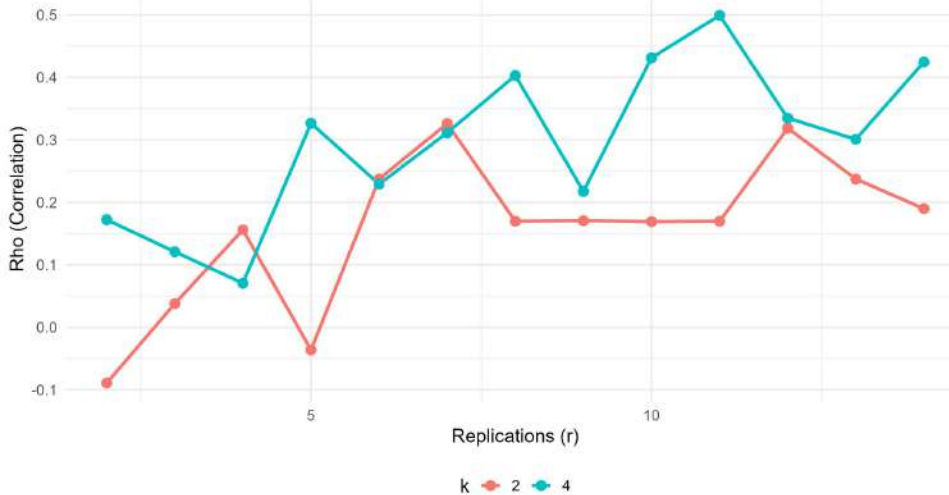
Link

Some simulation(although seems not so useful)

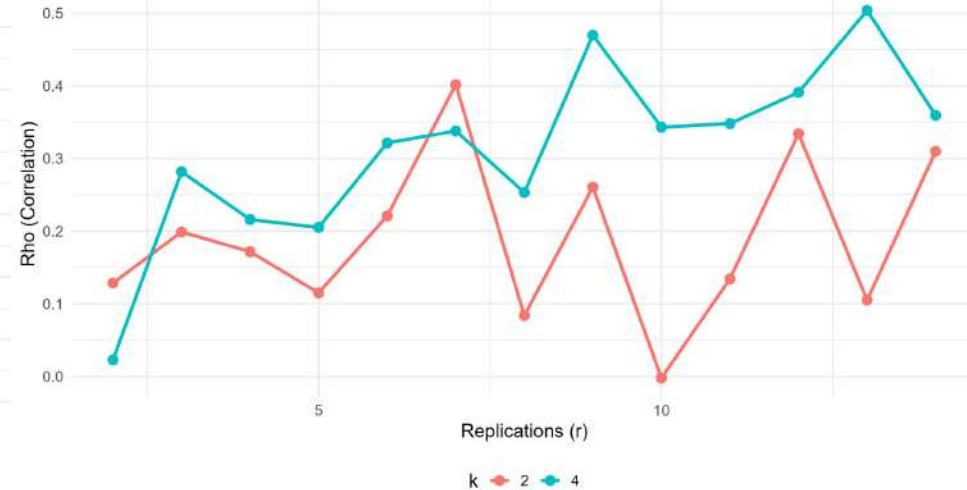
Temp = 200 , t100, k 2,4, r 14 , error in comparison, rho

Note, same side(left or right), same seed

Ranking Accuracy (Spearman's Rho)
Method: ranking | Error Loc: comparisons | Noise: 200



Ranking Accuracy (Spearman's Rho)
Method: ranking | Error Loc: comparisons | Noise: 200



Where to add error — rankings or comparisons

The Authoritative Rule: Add Error to Items (Rankings), Not Pairs (Comparisons)

- **The Golden Rule**

Add noise to items (Rankings) via the Random Utility Model (RUM) framework, expressed as $U_i = V_i + \epsilon_i$, rather than adding noise to each pairwise comparison.

- **Theoretical Authority**

According to Yellott (1977) and McFadden (1974), the Bradley-Terry model is mathematically derived from the assumption that each item possesses an underlying **Latent Utility** coupled with Gumbel-distributed noise.

Where to add error — rankings or comparisons

- **Transitivity vs. Intransitivity**

Adding error to rankings ensures **Transitivity** (logical consistency: if $A > B$ and $B > C$, then $A > C$). Conversely, adding error directly to comparisons allows for **Circular Triads** (e.g., $A > B > C > A$), which explicitly violates the fundamental assumptions of standard ranking models.

- **Luce's Choice Axiom**

Luce (1959) proves that for a ranking system to remain "probabilistically consistent," the final choice must depend on the relative strengths of the items themselves, rather than on independent stochastic errors applied to every single pair.

Link

amplify temperature or noise to create better variability

- For us, changing the location doesn't affect our result (the ranking) at all
But why we have to change the temp, not the scale?

The Mathematical Synergy: Gumbel-Max Trick and AI Temperature

- **Theoretical Alignment**

The concept of "Temperature" in Large Language Models (LLMs) is fundamentally rooted in the **Gumbel-Max Trick**, a core theorem at the intersection of machine learning and statistical mechanics.

- **Mathematical Equivalence**

The Gumbel-Max Trick demonstrates that adding Gumbel-distributed noise to a set of scores (log-probabilities) and selecting the maximum is mathematically equivalent to sampling directly from a **Softmax distribution**.

The Mathematical Synergy: Gumbel-Max Trick and AI Temperature

- **Isomorphic Structure**

The simulation formula directly mirrors the internal stochastic process of an LLM:

$$U = V + T \cdot \text{Gumbel}(0, 1)$$

In this framework, T (Temperature) serves to explicitly scale the entropy of the output probability distribution.

- **Methodological Validity**

This alignment ensures that the noise introduced in the R simulation functions as a theoretically sound and robust proxy for the actual variability produced by an AI agent during the empirical judging phase.

Questions

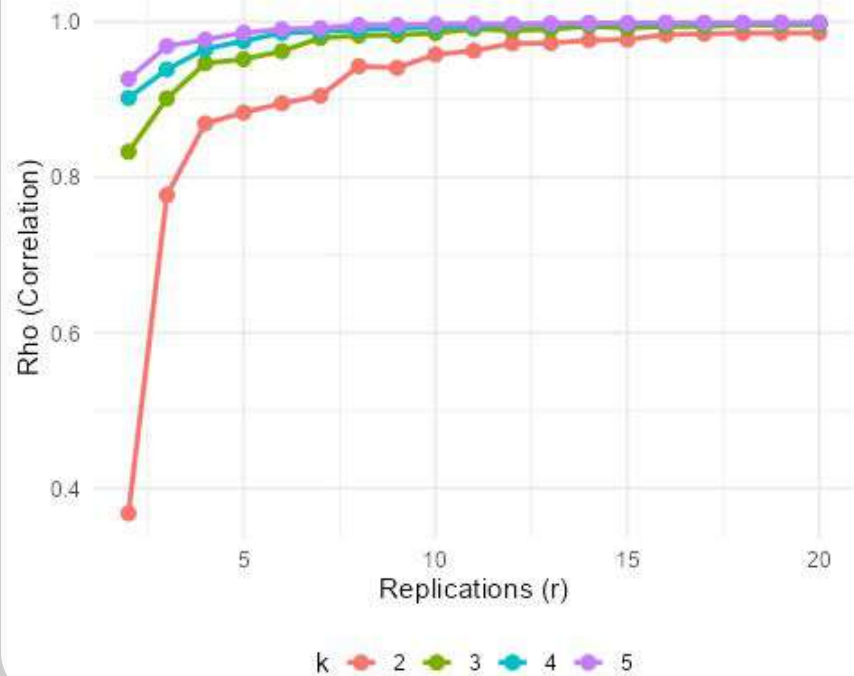
- Use AI : We can do ranking of > 2 treatment at once!
So, can change Bradley-Terry model to Plackett-Luce model (sounds like similar to BT, but have to dive deep into softmax and PL)
- Need to know gumbel max trick?
- Double noise: wonder if when we really getting the data, do we have to set the temperature of the LLM while telling it $U = V + \text{temp} * \text{gumbel}(0,1)$, if so, we will apply 2 noise (what I know from what I discuss with AI)

2026-04-14

Result

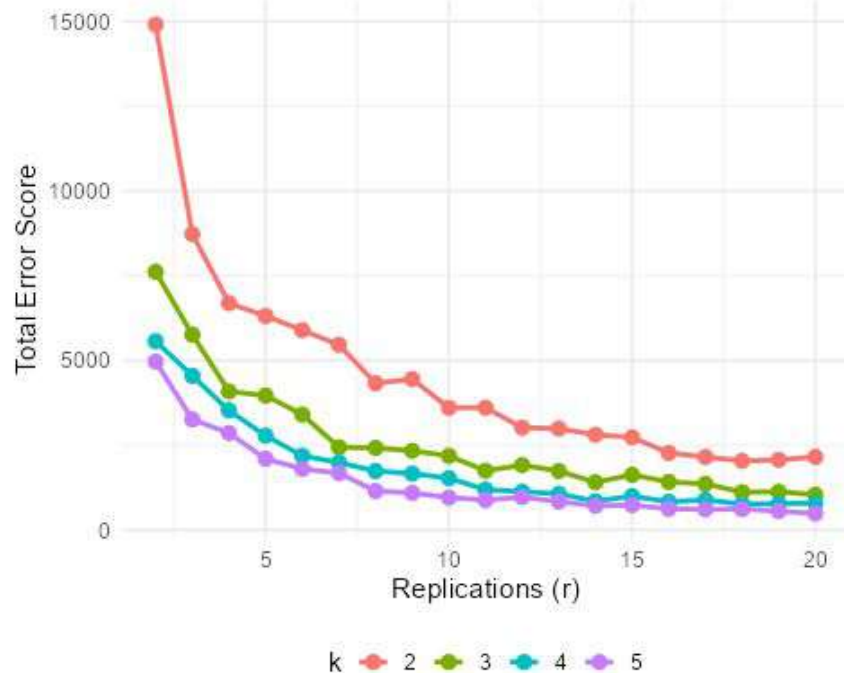
Ranking Accuracy (Spearman's Rho)

Method: ranking | Noise: 2

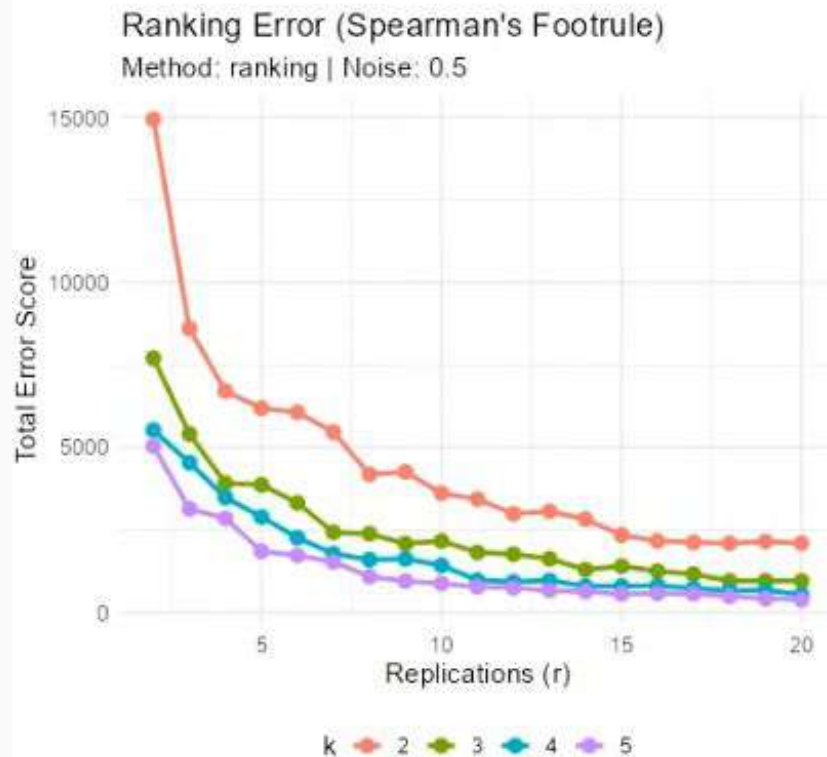
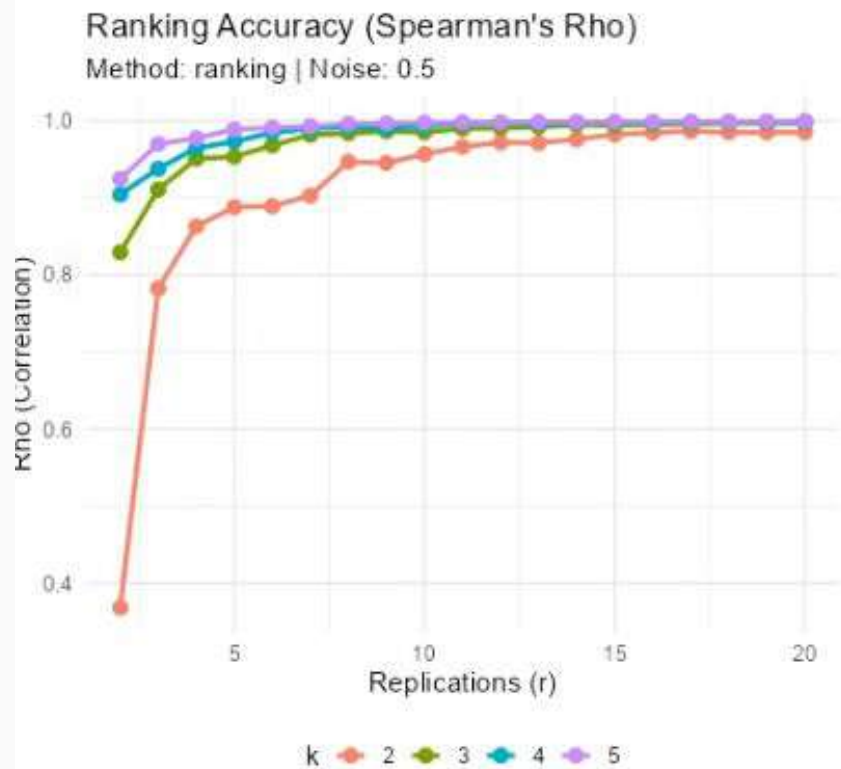


Ranking Error (Spearman's Footrule)

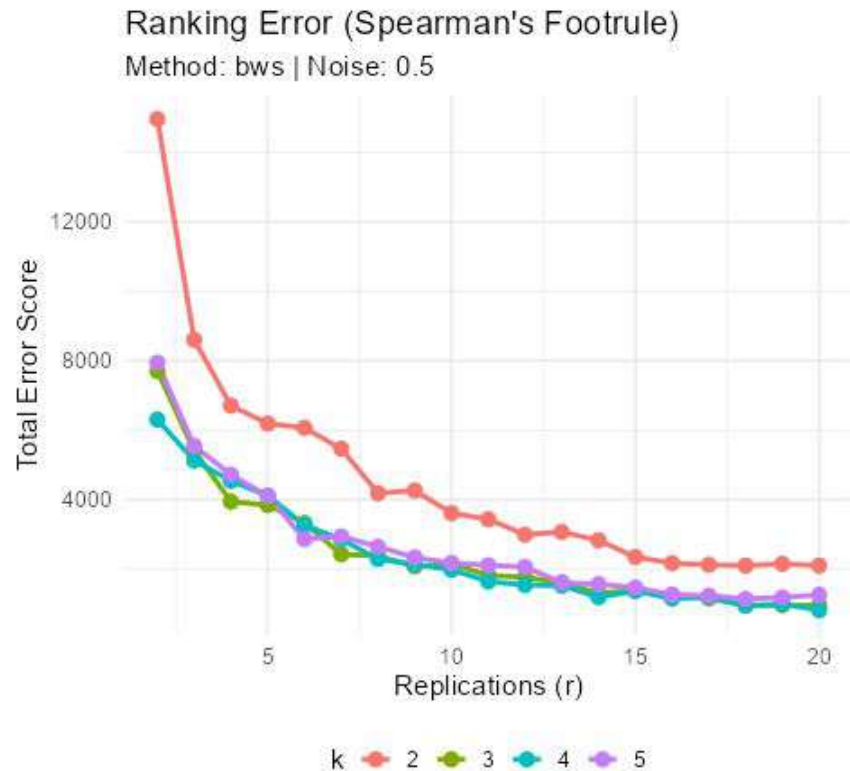
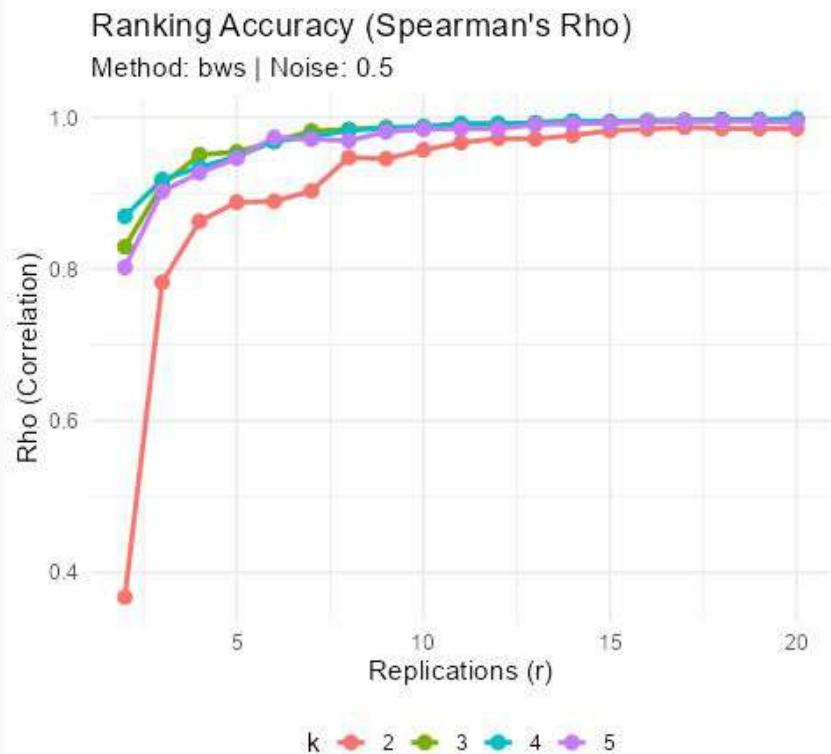
Method: ranking | Noise: 2



Result



Result



Link to code

- [code](#)

```
# --- Add Error to Rankings ---
u_scores <- block$TrueValue + temp * rgumbel(nrow(block), 0,
1)

pairs_mat <- combn(block$TREATMENT, 2)
u1 <- u_scores[match(pairs_mat[1, ], block$TREATMENT)]
u2 <- u_scores[match(pairs_mat[2, ], block$TREATMENT)]

winners <- ifelse(u1 > u2, pairs_mat[1, ], pairs_mat[2, ])
losers <- ifelse(u1 > u2, pairs_mat[2, ], pairs_mat[1, ])
return(data.frame(Winner = winners, Loser = losers))
```

```
# --- Add Error to Comparisons ---
error <- temp * rgumbel(nrow(block), 0, 1)

pairs_mat <- combn(block$TREATMENT, 2)
u1 <- block$TrueValue[match(pairs_mat[1, ],
block$TREATMENT)]
u2 <- block$TrueValue[match(pairs_mat[2, ],
block$TREATMENT)]

diff <- (u1 - u2) + error

winners <- ifelse(diff > 0, pairs_mat[1, ], pairs_mat[2, ])
losers <- ifelse(diff < 0, pairs_mat[2, ], pairs_mat[1, ])
return(data.frame(Winner = winners, Loser = losers))
```

Link to code

- [code](#)

```
# --- Add Error to Rankings ---
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block$TREATMENT)]

diff <- (u1 - u2) + error

winners <- ifelse(diff > 0, pairs_mat[1, ], pairs_mat[2, ])
losers <- ifelse(diff < 0, pairs_mat[2, ], pairs_mat[1, ])
return(data.frame(Winner = winners, Loser = losers))
```

2026-04-07

Our assumption

- We assume there is a “true rank for each solution ~~everyone~~”, but there are some errors (for example, our judge’s perception ~~that person’s current condition or his preference~~) caused by human when making judgements.

So that

Final rank = True rank + error

Gumbel distribution

- Type I extreme value distribution used to **model the maximum** (or minimum) of a number of samples.
- For example the maximum amount of rain within past 100 years.
- Because respondents are picking extremes, the unobserved random error in their evaluation is naturally modeled using the Gumbel distribution.
- So what is the proof that we can use this distribution?
Additionally, both discrete choice experiment and best-worst scaling interpret the aforementioned choice process with the error terms following Gumbel distribution and obtain a Multinomial (or conditional) logit model.

Temperature and noise

$$\tilde{q}_i = q_i + T \cdot \text{Gumbel}(0, 1)$$

- What is temperature and noise?

Temperature controls the degree of randomness in token selection.

Lower temperatures are good for prompts that require a less open-ended or creative response, while higher temperatures can lead to more diverse or creative results. A temperature of 0 means that the highest probability tokens are always selected. In this case, responses for a given prompt are mostly deterministic, but a small amount of variation is still possible.

Temperature and noise

$$\tilde{q}_i = q_i + T \cdot \text{Gumbel}(0, 1)$$

- What is temperature and noise?

Gumbel Distributed Random Noise is a stochastic perturbation rooted in **Extreme Value Theory**, specifically designed to model the distribution of **maximum values**. In machine learning and data analysis, it is utilized to transform deterministic selection into probabilistic sampling, allowing models to explore discrete choices with controlled variability while maintaining statistical rigor through its unique asymmetrical, **heavy-tailed distribution**.

Temperature and noise

$$\tilde{q}_i = q_i + T \cdot \text{Gumbel}(0, 1)$$

- The Gumbel-Max Trick with Temperature
- transform static numerical scores into stochastic (random) decisions
 - q_i (**Logits / Base Score**): Represents the "intrinsic strength" or preference of an option (e.g., the raw weight of an airport improvement suggestion).
 - $\text{Gumbel}(0, 1)$ (**Random Perturbation**): Noise drawn from a standard Gumbel distribution. it represents unobserved factors, latent individual biases, or random errors in real-world decision-making.
 - T (**Temperature / Scaling Factor**): This is your "control knob" for randomness.
 - $\tilde{q}_i$ (**Perturbed Score**): The final score after adding noise and adjusting for temperature.

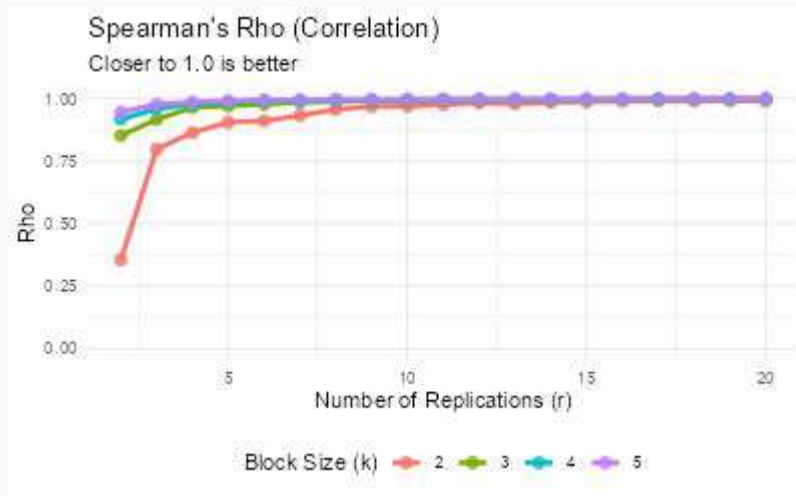
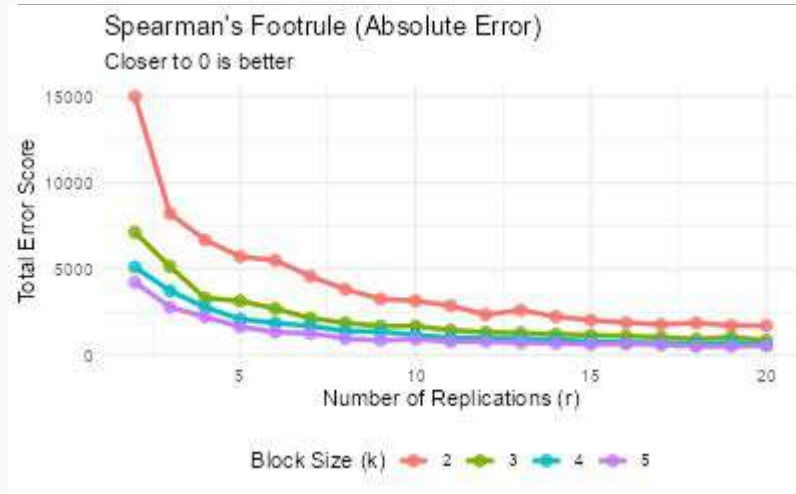
Temperature and noise

$$\tilde{q}_i = q_i + T \cdot \text{Gumbel}(0, 1)$$

- **Low T ($T \rightarrow 0$):** The noise is suppressed. The model becomes conservative and almost always picks the highest base score (Deterministic).
- **High T ($T \rightarrow \infty$):** The noise is amplified. The original scores (q_i) become irrelevant, and the choice becomes almost entirely random.

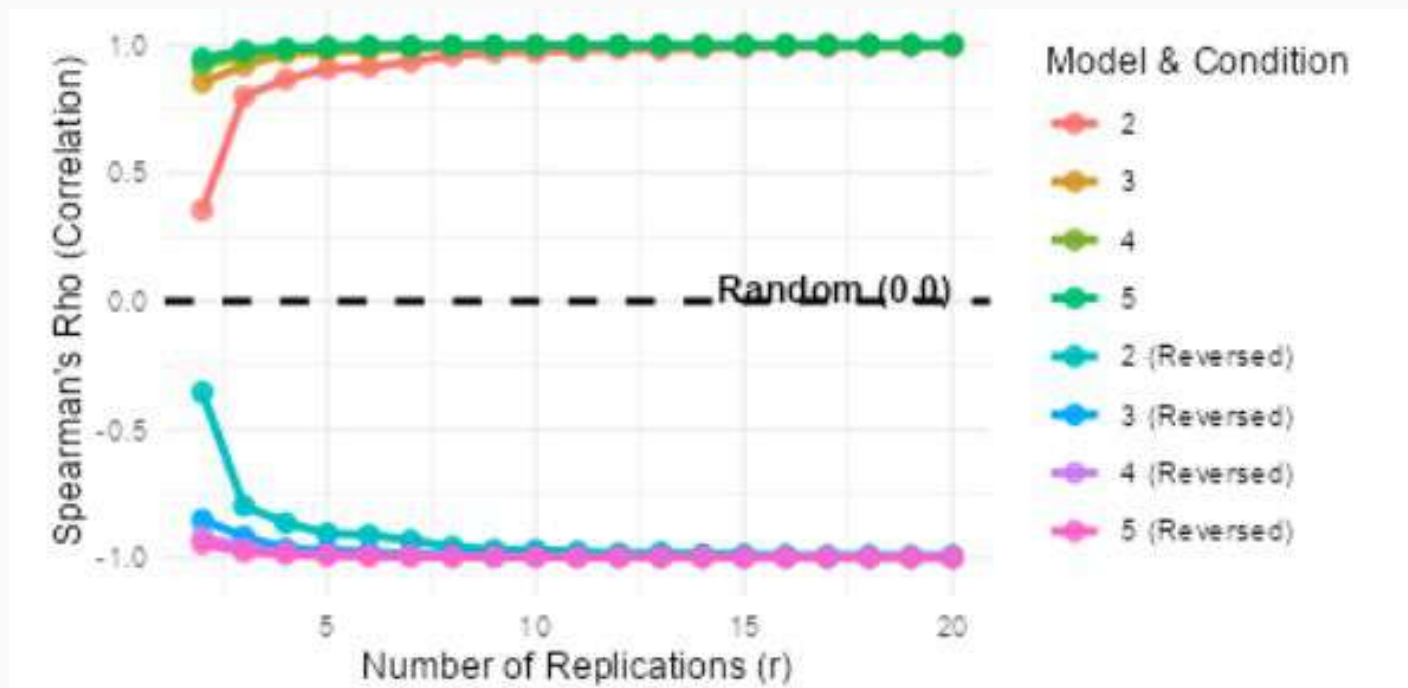
Link

Footrule vs ρ



[Link](#)

Reverse ranking



Q

- Should I mention the relationship of Gumbel & Random Utility Theory & logit model? —yes
- 400 Airport Items -> Design Experiment: Alpha Design (cognitive load reduction) -> Collect Data: Best-Worst Scaling (BWS) -> Model Utility: Random Utility Theory (RUT) & Gumbel Assumption -> Estimate Ranking: Bradley-Terry (BT) Model

2026-03-31

Spearman's ρ

- What does it do ?

Calculate the correlation between 2 sets of ranking

$$\rho = 1 - \{ 6 \sum d^2 / (n^3 - n) \} , \quad -1 < \rho < 1$$

- Where d = the differences between of the ranks of a treatment

treatment	Rank in 1 st replication	Rank in 2 nd replication	d
a	1	1	0
b	2	3	-1
c	3	2	1

Spearman's footrule

- Calculate the absolute difference between 2 set of rankings

$$R = 1 - \{ 3 \sum |d| / (n^2 - 1) \}$$

What can we use it to do

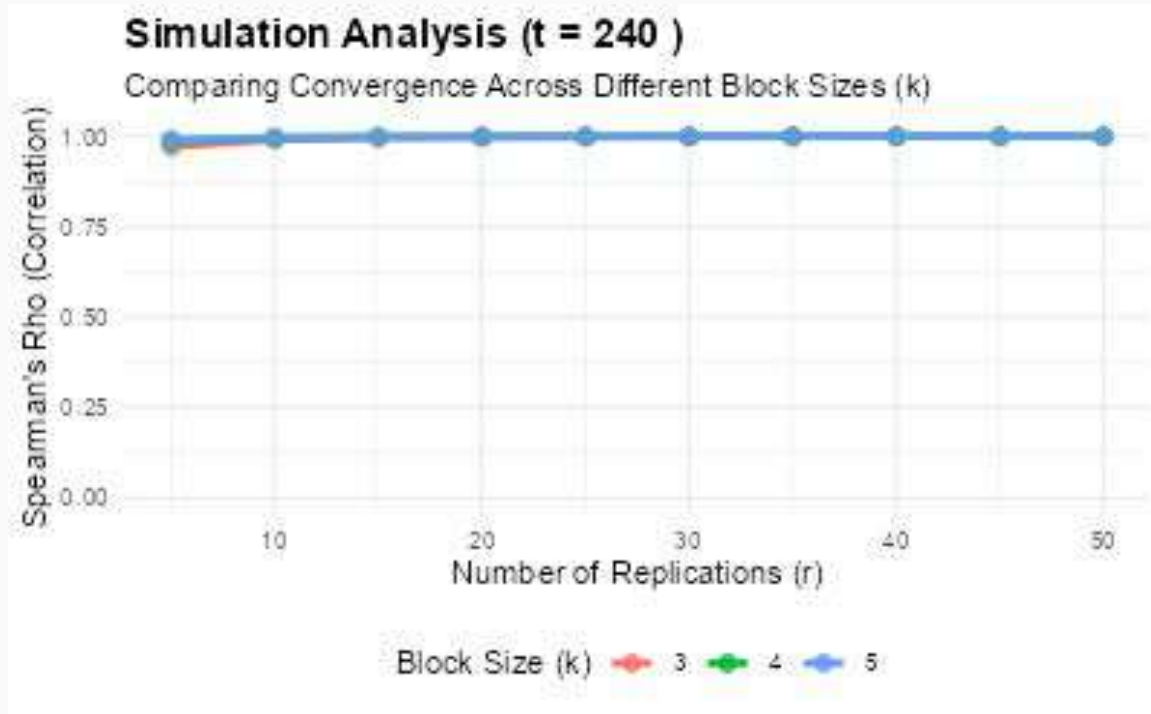
- Compare between 2 user segment
- Sample Stability Test (Split-Half Reliability) -> Check internal consensus within your sample group
- Model Robustness Check (Sensitivity Analysis) -> Run the BT model using the "raw data including all respondents," and then run the model again using the "clean data after removing questionable responses. If the p value remains very high, it demonstrates high robustness and are not significantly influenced by extreme values

result

Graph

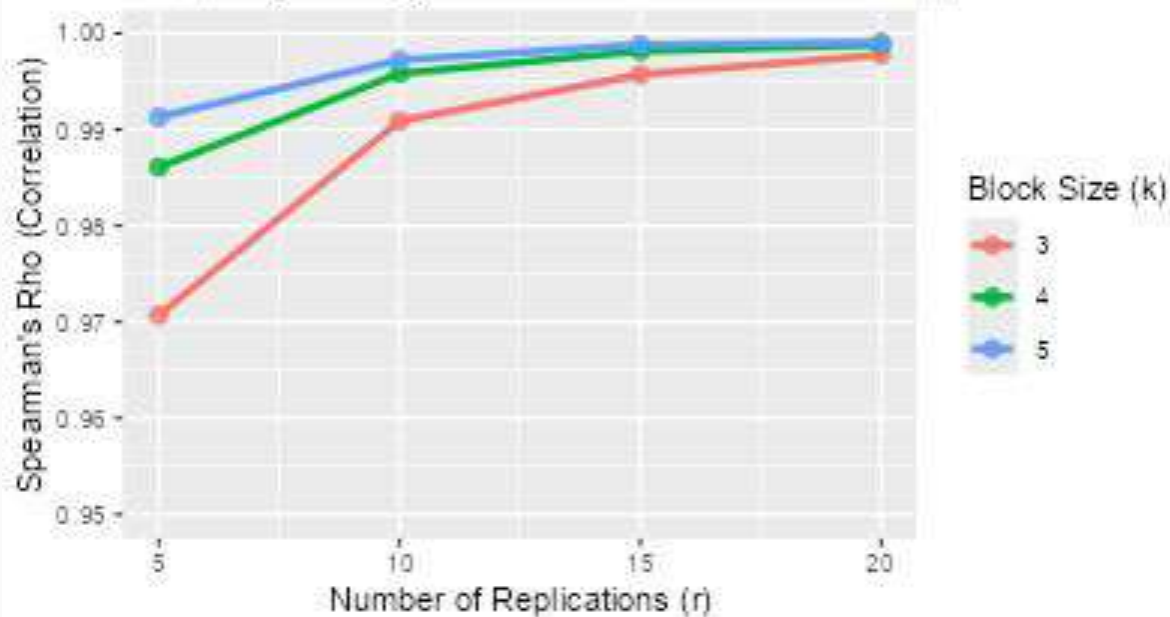
- [Github link](#)

-



Graph

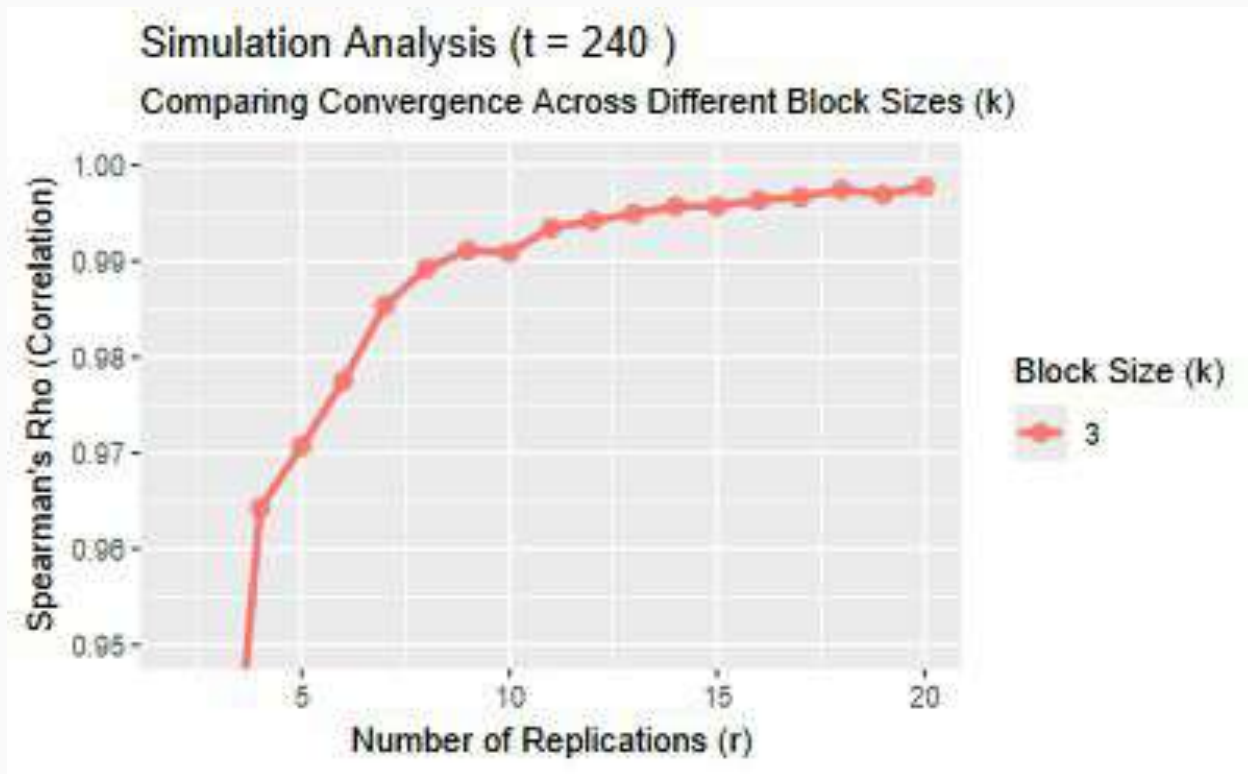
- Simulation Analysis ($t = 240$)
Comparing Convergence Across Different Block Sizes (k)



Link

Graph

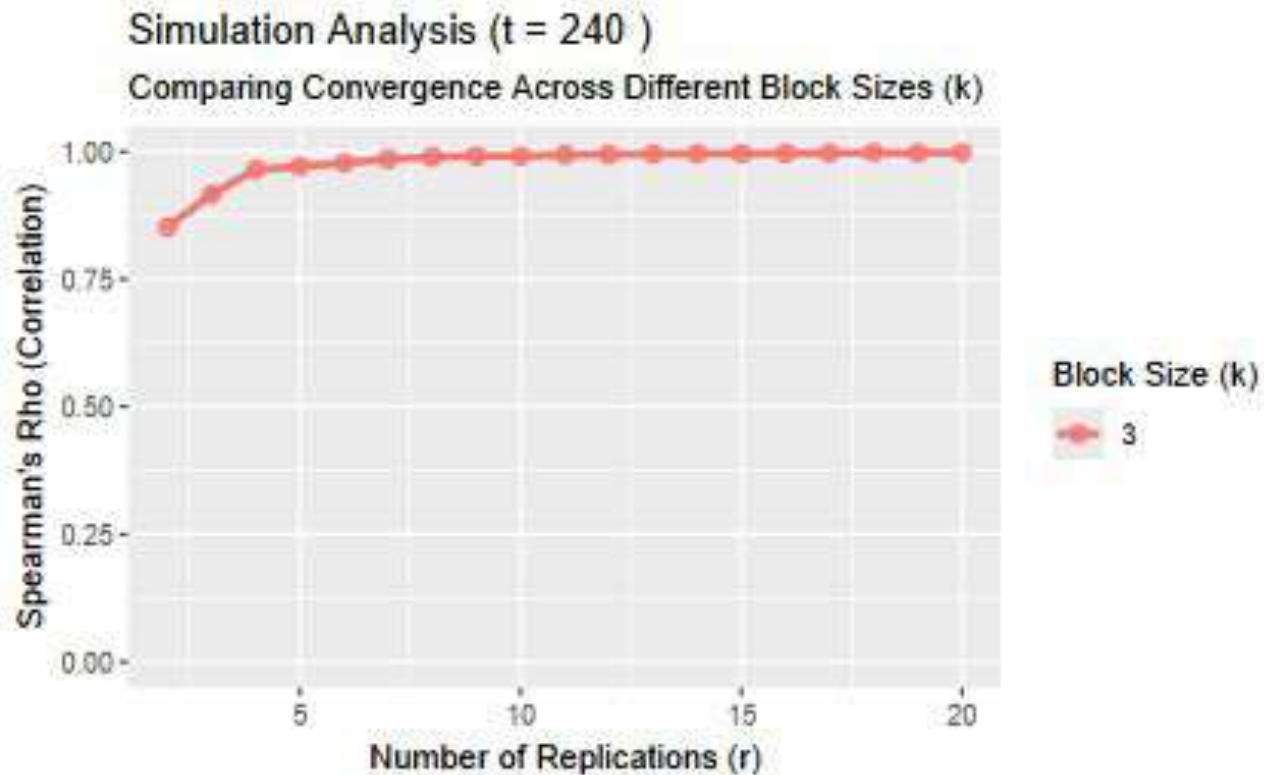
-



Link

Graph

-



2026-03-24

what is 'I' in alpha design

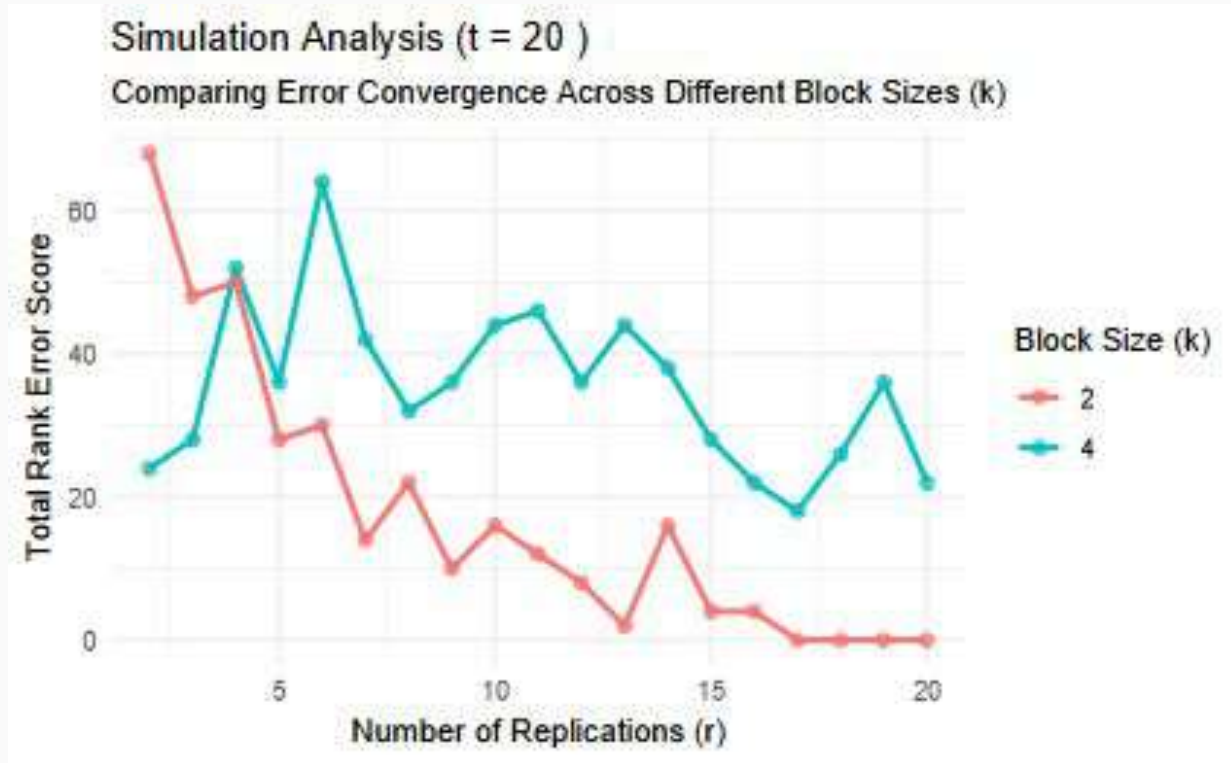
- Purpose of this parameter might can be explained by:
“field trials with large number of crop varieties cannot always be laid out in a single location or in a single season. Therefore, it is desired that variation due to location or time periods may also be controlled along with controlling variation within location or within time period.”
[Parsad, R., Gupta, V. K., Batra, P. K., Satpati, S. K., & Biswas, P. \(2007\). Monograph on \$\alpha\$ -designs. Indian Agricultural Statistics Research Institute \(ICAR\).](#)
- So **I think** we can use it as a way to mark how many groups of respondents we have, for example business travelers vs leisure travelers.
- For non agriculture situation, maybe season for time series analysis, different religions for political support analysis in Malaysia

L vs r

- r(Replication):Repeating the same question within the same group.
- L(Location):Separating the experiment into completely different environments / batches / kinds of people / time.

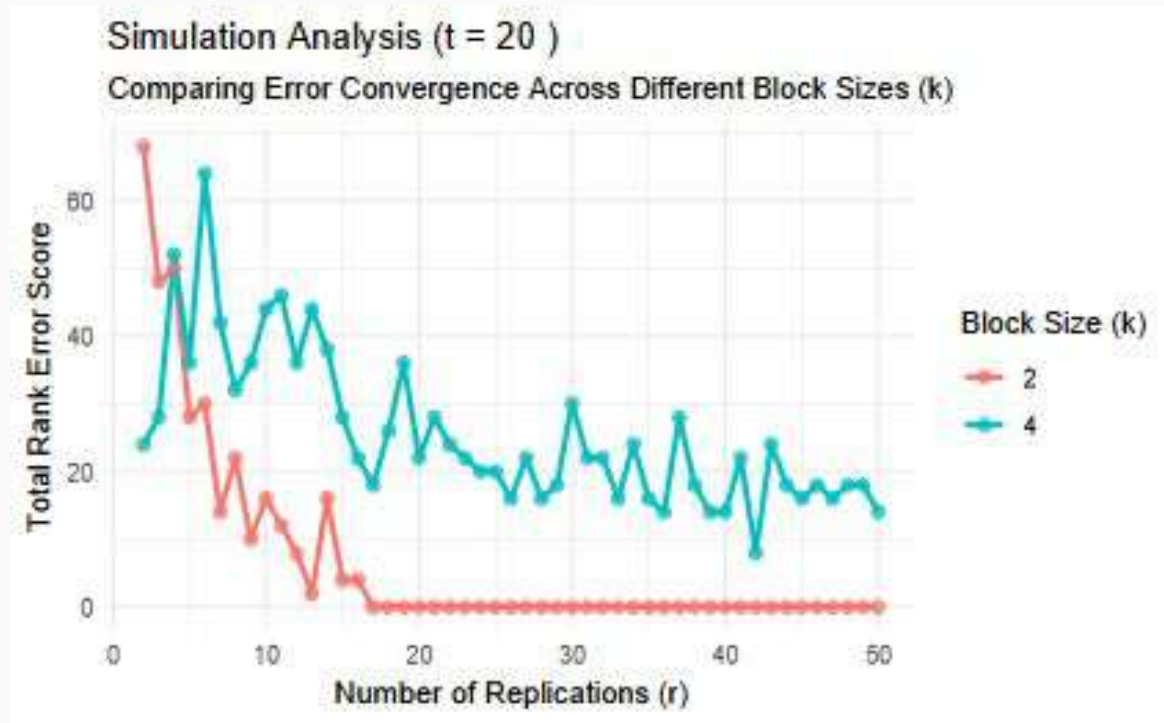
result

Error when $k = 2$ & 4



result

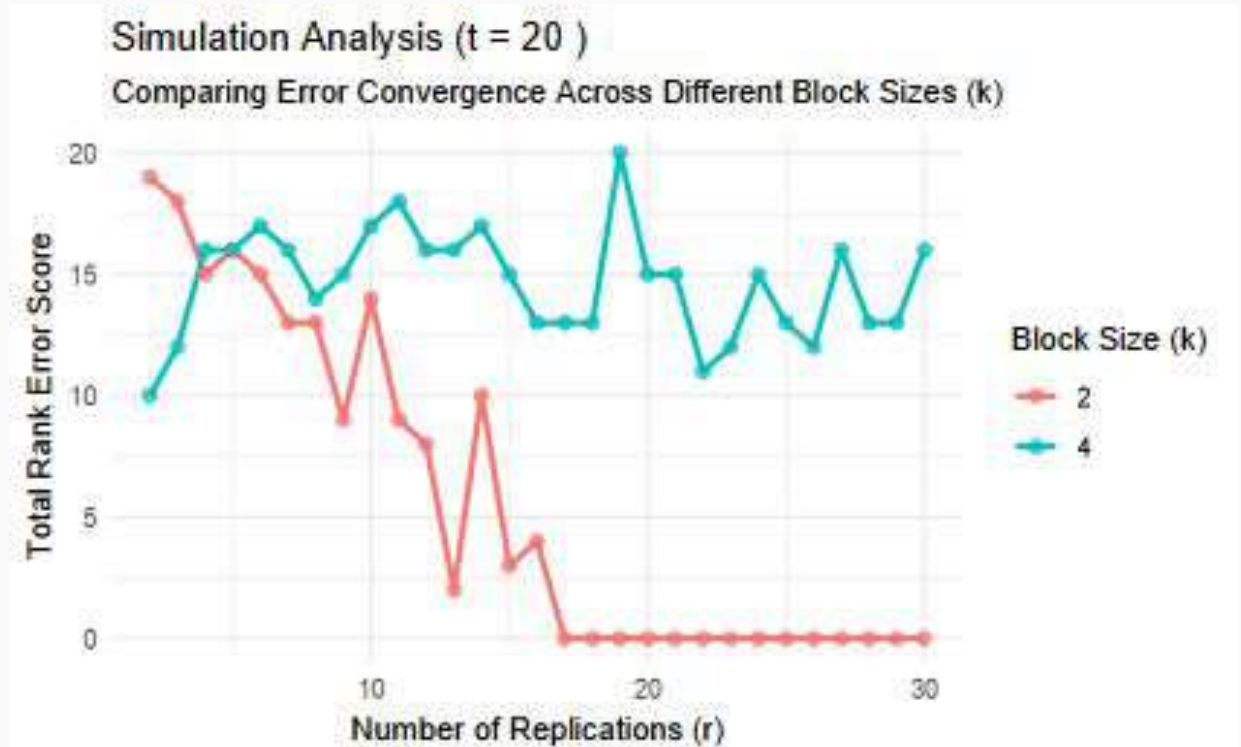
Error when $k = 2$ & 4



result

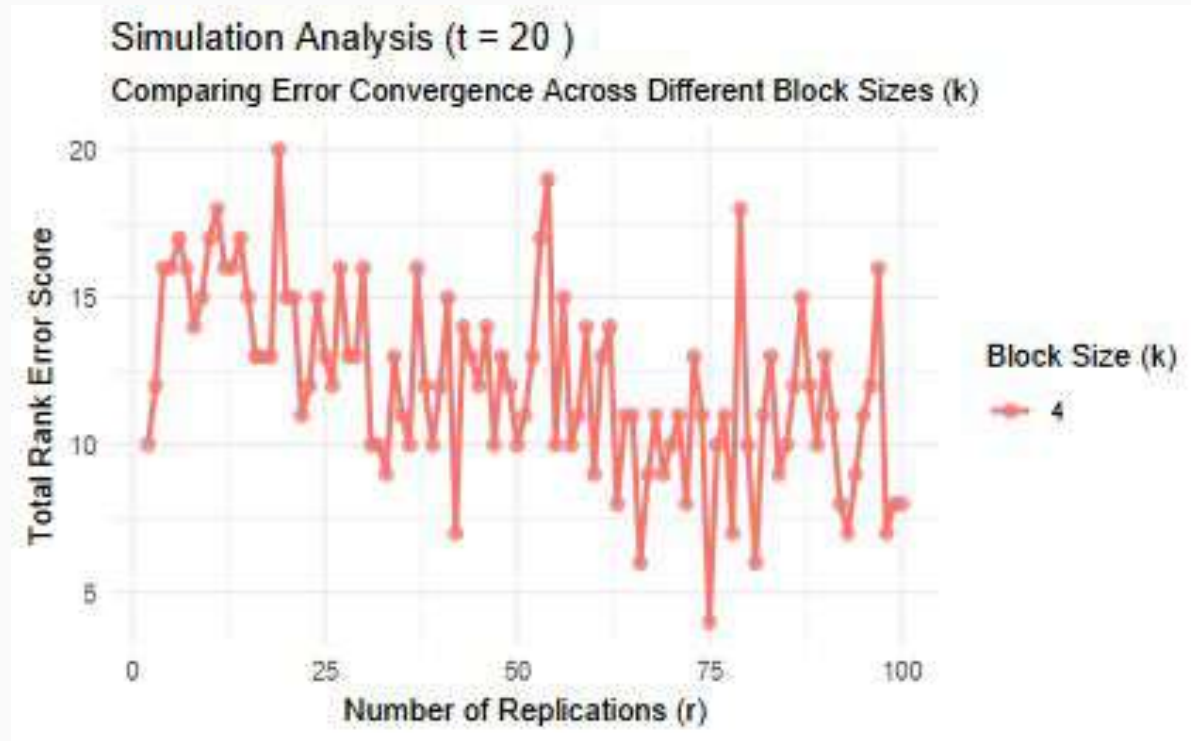
Error when $k = 2$ & 4 , $r = 30$

*show how
many treatment
not in right
place



result

Error when $k = 4$, $r = 100$



result

The code

[Link](#)

2026-03-17

what is 'l' in alpha design

- The [document](#), shows that l = location

But it didn't show why we need location for this function.

From the description, “FieldHub is an R Shiny design of experiments (DOE) app that aids in the creation of traditional, unreplicated, augmented and partially replicated (p-rep) designs **applied to agriculture, plant breeding, forestry, animal and biological sciences.**”

So *maybe* that's why we have l.

** need to generalize to non agriculture situation

** for our context , what did it do for us ?

definition of 'connection'

- “A design is ***disconnected*** if you can separate the treatments into two groups, with no treatment from the first group ever appearing in the same block with a treatment from the second group. A connected design is one that is not disconnected. In a connected design you can estimate all treatment differences. You cannot estimate all treatment differences in a disconnected design; in particular, you cannot estimate differences between treatments in different groups. Connectedness is obviously a very desirable property.”

Clarification

- show alphalattice result of simple example ($g=20$, $k=4$, $m=g/k=5$, $r=2$, etc.)
- Not like I expected
- Unlike traditional lookup tables that offer limited predefined configurations, FieldHub generates alpha-lattice designs algorithmically.

```
> simple_alpha$fieldBook
```

	ID	LOCATION	PLOT	REP	IBLOCK	UNIT	ENTRY	TREATMENT
1	1	SITE1	101	1	1	1	10	G-10
2	2	SITE1	102	1	1	2	4	G-4
3	3	SITE1	103	1	1	3	1	G-1
4	4	SITE1	104	1	1	4	5	G-5
5	5	SITE1	105	1	2	1	3	G-3
6	6	SITE1	106	1	2	2	16	G-16
7	7	SITE1	107	1	2	3	12	G-12
8	8	SITE1	108	1	2	4	9	G-9
9	9	SITE1	109	1	3	1	17	G-17
10	10	SITE1	110	1	3	2	2	G-2
11	11	SITE1	111	1	3	3	18	G-18
12	12	SITE1	112	1	3	4	13	G-13
13	13	SITE1	113	1	4	1	6	G-6
14	14	SITE1	114	1	4	2	8	G-8
15	15	SITE1	115	1	4	3	15	G-15
16	16	SITE1	116	1	4	4	20	G-20
17	17	SITE1	117	1	5	1	7	G-7
18	18	SITE1	118	1	5	2	19	G-19
19	19	SITE1	119	1	5	3	11	G-11
20	20	SITE1	120	1	5	4	14	G-14

simulation

File

```
> head(final_results_df)
  Match_ID REP IBLOCK Winner Loser
1         1   1      1     15    18
2         2   1      2      6    16
3         3   1      3     12    20
4         4   1      4      4    17
5         5   1      5      2    11
6         6   1      6      3    19
>
```

Algorithm and test

Algorithm

Test

```
print(sort(bt_abilities[, "ability"], decreasing = TRUE))
```

G-1	G-6	G-5	G-4	G-15	G-16	G-7	G-13	G-3	G-9	G-2	G-12
24.814426	22.057256	16.542968	2.757169	0.000000	-2.757170	-5.514341	-8.271458	-11.028681	-13.785798	-16.542968	-19.300138
G-8	G-18	G-17	G-19	G-11	G-14	G-10	G-20				
22.057256	-24.814426	-27.571596	-35.843107	-41.357394	-44.114564	-46.871682	-49.628852				

2026-03-10

Our challenge

- 200 data pre-collected, and we want to rank it
- Find out the preference toward the “solution” they gave, after we do the grouping (using method like alpha design/ RBIBD), we use pairwise comparison or BWS to get a brief knowledge of which is better than which
- After collecting the result we can use BT to calculate their “hidden power value”, finally rank them using it.

BT review

- BT:
 - BT model calculates **hidden power values** p_i for ranking items. Where Prob. i win j is calculated by $P(i>j) = p_i/(p_i+p_j)$
 - Model parameters are estimated using Maximum Likelihood Estimation (**MLE**) on pairwise data.
 - It works efficiently with incomplete comparisons, provided the comparison graph is **connected**.
- While the Elo system is a model designed for *dynamic, updating* based on the order of matches, the BT model treats all comparisons as a single dataset, making it *independent of time sequence*.

The grouping system(experiment design)

- We have a huge amount of data and want to make sure there are some connection between every data. Comparing every single data (with each other) will be too resource consuming.
- Thus we need a experiment design to make sure all the data somehow have connection with each other.

RIBD and Alpha design

- BIBD vs RIBD

BIBD emphasize on equal λ (times 2 treatments are paired w/e.o.)

RIBD emphasize on the completeness of data in one replication(r)

- We can use RBIBD to get a result with strong statistical relationship between the treatments (g) by dividing all the data into b blocks with size k. But again, RBIBD requires all the pair have the same λ , that is every pair of treatments must appear together in the same block exactly the same number of times.
- For RBIBD, It requires both properties of BIBD and RIBD
- So, we choose RIBD. But there is a problem with it.
- It didn't regulate how we arrange the group, we might meet a situation that a pair meet too many time with each other when we randomized it.

RIBD and Alpha design

- So, we can choose alpha design, a method that use a generating array to decide how we divide the treatments into groups
- It requires that the number of treatments be a multiple of the block size
 $g = mk$
- For example, we have $g = 20$, we want $k = 4$, $r = 2$, so total comparisons is 40, when we have $k = 4$, then we will have $b = 10$ and m is defined as the number of blocks within each replication. In this case, we have $m = g/k = 5$.
- Next, we go and find generating array for $m = 5$, $r = 2$ and $k = 4$

RIBD and Alpha design

- Generating array for $m = 5$, $r = 2$ (columns) and $k = 4$ (rows)

1	1
1	2
1	3
1	4

RIBD and Alpha design

- Take each column of the generating array and does cyclic substitution with 1,2,...m, from 1st column of the generating array, we have

1	2	3	4	5
1	2	3	4	5
1	2	3	4	5
1	2	3	4	5

- Now, 2nd column of generating array

1	2	3	4	5
2	3	4	5	1
3	4	5	1	2
4	5	1	2	3

(GPS: this results in a *shifting*, which will finally create good connections)

RIBD and Alpha design

- Finally, we take the intermediate array and add $m=5$ to the second row (and multiples of m in following rows)

1	2	3	4	5
1 + 5	2 + 5	3 + 5	4 + 5	5 + 5
1 + 10	2 + 10	3 + 10	4 + 10	5 + 10
1 + 15	2 + 15	3 + 15	4 + 15	5 + 15

- Now, 2nd column of generating array

1	2	3	4	5
2 + 5	3 + 5	4 + 5	5 + 5	1 + 5
3 + 10	4 + 10	5 + 10	1 + 10	2 + 10
4 + 15	5 + 15	1 + 15	2 + 15	3 + 15

RIBD and Alpha design

- So we arrive at:

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20

- And for the 2nd column of generating array we arrive at:

1	2	3	4	5
7	8	9	10	6
13	14	15	11	12
19	20	16	17	18

Show package results of previous
simple example

Example

```
alphalattice1 <- alpha_lattice(  
  t = 180,  
  k = 3,  
  r = 3,  
  l = 1,  
  plotNumber = 101,  
  locationNames = "group",  
  seed = 100  
)  
alphalattice1$infoDesign  
head(alphalattice1$fieldBook, 10)
```

```
> head(alphalattice1$fieldBook, 10)
```

	ID	LOCATION	PLOT	REP	IBLOCK	UNIT	ENTRY	TREATMENT
1	1	GROUP	101	1	1	1	50	G-50
2	2	GROUP	102	1	1	2	172	G-172
3	3	GROUP	103	1	1	3	178	G-178
4	4	GROUP	104	1	2	1	95	G-95
5	5	GROUP	105	1	2	2	164	G-164
6	6	GROUP	106	1	2	3	94	G-94
7	7	GROUP	107	1	3	1	8	G-8
8	8	GROUP	108	1	3	2	115	G-115
9	9	GROUP	109	1	3	3	90	G-90
10	10	GROUP	110	1	4	1	136	G-136

2026-02-23

Example of PBIBD

- Comparing treatment s - z, $g=8$ (no. of treatments), $k=4$ (block size), $r=3$ (no of times one treatment appears), $b = \text{no of blocks}$, $k < g$, $bk=gr$
- Why we want $bk = gr$?
- $bk = gr = \text{Total Number of Experimental Units}$
- cannot design an experiment with 3 blocks of size 4 if we have 5 treatments that we want to repeat 3 times

RBIBD (Resolvable-BIBD)

- Round Robin Tournament
- $g = 8, k = 2, r = 7, \lambda = 1$

Round (Set)	Block 1	Block 2	Block 3	Block 4
1	(1, 8)	(2, 7)	(3, 6)	(4, 5)
2	(2, 8)	(3, 1)	(4, 7)	(5, 6)
3	(3, 8)	(4, 2)	(5, 1)	(6, 7)
4	(4, 8)	(5, 3)	(6, 2)	(7, 1)
5	(5, 8)	(6, 4)	(7, 3)	(1, 2)
6	(6, 8)	(7, 5)	(1, 4)	(2, 3)
7	(7, 8)	(1, 6)	(2, 5)	(3, 4)

BIBD

- $g = 5, k = 2, r = 4, \lambda = 1$

Round 1	(1, 2)	(3, 4)	Player 5
Round 2	(1, 3)	(4, 5)	Player 2
Round 3	(1, 4)	(2, 5)	Player 3
Round 4	(1, 5)	(2, 3)	Player 4
Round 5	(2, 4)	(3, 5)	Player 1

Types of Group Divisible PBIBD

- Singular Group Divisible: Treatments in the same group are inseparable. If one treatment appears in a block, all its group-mates must also appear in that same block.
- Semi-Regular Group Divisible : The design enforces strict fairness. Every block must contain an exactly equal number of treatments from each group.
- Regular Group Divisible: The general case of group divisible designs. Treatments are grouped, but the design does not impose the strict "inseparable" or "equal distribution" rules found in S-type or SR-type.

Types of Group Divisible PBIBD

- 8 Treatment, 4 groups:
 - Group 1: {1, 2}, Group 2: {3, 4}, Group 3: {5, 6}, Group 4: {7, 8}
- ~~Singular~~
 - must take the entire group or none of it
 - Valid Block: [1, 2, 3, 4]
 - Invalid Block: [1, 3, 5, 7]
- ~~Semi-Regular~~
 - must take the exact same number of treatments from every group.
 - Valid Block: [1, 3, 5, 7]
 - Invalid Block: [1, 2, 3, 4]

Types of Group Divisible PBIBD

- 8 Treatment, 4 groups:
 - Group 1: {1, 2}, Group 2: {3, 4}, Group 3: {5, 6}, Group 4: {7, 8}
- Regular
 - allows for uneven contributions from groups, unlike the strict rules of the previous
 - Valid Block: [1, 2, 3, 5]

Types of Group Divisible PBIBD

- Suitable for us ?
- Singular : No, need the whole group when comparing, overemphasizing within-group comparisons
- Semi-Regular: Too Strict. Hard to design.
- Regular : better

Other Types of PBIBD

- Triangular : Relationships are defined by conceptually arranging treatments in a triangular or matrix pattern (specifically, the positions above the diagonal of a square matrix).
- Latin Square type : Relationships are determined by arranging treatments in a square grid. Two treatments are related if they share the same row or column (similar to a Sudoku structure).
- Cyclic : Relationships are determined by the numerical difference between treatment labels. Two treatments are considered related (associates) if the difference between their numbers belongs to a specific, fixed set of values.

Other Types of PBIBD - Triangular

- Triangular
: Requirement:
 $g = n(n-1)/2$,
we use $g = 10$ here

*But for our 400
treatment, it didn't suit
 $n(n-1)/2$

	Col 1	Col 2	Col 3	Col 4	Col 5
Row 1	x	1	2	3	4
Row 2	1	x	5	6	7
Row 3	2	5	x	8	9
Row 4	3	6	8	x	10
Row 5	4	7	9	10	x

匯出到試算表

The Rules:

- **Associates (Friends):** Two treatments are "friends" if they lie in the **same row** or the **same column**.
- **Example for Treatment 1:**
 - It is in **Row 1** and **Column 2**.
 - **Friends (Same Row):** 2, 3, 4
 - **Friends (Same Col):** 5, 6, 7 (See the symmetric positions)
 - **Strangers:** 8, 9, 10 (They share no row or column with 1).

Other Types of PBIBD - Triangular

	Col 1	Col 2	Col 3	Col 4	Col 5
Row 1	x	1	2	3	4
Row 2	1	x	5	6	7
Row 3	2	5	x	8	9
Row 4	3	6	8	x	10
Row 5	4	7	9	10	x

Other Types of PBIBD - Latin Square

- Latin Square type : g is a perfect square, for $g = 9$

	Col 1	Col 2	Col 3
Row 1	1	2	3
Row 2	4	5	6
Row 3	7	8	9

 匯出到試算表 

The Rules:

- **Associates (Friends):** Two treatments are friends if they share the **same row** OR the **same column**.
- **Example for Treatment 1:**
 - **Row Friends:** 2, 3
 - **Column Friends:** 4, 7
 - **Strangers:** 5, 6, 8, 9 (Diagonal positions).

Other Types of PBIBD - Cyclic

- Cyclic :Arrange treatments 1 to 8 in a circle

Long path length,
weak statistical
relation between
front and the middle
treatment.

The Logic: Arrange treatments 1 to 8 in a circle (like a clock face). We define "friendship" based on **distance**.

Rule: Let's say "Friends" are those with a **difference (distance) of 1**.

The Layout: (1) – (2) – (3) – (4) – (5) – (6) – (7) – (8) – (1)...

The Rules:

- **Example for Treatment 1:**
 - **Friends (Distance 1):** 2 and 8 (8 is connected to 1 in a cycle).
 - **Strangers:** 3, 4, 5, 6, 7 (Too far away).
- **Example for Treatment 4:**
 - **Friends:** 3 and 5.
 - **Strangers:** Everyone else.

Alpha designs and RIBD -- which is feasible/better?

Experimental Design: Why Alpha Design?

- The Constraint:
 - High Item Count $N = 200$.
 - Small Block Size $k = 5$ for BWS.
- Why NOT RIBD:
 - **Infeasible:** Combinatorial explosion makes $\lambda=1$ impossible.
 - **Resource Heavy:** Requires unrealistic sample sizes.
- Why Alpha Design:
 - **Feasible:** Optimized for large N .
 - **Efficient:** Uses **Partial Balance** to reduce survey complexity.

Link

FieldHub alpha design

- [FieldHub](#)

2026-01-05

A first course in design and analysis of experiments.

Oehlert, G. W. (2010).
A first course in design and analysis of experiments.
Gary W. Oehlert.

Example of PBIBD

- Comparing treatment s - z, $g=8$ (no. of treatments), $k=4$ (block size), $r=3$ (no of times one treatment appears), $k < g$, $bk=gr$, ~~for now, $r=1$~~

Block ID	Treatments			
1	s	w	t	x
2	u	y	v	z
3	s	w	u	y
4	t	x	v	z
5	s	w	v	z
6	t	x	u	y

Example of PBIBD

Obs A: Frequent Encounters

- Look at Pair (s, w). (in red)
- They appear together in Blocks 1, 3, and 5.
- Total Co-occurrence: 3 times.

Obs B: Occasional Encounters

- Look at Pair (s, t). (in blue)
- They appear together only in Block 1.
- Total Co-occurrence: 1 time.

Block ID	Treatments			
1	s	w	t	x
2	u	y	v	z
3	s	w	u	y
4	t	x	v	z
5	s	w	v	z
6	t	x	u	y

Unlike a BIBD, not all pairs meet with the same frequency. Relationships are unequal.

We see, for example, that treatment 1 occurs three times with treatments 5 and 9, and four times with all other treatments.

The design rules for a PBIBD are fairly complicated:

1. There are g treatments, each used r times. There are b blocks of size $k < g$. Of course, $bk = gr$. No treatment occurs more than once in a block.
2. There are m associate classes. Any pair of treatments that are i th associates appears together in λ_i blocks. We usually arrange the λ_i values in decreasing order, so that first associates appear together most frequently.
3. All treatments have the same number of i th associates, namely ρ_i .
4. Let A and B be two treatments that are i th associates, and let p_{jk}^i be the number of treatments that are j th associates of A and k th associates of B. This number p_{jk}^i does not depend on the pair of i th associates chosen. In particular, $p_{jk}^i = p_{kj}^i$.

Requirements for
PBIBD

Associate classes

ρ_i i th associates

The PBIBD is partially balanced, because the variance of $\hat{\alpha}_i - \hat{\alpha}_j$ depends upon whether i, j are first, second, or m th associates. The randomization of a PBIBD is just like that for a BIBD.

Randomize
PBIBD like BIBD

Let's check the design given above and verify that it is a PBIBD. First note that $g = 12$, $k = 7$, $r = 7$, $b = 12$, and no treatment appears twice in a block. Next, there are two associate classes, with first associates appearing together four times and second associates appearing together three times. The pairs (1,5), (1,9), (2,6), (2,10), (3,7), (3,11), (4,8), (4,12), (5,9), (6,10), (7,11), and (8,12) are second associates; all other pairs are first associates. Each treatment has nine first associates and two second associates. For any pair of

Example of PBIBD

- The properties
 - **Associate Classes:** Pairs are classified into m classes. i^{th} associates appear together λ_i times.
 - (s,w) meet each other 3 times, but individually, they pair with others 1 time; the same is true for (u,y) and so on
 - **Uniformity:** All treatments have the same number of i^{th} associates n_i .
 - Same as above, but emphasis on “every one have to be the same”, as when comparing, for any treatment, it only meet 1 treatment 3 times, and others 1 time.
 - **Structural Symmetry:** No matter which treatment you look at, the "web of connections" looks identical. This ensures the statistics are valid.

Example of RBIBD (Resolvable-BIBD)

- Round Robin Tournament
- $g = 8, k = 2, r = 7, \lambda = 1$

Round (Set)	Block 1	Block 2	Block 3	Block 4
1	(1, 8)	(2, 7)	(3, 6)	(4, 5)
2	(2, 8)	(3, 1)	(4, 7)	(5, 6)
3	(3, 8)	(4, 2)	(5, 1)	(6, 7)
4	(4, 8)	(5, 3)	(6, 2)	(7, 1)
5	(5, 8)	(6, 4)	(7, 3)	(1, 2)
6	(6, 8)	(7, 5)	(1, 4)	(2, 3)
7	(7, 8)	(1, 6)	(2, 5)	(3, 4)

- r groups, with each group representing a complete set of treatments.
- all treatments appear exactly once.

Types of PBIBD

- Singular Group Divisible: Treatments in the same group are inseparable. If one treatment appears in a block, all its group-mates must also appear in that same block.
- Semi-Regular Group Divisible : The design enforces strict fairness. Every block must contain an exactly equal number of treatments from each group.
- Regular Group Divisible: The general case of group divisible designs. Treatments are grouped, but the design does not impose the strict "inseparable" or "equal distribution" rules found in S-type or SR-type.

Types of PBIBD

- 8 Treatment, 4 groups:
 - Group 1: {1, 2}, Group 2: {3, 4}, Group 3: {5, 6}, Group 4: {7, 8}
- Singular
 - Valid Block: [1, 2, 3, 4]
 - Invalid Block: [1, 3, 5, 7]
 - must take the entire group or none of it
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Types of PBIBD

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 $g = n(n-1)/2$,
we use $g = 10$ here

*But for our 400
treatment, it didn't suit
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匯出到試算表

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Types of PBIBD

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 匯出到試算表 

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Long path length,
weak statistical
relation between
front and the middle
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- **Example for Treatment 4:**
 - **Friends:** 3 and 5.
 - **Strangers:** Everyone else.

Example of RBIBD

- Square Lattice Design : Treatments are arranged conceptually in a perfect square grid. Blocks are formed by taking an entire Row (in the first replication) or an entire Column (in the second replication).

Rigid constraint where the total number of treatments must be a perfect square. So using this, our block size will have to be $k = 20$. It is too big!

Example of RBIBD



The Setup: The Grid Arrangement of Treatments 1 to 9:

1	2	3
4	5	6
7	8	9

Square Lattice Example

- **Logic:** "Straight Lines Only." You can only group treatments that form a straight horizontal line (Row) or a straight vertical line (Column).
- **Valid Block:** [1, 2, 3]
 - **Why?** This is **Row 1**. It is a complete straight line in the grid.
 - *Also Valid:* [1, 4, 7] (This is **Column 1**).
- **Invalid Block:** [1, 2, 4]
 - **Why?** This forms an **"L" shape** (1 and 2 are Row 1, but 4 is Row 2). You cannot turn corners; you must stick to one straight line.

Package 'agricolae'

Alpha design example

- [Rmd](#)
- Not found, BT model use with alpha design

2025-12-01

A first course in design and analysis of experiments.

Oehlert, G. W. (2010).
A first course in design and analysis of experiments.
Gary W. Oehlert.

Why Choose RIBD over BIBD?

- Operational Flexibility
 - The experiment is modular (grouped into complete replicates).
 - Can be run in distinct batches
 - Risk Management: If resources run out, entire replications can be dropped without ruining the whole experiment.
- Feasibility for Large Scale
 - For large number of treatments (e.g., $g=400$), a perfect BIBD requires an impractical number of blocks/samples
 - RIBD allows for valid statistical analysis with a manageable sample size.

Why Choose RIBD over BIBD?

- Hierarchical Error Control
 - **Blocks** control local variation; **Replicates** control global variation.
 - Allows the researcher to account for large-scale shifts (e.g., Week 1 vs. Week 2 effects) which non-resolvable BIBDs cannot easily separate.

Example: Imagine you run an experiment over two weeks.

Week 1 (Replicate 1): The weather was sunny.

Week 2 (Replicate 2): The weather was rainy.

The Benefit: Because Replicate 1 and Replicate 2 are separate complete sets, the math can easily calculate and remove the "Weather Effect" (Week 1 vs. Week 2) from the results. A non-resolvable BIBD mixes everything up, making it very hard to separate these large-scale environmental shifts.

RIBD (**Resolvable** incomplete block design)

- A resolvable design is one in which the blocks can be arranged into r groups, with each group representing a complete set of treatments.
- Example, comparing A,B,C,D
- Total no. of treatment = 4, we set block size = 2
- Rep 1
 - Block 1 : A vs B
 - Block 2 : C vs D
- Rep 2
 - Block 1 : A vs C
 - Block 2 : B vs D

RIBD (Resolvable incomplete block design)

- Rep 1
 - Block 1 : A vs B
 - Block 2 : C vs D
- Rep 2
 - Block 1 : A vs C
 - Block 2 : B vs D
- Now we have $\lambda_{AB} = \lambda_{CD} = 1$ in rep 1, but $\lambda_{AC}, \lambda_{AD}, \lambda_{BC}, \lambda_{BD} = 0$
- Similarly rep 2, we have $\lambda_{AC} = \lambda_{BD} = 1$, others = 0

RIBD (Resolvable incomplete block design)

- In each Rep, all treatments (A, B, C, D) appear exactly once.
- This forms a Complete Replicate (Resolvable).
- Contrast: An Incomplete Replicate would fail to cover all treatments (e.g., Block 1: {A, B}, Block 2: {A, D} — C is missing).

PBIBD (Partially Balanced Incomplete Block Design)

- Solving BIBD huge comparison requirement.
- 1st we need to go and find a table (like in alpha design)
- So how they design the table ?
 - The sample size is consistent $bk = gr$
 - Co-occurrence frequencies λ_i are fixed for each associates
 - Every treatment has the exact same number of partners p_i in each associates
 - The intersection of relationships p_{jk}^i is constant and symmetric.

PBIBD (Partially Balanced Incomplete Block Design)

- [ref](#)

2025-11-24

Design of Experiments

A first course in design and analysis of experiments.

Oehlert, G. W. (2010).
A first course in design and analysis of experiments.
Gary W. Oehlert.

Parameter for BIBD

Parameter	Meaning	In Our Study
v / g / t	Total number of items/treatments	~400
k	Block size, how many treatments in the block ** $k < g$	
b	Total number of blocks or , simply said , total no of respondent	
r	how many times one treatment appears	
λ (lambda)	How many times a pair appears together	

BIBD: Core Concepts

1. Incomplete Block:

Definition: A design where the **block size (k)** is **smaller than the total number of treatments (g)**.

Why:

- Used when a single block cannot physically accommodate all treatments.
- Including too many treatments causes subject fatigue or inaccurate responses.

2. Balanced:

Definition: The **number of blocks** in which a **specific pair of treatments appear together**.

Benefit: For a design to be a BIBD, the pairwise occurrence λ must be identical for every single possible pair of treatments. This symmetry ensures that all pairwise comparisons (Treatment A vs. B) have the same variance/precision.

Two Golden Rules $N = bk = rg$

- Total Count $N = bk = rg$

To ensure the Total Response Volume (answers provided by all respondents) matches the Total Data Requirement (ratings needed for all items).

- Example

We need 50 ratings r for each of the 10 products g . Total = 500 ratings.

Each respondent b only rates 5 products k .

$$bk = rg$$

We must recruit exactly 100 respondents $100 * 5 = 10 * 50$ to satisfy the demand.

Two Golden Rules $\lambda(g - 1) = r(k - 1)$ —part 1

- The Balancing $\lambda(g - 1) = r(k - 1)$

To ensure every pair of items is compared directly by the same number of respondents.

Example:

Consider a specific treatment, Product X, within the set of g total treatments.

$r(k - 1)$: Product X appears in exactly r blocks (questionnaires). In each of these blocks, there are $(k - 1)$ other experimental units available for other treatments. Consequently, Product X co-occurs with other treatments a total of $r(k - 1)$ times across the entire experiment.

Two Golden Rules $\lambda(g - 1) = r(k - 1)$ —part 2

- The Balancing $\lambda(g - 1) = r(k - 1)$

To ensure every pair of items is compared directly by the same number of respondents.

$\lambda(g - 1)$: There are $(g - 1)$ other distinct treatments in the experiment (excluding Product X). For the design to be balanced, Product X must co-occur with each of these $(g - 1)$ treatments exactly λ times. Thus, the sum of co-occurrences required to pair Product X with every rival is $\lambda(g - 1)$.

Conclusion: By equating the total co-occurrence slots to the required balanced distribution, we derive the identity $\lambda(g - 1) = r(k - 1)$.

Resolvable Incomplete Block Design

- A design where blocks can be grouped into separate sets called Replicates
- Each Replicate contains exactly one complete set of all treatments.
- For 500 treatments, Replicate 1 consists of 50 blocks that cover all 500 items once.
- Ideal for management. Replicates can be run at different times, locations, or dropped if resources fail.

Alpha Designs

- A method to construct Resolvable Incomplete Block Designs (RIBD)
- When the number of treatments g is large and is a multiple of the block size k (i.e., $g=mk$)
- Uses a "Generating Array" and "Cyclic Expansion" to mathematically ensure that treatments are mixed thoroughly across the replicates.

Alpha Designs

- Example
- Total Treatments g : 20
- Block Size k : 4 (4 treatments per block)
- Replicates r : 2 (We will create 2 full replications)
- Number of Groups m : g/k
= $20/4$
= 5

Alpha Designs

C.4 Alpha Designs

Alpha Designs are resolvable block designs for $g = mk$ treatments in $b = mr$ blocks of size k . These tables give the initial alpha arrays for $5 \leq m \leq 15$, block sizes from 4 up to the minimum of m and $100/m$, and up to four replications. These tables are adapted from Table 2 of Patterson, Williams, and Hunter (1978).

$m = 5$ $4 \leq k \leq 5$				$m = 6$ $4 \leq k \leq 6$				$m = 7$ $4 \leq k \leq 7$			
1	1	1	1	1	1	1	1	1	1	1	1
1	2	5	3	1	2	6	5	1	2	4	3
1	3	4	5	1	4	3	6	1	3	7	5
1	4	3	2	1	3	4	2	1	5	6	2
1	5	2	4	1	5	2	3	1	4	3	7
				1	6	2	4	1	6	2	4
								1	7	5	6

$m = 8$ $4 \leq k \leq 8$				$m = 9$ $4 \leq k \leq 9$				$m = 10$ $4 \leq k \leq 10$			
1	1	1	1	1	1	1	1	1	1	1	1
1	2	3	7	1	2	9	8	1	2	10	6
1	4	8	2	1	4	7	5	1	4	7	10
1	6	4	5	1	8	3	4	1	6	8	3
1	3	6	4	1	3	4	6	1	5	6	7
1	5	2	7	1	5	2	7	1	7	4	2
1	7	1	3	1	6	8	3	1	8	3	5
1	8	7	6	1	7	6	2	1	9	5	8
				1	9	5	8	1	10	9	3
								1	3	7	4

C.5 Two-Series Confounding and Fractioning Plans

617

$m = 11$ $4 \leq k \leq 9$				$m = 12$ $4 \leq k \leq 8$				$m = 13$ $4 \leq k \leq 7$			
1	1	1	1	1	1	1	1	1	1	1	1
1	2	7	8	1	2	3	4	1	2	5	11
1	5	9	2	1	8	6	2	1	4	9	12
1	10	8	6	1	10	7	5	1	10	3	2
1	3	4	7	1	5	12	9	1	13	11	7
1	6	2	4	1	12	4	11	1	9	6	13
1	7	6	11	1	11	5	8	1	7	8	9
1	4	10	5	1	6	2	7				
1	8	5	2								

$m = 14$ $4 \leq k \leq 7$				$m = 15$ $4 \leq k \leq 6$			
1	1	1	1	1	1	1	1
1	2	9	11	1	2	9	8
1	10	11	8	1	4	13	15
1	12	14	3	1	8	3	6
1	3	7	2	1	11	14	12
1	6	12	13	1	15	4	9
1	4	2	12				

Alpha Designs

Step 2: Cyclic Expansion

Let's construct the **Second Replication (Rep 2)**. We use the seed column: [1, 2, 3, 4]. **Rule:** For each subsequent block, add +1 to the numbers (cycling within the range 1 to 5).

	Block 1 (Seed)	Block 2 (+1)	Block 3 (+1)	Block 4 (+1)	Block 5 (+1)
Row 1	1	2	3	4	5
Row 2	2	3	4	5	1
Row 3	3	4	5	1	2
Row 4	4	5	1	2	3



EXPORT TO SHEETS



Alpha Designs

Step 3: Adding Offsets (Finalizing IDs)

Now, convert these numbers (1-5) into actual Treatment IDs (1-20). **Offset Rule ($m = 5$):**

- Row 1: Add 0
- Row 2: Add 5 (m)
- Row 3: Add 10 ($2m$)
- Row 4: Add 15 ($3m$). 

Let's calculate the contents for **Block 1** and **Block 2**:

Calculation for Block 1:

- Row 1: $1 + 0 = \text{Treatment 1}$
- Row 2: $2 + 5 = \text{Treatment 7}$
- Row 3: $3 + 10 = \text{Treatment 13}$
- Row 4: $4 + 15 = \text{Treatment 19} \rightarrow \text{Block 1} = \{1, 7, 13, 19\}$

Alpha Designs

Calculation for Block 2:

- Row 1: $2 + 0 = \text{Treatment } 2$
- Row 2: $3 + 5 = \text{Treatment } 8$
- Row 3: $4 + 10 = \text{Treatment } 14$
- Row 4: $5 + 15 = \text{Treatment } 20 \rightarrow \text{Block } 2 = \{2, 8, 14, 20\}$

Repeat this for all 5 blocks, and your design for Replication 2 is complete!

2025-11-17

Bradley Terry Scalable

Ella Kaye, David Firth
Bradley Terry Scalable
<https://ellakaye.github.io/BradleyTerryScalable/>

Link

PDF & RMD file

- [PDF](#)
- [RMD](#)

Best-worst scaling improves measurement of first impressions

Burton, N., Burton, M., Rigby, D. et al.
Best-worst scaling improves measurement of first impressions.
Cogn. Research 4, 36 (2019). <https://doi.org/10.1186/s41235-019-0183-2>

Best-Worst Scaling (BWS) for Measuring Facial First Impressions

- The Challenge: Likert ratings are simple but have response issues and limited reliability for subjective impressions.
- The Alternative: Best-Worst Scaling (BWS), a forced-choice method, was examined as a potential high-reliability alternative.
- Group Level: BWS scores correlated almost perfectly with Likert scores, confirming they measure the same impressions.
- Individual Level (Superiority): BWS significantly outperforms Likert ratings, showing greater:
 - Test-Retest Reliability
 - Predictive Validity

Study 1: Validity (Attractiveness)

- Core Goal: Check group-level consistency between BWS and Likert, and test their validity (predictive power).
- BWS Method: 30 items (faces), 30 trials.(use Sawtooth Software's Lighthouse Studio)
 - 5 items per subset.
 - Select "Most Attractive" and "Least Attractive."
- Criterion Task: Rank 3 faces from most attractive to least attractive.
- Results: BWS scores were a more accurate predictor of participants' rankings in the criterion task (AICc difference > 10).

Study 2: Test-Retest Reliability (Distinctiveness)

- Core Goal: Directly compare the Test-Retest Reliability of BWS and Likert.
- Design: Two participant groups (BWS vs. Likert), tested twice with an interval of approximately 3-5 days.
- BWS Method: 30 items, 30 trials, 5 items per subset.
 - Select "Most Distinctive" and "Least Distinctive."
- Consistency Analysis: Calculated correlation (Pearson's r) between Session 1 and Session 2 scores.
- Results: BWS correlation ($r = .66$) was significantly higher than Likert ($r = .53$), showing greater BWS reliability.

Study 3: Reliability Replication (Trustworthiness)

- Core Goal: Replicate Study 2's high reliability findings and validate the feasibility of Open-Source Tools (R + Qualtrics).
- Materials: 31 "ambient" faces (from the US 10k database).
- BWS Design (BIBD): 31 items, 31 trials.
 - 6 items per subset.
 - Select "Most Trustworthy" and "Least Trustworthy."
- Results: BWS reliability ($r = .76$) was again significantly higher than Likert ($r = .63$).

Summary: Why is BWS Superior?

- Group Level: BWS and Likert scores were highly correlated ($r > .86$), confirming both measure the same impressions.
- Individual Level: BWS clearly outperformed Likert:
 - **Higher Validity (Study 1):** BWS more accurately predicted individual preferences in the ranking task.
 - **Higher Reliability (Studies 2 & 3):** BWS showed higher consistency in test-retest scores, indicating more stable participant responses.
- Conclusion: BWS reduced response bias and noise from participants, making it a more reliable tool for measuring individual differences.

2025-11-02

What is the largest BT model we can have

- theoretically unlimited but is constrained primarily by computational power and data quality in practice

What is the largest BT model we can have

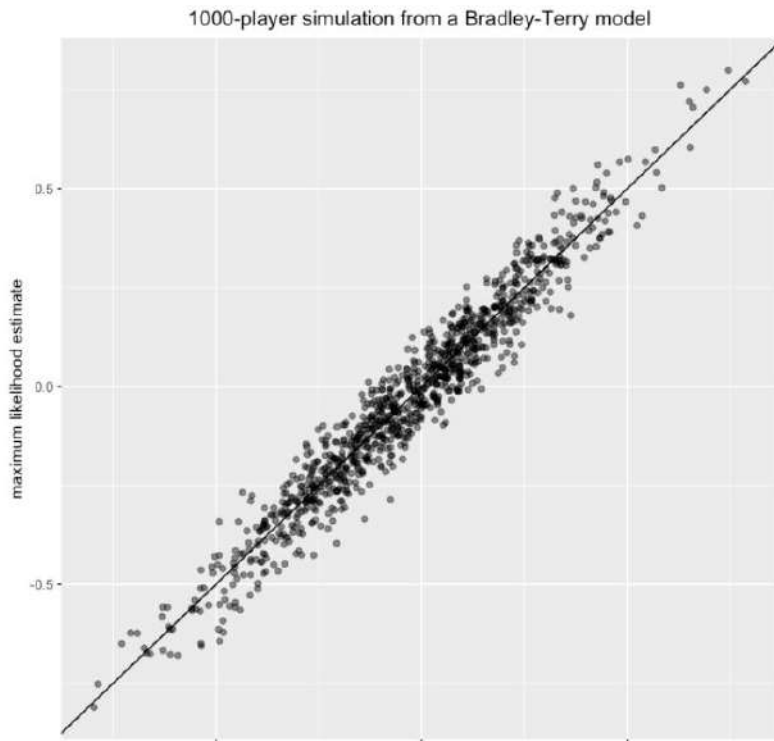
- theoretically unlimited but is constrained primarily by computational power and data quality in practice

how can we make a design with minimum number of comparisons?

- Will included in last slide, but not sure it is suitable to use as explanation or not

Huge data example

- As a bigger example, let's simulate a single instance of a fairly sparse tournament with 1000 items (or 'players') and then fit the Bradley-Terry model to the resulting data:
- [BradleyTerryScalable](#)



Best-Worst Scaling: Theory, Methods and Applications.

Louviere, J. J., Flynn, T. N., & Marley, A. A. J. (2015).
Best-Worst Scaling: Theory, Methods and Applications.
Cambridge: Cambridge University Press.

BWS

- Best-Worst Scaling (BWS), also known as "BWS," is a choice modeling technique or discrete choice experiment (DCE) invented by Jordan Louviere in 1987.
- Steps:
 - Present Items: Respondents are shown a subset of items (e.g., brands, statements, features) taken from a master list.
 - Make Choices: They are asked to choose the single "best" (e.g., most important, most appealing, most preferred) item and the single "worst" (e.g., least important, least appealing, least preferred) item from that specific subset.

BWS

- Repeat Task: This task is repeated several times, with the items in the subset varying systematically according to a statistical design.
- The purpose of BWS is to measure subjective quantities (like preferences, attitudes, or perceived importance) on a measurement scale with known properties.
- Extension of the paired comparison method.
- Forces respondents to make trade-offs, provides more information than "best-only" tasks, and avoids many of the biases associated with traditional rating scales (like Likert scales).

Core Assumptions 1

1. The BWS Task and some definition

- BWS models are statistical models that **treat choices as probabilities**, not certainties.
 - "All BWS models are probabilistic discrete choice models."
- **"Best" and "worst" are labels for two extremes** of any scale (e.g., most/least important, heaviest/lightest).
 - "Here 'best' and 'worst' simply constitute a metaphor for any appropriate terms that define the extremes of a latent, subjective continuum."

Core Assumptions 2

2.Theoretical Framework Assumptions (RUT & FUT)

- Users must be able to pick the top and bottom items from a set of 3 or more.
 - "Best-worst scaling is based on the idea that a person faces choices among collections ("sets") of three or more items or options, and can identify the best and the worst options in the collection."
- Assumes a person's underlying preference (utility) is somewhat random or unstable 【RUT (Random Utility Theory)】.
 - "Random utility theory assumes that the decision process underlying the choices is deterministic, but the utilities have a random component."
- Assumes the preference is stable, but the act of choosing is random or imperfect.
 - "Fixed utility theory"...

Core Assumptions 2

2.Theoretical Framework Assumptions (RUT & FUT) continued

- "The two theoretical perspectives can lead to choice models that are mathematically equivalent, even though they may imply potentially different economic or psychological processes (see Chapter 5)."

Core Assumptions 3

3.About Aggregating Data (related to our RESEARCH)

- The book's models are designed to **analyze one person at a time**.
- Trying to combine, or aggregate, data from many individuals to summarize a whole group is complex. This **aggregation requires making "strong assumptions" which might be incorrect and can create significant problems** for the analysis.

Preferences for tongue swab-based versus sputum-based testing in the context of TB care: a best-worst scaling exercise in Vietnam and Zambia.

Castro, M. D. M., Le, H., Muzazu, S., Pham, N., Trinh, T., Nyirenda, H. C., Shabalu, P., West, N., Phan, H., Cattamanchi, A., Denking, C. M., Muyoyeta, M., & Kerkhoff, A. D. (2025). Preferences for tongue swab-based versus sputum-based testing in the context of TB care: a best-worst scaling exercise in Vietnam and Zambia. **BMJ Global Health**, 10(10), e019092. <https://doi.org/10.1136/bmjgh-2025-019092>

Example of BWS design with BIBD

- [Castro MDM, Le H, Muzazu S, Pham N, Trinh T, Nyirenda HC, et al. Preferences for tongue swab-based versus sputum-based testing in the context of TB care: a best-worst scaling exercise in Vietnam and Zambia. BMJ Global Health. 2025;10:e019092. <https://doi.org/10.1136/bmjgh-2025-019092>](https://doi.org/10.1136/bmjgh-2025-019092)
- Used Best-Worst Scaling (BWS) to find out which of 16 features for TB testing were most important to patients and used a Balanced Incomplete Block Design (BIBD) to create the survey.
 - The **total number of features** (for other case, item / treatment / solution) being evaluated. $v = 16$
 - The **number of features shown in each block** (here, they present the block in the form of question). $k = 4$
 - The **total number of blocks** (questions) each participant completed. $b = 12$
 - The **number of replications** (how many times each feature appeared for each participant). $r = 3$
 - The paper calls this a "**near-BIBD**," meaning this pairing might not be perfectly balanced for every single pair, so no λ (I think)

Example of BWS design with BIBD

- [Supplementary File for Castro MdM, et al](#)
- Shows how to calculate minimum sample size.

$$N \geq \left(\frac{z_{\alpha/2}}{\delta} \right)^2 \frac{2.12n}{mt}$$

Where:

- N = minimum sample size required to achieve the specified margin of error for estimated preference weights
- $Z_{\alpha/2} = 1.96$ for 95% confidence
- δ = is the margin of error (MOE)
- n = 16 is the total number of features
- m = 4 is the number of features per task
- t = 12 is the number of tasks per respondent
- 2.12 = simulation-derived constant (Lipovetsky et al. 2015)

Using a margin of error $\delta = 0.143$, the minimum sample size required per country is:

$$N \geq (1.96 / 0.143)^2 * (2.12 \times 16) / (4 \times 12) \approx 133$$

Thus, a sample size of 133 participants per country is sufficient to achieve an approximate MOE of 0.143, ensuring acceptable precision in the estimation of utility scores derived from the BWS design. Figure 1 presents the estimated margin of error for a range of sample sizes using the current design.

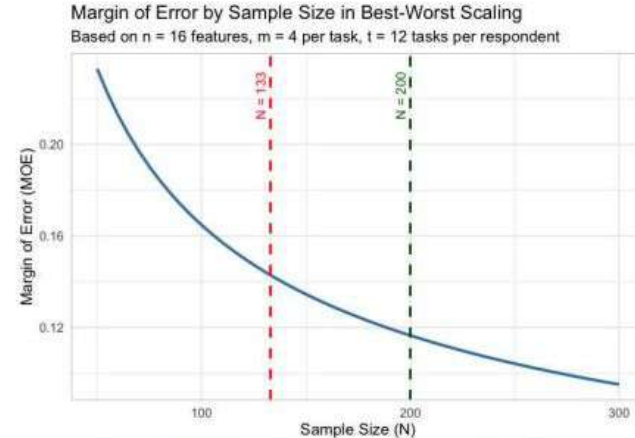


Figure 1. Margin of error by sample size estimations.

2025-10-27

RANK ANALYSIS OF INCOMPLETE BLOCK DESIGN

THE METHOD OF PAIRED COMPARISON

Bradley, R. A., & Terry, M. E. (1952). Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons. *Biometrika*, 39(3/4), 324–345. <https://doi.org/10.2307/2334029>

Group vs block vs repetition

- Group: **a set of experiment repetitions**, which often represents a distinct source of variation like different judges or different times. The primary statistical constraint is that the model's design requires these groups to be defined such that the true treatment ratings ($\pi_{1u}, \dots, \pi_{tu}$) exist within each u^{th} group, but are not necessarily identical from one group to the next. (**no. of judge not necessary = no. of group)
- ref: In certain cases, as noted above, repetitions of the design may be performed by different judges or at different times. We shall discuss, in a subsequent section, the analysis when true ratings $\pi_{1u}, \dots, \pi_{tu}$ exist in the u^{th} of y groups not necessarily identical from group to group
- no. of judge not necessary = no. of group, same kind of judges can form into 1 group
- ref: 7. THE COMBINATION OF EXPERIMENTS, where we discuss experiment is performed in groups of repetition of sizes n

Group vs block vs repetition

- Block: is the **fundamental unit of comparison** where two treatments, i and j , are presented together.

ref: Further definition follows with the assumption that, when treatment i appears with treatment j in a block, the probability that treatment i obtains top rating (or a rank of 1) is $\pi_i / (\pi_i + \pi_j)$.

ref: One of the questions asked in connexion with this work pertains to the possibility of extending the analysis to incomplete block designs with larger block sizes. We are proceeding with a consideration of such extensions. The method of paired comparisons becomes inefficient where it is possible to rank more than two treatments at a time and where more than a few items or treatments are considered.

Group vs block vs repetition

- Repetition : A single complete running of the entire comparison design. This means enough comparisons are made to ensure all treatments have been compared and are connected within that data set.
- Ref : and n will be reserved to denote the number of repetitions of the design when a repetition is defined to be a set of all pairs of treatments

Can we only have 1 repetition?

- Yes
- Ref : If three items are compared in a single repetition, the possible sets of rank sums are 2, 3, 4 and 3, 3, 3.
- But, the approximation appears to be fairly good for practical purposes if the number of repetitions is not too small (say $n > 15$).

How many judges can we use?

- No theoretical upper limit, but it is tied to the total number of repetitions n and number of groups g ($n \leq \text{no. of judge}$, $g \leq \text{no. of judge}$)
- $n \geq \text{no. of judge}$, as we want to have all the data from a judge for every repetition
- $g \leq \text{no. of judge}$, as if g were greater than the number of judges, the model would fail because some analytical groups would lack any corresponding data source.

Example from paper

Table 4. *Rankings for two judges in the pork experiment*

Repetition ...	1	2	3	4	5
	C Cp CP	C Cp CP	C Cp CP	C Cp CP	C Cp CP
Pair	Judge 1				
C, Cp C, CP Cp, CP	2 1 — 2 — 1 — 2 1	2 1 — 1 — 2 — 1 2	2 1 — 2 — 1 — 2 1	2 1 — 2 — 1 — 1 2	2 1 — 2 — 1 — 2 1
	Judge 2				
C, Cp C, CP Cp, CP	2 1 — 1 — 2 — 1 2	2 1 — 1 — 2 — 1 2	1 2 — 1 — 2 — 2 1	1 2 — 1 — 2 — 2 1	1 2 — 2 — 1 — 1 2

Example from paper

1. Block

Example: When a judge compares the Corn ration (C) pork against the Corn plus Peanut supplement (Cp) pork, that single comparison is a block.

2. Repetition (重複)

Example: Completing the comparison of C vs Cp, C vs CP, and Cp vs CP forms one repetition.

In the paper, each judge performed five repetitions ($n=5$).

3. Group (組別)

Example: Judge 1 is Group 1, and Judge 2 is Group 2. They are separate groups because their true ratings may be different.

2025-10-21

RANK ANALYSIS OF INCOMPLETE BLOCK DESIGN

THE METHOD OF PAIRED COMPARISON

Bradley, R. A., & Terry, M. E. (1952). Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons. *Biometrika*, 39(3/4), 324–345. <https://doi.org/10.2307/2334029>

Why develop BT

- Overcame distributional and structural limitations
- Earlier Method:
 1. assumptions of normality and equal variances; estimation is mathematically complex
 2. depends on sensation
 3. No explicit probabilistic model; cannot estimate underlying “true scores.”

The setting

1. “ t ” treatments (item, or suggestion in our case)
2. Treatments have true ratings $\pi_1, \dots, \pi_t$
3. Every $\pi_i \geq 0$ and that $\sum \pi_i = 1$ (for convenience)
4. When comparing i to j , probability that treatment i obtain top rating (win) is $\pi_i / (\pi_i + \pi_j)$
5. r_{ijk} is rank of the i^{th} treatment in the k^{th} repetition of the block in which treatment i vs treatment j , $r_{ijk} = 3 - r_{jik}$ (rank for win = 1, lose = 2)
6. $p_1, \dots, p_t$ are estimates of $\pi_1, \dots, \pi_t$
7. n is number of repetitions of the design when a repetition is defined to be a set of all pairs of treatments

*why the title said incomplete block design but they said “a set of all pairs of treatments” in definition of n ?

Each block only observes a subset of treatments. However, when all blocks are aggregated, the overall data set may still represent one or more complete repetitions of all possible treatment pairs. The parameter n thus denotes how many full sets of pairwise comparisons are represented in total.

The setting 2

- true ratings $\pi_{1u}, \dots, \pi_{\square u}$ exist in the u^{th} of y groups not necessarily identical from group to group

THE LIKELIHOOD FUNCTION - L

- Assuming probability independence between blocks or pairs of treatments
- When we are comparing i and j, the probability for i or j win will be

$$\left(\frac{\pi_i}{\pi_i + \pi_j} \right)^{2-r_{ijk}} \left(\frac{\pi_j}{\pi_i + \pi_j} \right)^{2-r_{jik}}$$

- Note: r_{ijk} is rank of the i^{th} treatment in the k^{th} repetition of the block in which treatment i vs treatment j, $r_{ijk} = 3 - r_{jik}$ (rank for win = 1, lose = 2)
- So, when i win, it becomes $\pi_i / (\pi_i + \pi_j)$, if j win it becomes $\pi_j / (\pi_i + \pi_j)$
- Repeat for all treatment and multiply them together, we get the likelihood function

$$L = \prod_i \pi_i^{2n(t-1) - \sum_{j \neq i} \sum_k r_{ijk}} \prod_{i < j} (\pi_i + \pi_j)^{-n}.$$

LIKELIHOOD RATIO TESTS AND ESTIMATION

- Their null hypothesis (H_0): $\pi_i = 1/t$ ($i = 1, \dots, t$)
- Alternative hypothesis (H_α): $\pi_i = \pi(h)$ ($h = 1, \dots, m$);
 $i = s_{h-1} + 1, \dots, s_h$ where
 $s_0 = 0, s_m = t$ and
 $\sum (s_h - s_{h-1})\pi(h) = 1$
- where treatments have identical ratings within each of m groups of treatments while the groups themselves may differ
- But our case is a special case with
 H_1 : no π_i is assumed equal to any π_j ($i \neq j$).
Meaning that $m = t$ in H_α

LIKELIHOOD RATIO TESTS AND ESTIMATION 2

- If $p(h)$ is the maximum-likelihood estimate of $\pi(h)$, we have

$$\frac{a_i}{p_i} - n \sum_{j \neq i} (p_i + p_j)^{-1} = 0 \quad (i, j = 1, \dots, t),$$

and

$$\sum_i p_i = 1.$$

We define

$$a_i = 2n(t-1) - \sum_{j \neq i} \sum_{k=1}^n r_{ijk}.$$

- The test statistic become

$$B_1 = n \sum_{i < j} \log (p_i + p_j) - \sum_i \left\{ 2n(t-1) - \sum_{j \neq i} \sum_{k=1}^n r_{ijk} \right\} \log p_i.$$

Table 4. *Rankings for two judges in the pork experiment*

Repetition ...	1	2	3	4	5
	C Cp CP	C Cp CP	C Cp CP	C Cp CP	C Cp CP
Pair	Judge 1				
C, Cp C, CP Cp, CP	2 1 — 2 — 1 — 2 1	2 1 — 1 — 2 — 1 2	2 1 — 2 — 1 — 2 1	2 1 — 2 — 1 — 1 2	2 1 — 2 — 1 — 2 1
	Judge 2				
C, Cp C, CP Cp, CP	2 1 — 1 — 2 — 1 2	2 1 — 1 — 2 — 1 2	1 2 — 1 — 2 — 2 1	1 2 — 1 — 2 — 2 1	1 2 — 2 — 1 — 1 2

Things to decide

- Pooled analysis
- Combined analysis
- Affect how B_1 is calculated

To Be Done

- Pooled analysis / Combined analysis
- LARGE SAMPLE DISTRIBUTION
- [VS my report last time](#)

2025-10-13

ELO VS BT - Conceptual Part

	ELO	BT
purpose	Shows the real-time skill change using dynamic score	Estimate the static latent ability from all pairwise outcomes statistically
Nature of model	A system that learns step by step, updating scores after every match based on how players perform.	A probabilistic model based on maximum likelihood estimation (MLE).
Data requirement	Works incrementally, the sequence of the match results matters.	Requires the full set of pairwise comparisons or a connected network , no problem from sequence.

ELO VS BT - Conceptual Part

	ELO	BT
Assumption	Win probability depends on rating difference.	Win probability depends on the logit difference of latent ability.
Academic Difference	A heuristic updating rule (rule of thumb), designed for practical use rather than for formal statistical inference .	Grounded in probability theory and estimated through maximum likelihood estimation (MLE), making its parameters interpretable and academically defensible .

ELO VS BT Math Part

In the Bradley–Terry model, the derivative of the log-likelihood with respect to each player's ability β_i turns out to be $S_{ij} - P_{ij}$, where S_{ij} is the actual result (1 if player i wins, 0 otherwise) and P_{ij} is the predicted win probability. This means the model updates each player's score in the direction of the difference between actual and expected outcomes — if someone wins more than expected, their score increases, and if they lose more than expected, it decreases. This update rule is exactly what the Elo system uses, just applied one match at a time.

S_{ij} : i = player we are interested in; j = comparative player;
S = score of comparison (player i wins vs. player j: 1, 0 otherwise)

P_{ij} : probability (between 0.00 and 1.00) of player i winning over player j

What is a BIBD (Balanced Incomplete Block Design) for BWS

A scientific blueprint for survey design.

We often need to rank numerous items.

But, asking respondents to rank all items at once imposes a significant cognitive burden and results in poor data quality. The BWS solution is to present respondents with only a small subset of items at a time (a "block") and ask them to choose the best and worst from that subset.

What is a BIBD

BIBD's role is to provide a systematic and unbiased way to construct these small subsets. Its ultimate goal is to ensure the "fairness" and "balance" of the overall survey design. This prevents any item from gaining an unfair advantage or disadvantage due to poor design, thereby ensuring the final preference ranking is reliable and valid.

What is a BIBD

A method rooted in the field of Experimental Design, used to systematically structure comparison tasks to collect the most accurate data with the highest efficiency.

Core Assumptions

1. Choice Reflects True Preference: What they choose is what they mean. The choice is not a random act.
2. Independence of Choices : Respondent's choice in each block is made independently.
3. No Context Effect: The appeal of an item is independent and does not fundamentally change based on the other "competing" items it is presented with.

Parameter	Term	Meaning	Explanation in the Context of a BWS Survey
v	varieties / points	Total number of items	The total number of things we want to rank (e.g., 10 items).
k	block size	Block size	The number of options presented to the respondent in each single question (e.g., 4 items are shown per task).
b	blocks	Total number of blocks	The total number of unique questions (or sets) in the entire survey design.
r	replication number	Replication number	The number of times each individual item appears across the entire survey. In a BIBD, this number is identical for all items.
λ (lambda)	Lambda	Pairwise co-occurrence	The number of times any given pair of distinct items appears together in the same question. In a BIBD, this number is also identical for all possible pairs.

Summary

1. A perfect BIBD provides the mathematically rigorous blueprint for our study.
2. It allows us to take our
3. v total items (varieties) and systematically organize them into b unique questions, or blocks.
4. Crucially, each of these blocks contains a smaller, manageable subset of k items for the respondent to evaluate.
5. The true power of this design is that by following this structure, we can guarantee two levels of perfect fairness: every single item appears exactly r times throughout the entire survey (replications), and every possible pair of items appears together in a question exactly λ times (Lambda).
6. It is this complete balance, defined by all five parameters, that gives us confidence that our final results will reflect true preferences, free from design-induced bias.

How does Elo get around connectedness problem that BT and BWS are worried about?

The problem still exist, If there is no connection at all, Elo cannot infer global rankings.

But as long as there is any minimal connection between groups, Elo will still work—it will converge correctly, only more slowly as connectivity weakens.

Olesker-Taylor, Sam & Zanetti, Luca. (2024). An Analysis of Elo Rating Systems via Markov Chains. 10.48550/arXiv.2406.05869.

Others major pairwise comparison methods

1. Analytic Hierarchy Process (AHP) : likert scale + weighted result
2. PROMETHEE (Preference Ranking Organization METHod for Enrichment Evaluations) : Compares alternatives pairwise under each criterion, converts performance differences into preference intensities, and aggregates them into an overall outranking flow to rank the options.

2025-10-07

ELO: score & rank

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}.$$

$$E_B = \frac{1}{1 + 10^{(R_A - R_B)/400}}.$$

$$R'_A = R_A + K(S_A - E_A).$$

CSDN @学软件的小垃圾

E_A prob of a win

R_A current ELO score of player A

R'_A ELO score after the match

K-factor one of the factor that determine how violently the score change, usually high if the player have high **uncertainty** (new players, old player who didn't play for a while)

- We can use K factor= 40 to make it converge quicker
- S_A : game score (lose=0, tie=0.5, win=1)

Starting Match:

- Everyone starts with baseline 1500 in chess, but actually it doesn't matters

What is the min “N” comparison needed?

- There is no certain “N”, but we can do it until the score become fixed.

About video on ELO for smartphone comparisons

No math details, but similar to our situation, the testing subject (picture showed) didn't change at all.

They give all the participant exact same picture to compare with and the “experiment” only consist one round.

It means they didn't “give every testing subject a second chance”, similar to us as the picture used in the test is same and never change, thus no uncertainty from the picture showed to participant.

Why 400 ?

Elo suggested a scaling constant such that **200 points difference in favor of player A corresponds to a 0.75 expected score for A.**

- 200 points corresponds to chess levels (masters?)

400 is an empirical scaling factor in the Elo formula, ensures the formula matches observed win probabilities (e.g., 200-point difference –

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}.$$

$$E_B = \frac{1}{1 + 10^{(R_A - R_B)/400}}.$$

Pros and cons

Pros:

- Simple and Effective
- Ability to predict the expected outcome of a match based on the rating difference between two players

Cons:

- Cannot accurately assess individual contributions in *team-based games*
- *High-ranked players are incentivized not to play against lower-ranked opponents* to avoid the risk of losing a large number of rating points from a single upset

Example

- Ranks scientific journals by adapting the Elo rating system
- Lehmann, R., & Wohlrabe, K. (2017). Who is the 'Journal Grand Master'? A new ranking based on the Elo rating system. *Journal of Informetrics*, 11, 800–809. <https://doi.org/10.1016/j.joi.2017.05.004>

Pros and cons - BT

Pros

1. Interpretability.

The model is $P(i > j) = \text{logit}^{-1}(\beta_i - \beta_j)$

So β is a latent “ability.”

A one-unit gap $(\beta_i - \beta_j = 1)$

means i beats j with $\approx 73\%$ probability—making rankings and upset probabilities easy to explain.

2. Works with incomplete/uneven comparisons.

Don't have to compare every pair of items; MLE uses whatever pairwise data we have as long as the comparison graph is (strongly) connected, so it's practical for real-world

Pros and cons - BT

cons

1. **Strong assumptions**

The BT model assumes each outcome is independent and context-free, but real data often show context effects

2. **Estimation pathologies with separation/sparsity**

If an item never loses (or never wins) or the comparison graph isn't well connected, the MLE may not exist or be unstable.

Typical fixes include adding regularization/Bayesian priors, constraints, or explicit tie/home-field extensions. (thx gpt)

BWS - how if the item is never the best or worst?

As long as the BWS design is **well connected**, every item can be indirectly compared through others, so it's not a problem if some items are never chosen as best or worst.

A well-designed **Balanced Incomplete Block Design** (BIBD) ensures that:

- each item appears several times across different sets;
- each item co-occurs with other items roughly the same number of times;

therefore, even if an item is never directly chosen as the best or worst, it can still be positioned on the overall preference scale through indirect (transitive) comparisons.

Pros and cons - BWS

Pros

Produces ratio-scaled, discrimination-friendly data, allowing clear comparisons between items.

Reduces response bias found in rating scales by forcing trade-offs (best vs. worst choices).

Cons

Cognitively demanding when many items are compared or sets are large.

Measures are relative, not absolute — results depend on the specific items included.

Example - BWS

- Applied it to a small set of items by measuring consumer preferences for six different weekend getaway destinations.
- ranking the perceived quality of 72 academic marketing journals.
- Louviere, J., Lings, I., Islam, T., Gudergan, S., & Flynn, T. (2013). An introduction to the application of (case 1) best-worst scaling in marketing research. *International Journal of Research in Marketing*, 30(3), 292–303.
<https://doi.org/10.1016/j.ijresmar.2012.10.002>

Rank Centrality

Stuck at stationary distribution...

2025-09-22

Scoring two content creation solutions

1. LLM (e.g., ChatGPT) do pairwise comparisons and provides scores
2. Human raters pairwise comparisons and provides scores
3. Human raters do pre-rating and given qualitative feedback on why they prefer some solutions over others — we can then feed these qualitative assessments into an LLM to be an automated rater
 - a. Un-prompted LLM might have different preferences than human raters (e.g., structure > content)

ELO: score & rank

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}.$$

$$E_B = \frac{1}{1 + 10^{(R_A - R_B)/400}}.$$

$$R'_A = R_A + K(S_A - E_A).$$

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- S_A : game score (lose=0, tie=0.5, win=1)

Starting Match:

- Everyone starts with baseline 1500 in chess, but actually it doesn't matters

What is the min “N” comparison needed?

- There is no certain “N”, but we can do it until the score become fixed.

Bradley-Terry model

- Calculate hidden “power value” p_i , using the existing comparison result and Maximum Likelihood estimation method
- So, how we find out each person's p ?
Step
 - 1) Recording all the comparison result
 - 2) Use MLE to estimate each player's ability parameter p_i based on the observed match outcomes, such that the likelihood of these outcomes under the model is maximized.
(We can use Python `choix` or R `BradleyTerry2`)

Bradley-Terry model

The Math part

- Prob. i win j is calculated by $P(i > j) = p_i / (p_i + p_j)$
- Let $\beta_i = \ln(p_i)$. Thus $p_i = e^{\beta_i}$
- So, $P(i > j) = e^{\beta_i} / (e^{\beta_i} + e^{\beta_j})$
- We define odds as $\text{Odds}(i > j) = P(i > j) / P(j > i)$
- Thus $\text{Odds}(i > j) = [p_i / (p_i + p_j)] / [p_j / (p_i + p_j)]$
- $\text{Odds}(i > j) = p_i / p_j$
- $\ln(\text{Odds}(i > j)) = \ln(p_i / p_j)$
- $\ln(\text{Odds}(i > j)) = \ln(p_i) - \ln(p_j)$

Now ! the hard part

- We know for a single comparison, $P(i > j) = p_i / (p_i + p_j)$
- If i beat j for w_{ij} times, and j win i for w_{ji} times, the prob. that we will get the result mentioned will be:

$$L = [p_i / (p_i + p_j)]^{w_{ij}} * [p_j / (p_i + p_j)]^{w_{ji}}$$

Where L is Likelihood Function for observing i beat j for w_{ij} times, and j win i for w_{ji} times

Bradley-Terry model

The Math part

- Expanding this mindset to other's player result, we can say:

$$L(p_1, p_2, \dots | \text{data}) = \prod (p_i / (p_i + p_{-i}))^{w_i}$$

- Using $\ln()$

$$\ln(L) = \ln \left[\prod (p_i / (p_i + p_{-i}))^{w_i} \right]$$

$$\ln(L) = \sum [w_i * \ln(p_i / (p_i + p_{-i}))]$$

$$\ln(L) = \sum [w_i * (\ln(p_i) - \ln(p_i + p_{-i}))]$$

- Finally, the not for human part

We need to max. $\ln(L)$

By finding all the partial derivatives ☺

So.... thx python and R ☺

Bradley-Terry model

What is the min “N” comparison needed?

Still cant find reliable resource for this

~~AI: 15~20, but inconsistent because it also said 10~20~~

Best Worst Scaling (BWS)

- Raters compare between **3 or more options** and choose the **best and the worst** from it and do several groups of comparisons
- Calculate how many times each option was voted as best and as worst
- Subtract the number of worst votes from the number of best votes for each option, and then rank the options by that result
- **Thurstone – only compare 2 options, rank by win rate

BWS

- What is the min “N” comparison needed?

No absolute number that is needed for this

But need to follow:

a) **Item Frequency Balance**: Each item is presented to respondents the same number of times, or as close as possible, across the entire survey design (summed across all versions).

b) **Pairwise Balance**: Any two items (e.g., "Price" and "Quality") appear together in the same set the same number of times, or as close as possible, as any other pair of items (e.g., "Price" and "Ambiance").

BWS

- What is the min “N” comparison needed?
 - c) Ensure the design has "**Connectivity**." This means that even if item A and item B never appear in the same set together, the model can still indirectly infer the relative importance between A and B through their respective comparisons with other items like C, D, and E.

The “I still dont really can explain part”

Rank Centrality

- Assigns global scores to items from pair-wise comparison data
- Using **random walk** on a comparison data, where an item's final score is its stationary probability
- In short :using the win rate of the option to distribute a fix total score multiple times until the result converged
 - 1) Build a matrix using the outcome of comparison
 - 2) Create a probability matrix that describes the flow of "rank" from losers to winners.
 - 3) Repeatedly update a score vector ($s_{\text{new}} = M * s_{\text{old}}$) until it stabilizes. This process finds the principal eigenvector of the matrix.
 - 4) The final, stable score vector provides the ranks. The higher the score, the higher the rank.

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Scoring two content creation solutions

1. LLM (e.g., ChatGPT) do pairwise comparisons and provides scores
2. Human raters pairwise comparisons and provides scores
3. Human raters do **pre-rating** and given **qualitative feedback** on why they prefer some solutions over others — we can then feed these qualitative assessments into an LLM to be an automated rater
 - a. Un-prompted LLM might have different preferences than human raters (e.g., structure > content)

pairwise comparison

determine their relative importance

ELO: score & rank

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}.$$

E_A prob of a win

R_A current ELO score of player A

$$E_B = \frac{1}{1 + 10^{(R_A - R_B)/400}}.$$

R'_A ELO score after the match

$$R'_A = R_A + K(S_A - E_A).$$

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K-factor one of the factor that determine how violently the score change, usually high if the player have high **uncertainty** (new players, old player who didn't play for a while)

- S_A : score (0, 0.5, 1)

Starting Match:

- Everyone starts with baseline 1500 in chess

Bradley-Terry model

- Calculate hidden “power value” p_i , using the existing comparison result and Maximum Likelihood estimation method

$$P(i > j) = p_i / (p_i + p_j)$$

$$\ln(P(i > j) / P(j > i)) = \beta_i - \beta_j$$

$$\beta_i = \ln(p_i)$$

Rank Centrality

- Assigns global scores to items from pair-wise comparison data
- Using random walk on a comparison data, where an item's final score is its stationary probability
- In short :using the win rate of the option to distribute a fix total score multiple times until the result converged

BWS

- Participant compare between **3 or more options** and chose the best and the worse from it and do severe groups of comparison
- Calculate how many times each option was voted as best and as worst
- Subtract the number of worst votes from the number of best votes for each option, and then rank the options by that result
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